Competing Risks and Multi-State Models

Concepts, methods and software

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Lecture notes competing risks (day 1 and day 2)

Ronald Geskus (competing risks),
Oxford University Clinical Research Unit (OUCRU),
Hospital for Tropical Diseases, Ho Chi Minh City, Viet Nam

Hein Putter (multi-state models)
Department of Biomedical Data Sciences
Leiden University Medical Center, Leiden, the Netherlands

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Abstract.
The role of competing risks and intermediate states in the analysis of time-to-event data is increasingly acknowledged. Regarding competing risks, confusion remains about the proper analysis. The most important reason for the confusion is conceptual. What do the different quantities represent, which one is best suited for answering our study question, and which assumptions do we need to make with respect to censored data? Once a model that gives interpretable results has been chosen and the assumptions are satisfied, estimation is relatively straightforward with right censored and/or left truncated data using readily available software. Extensions to the Markov multi-state setting are relatively straightforward if transition times are observed exactly.

We first cover competing risks analysis. We explain when and how we need to take them into account. We define the main concepts: cause-specific cumulative incidence, cause-specific hazard and subdistribution hazard and explain estimation of these quantities. We contrast cause-specific and marginal analyses and discuss the assumption of independent competing risks. We explain regression models on each of the three quantities. We show how analyses can be performed with standard software.

In the second part of the course, the extension to (Markov) multi-state models is discussed. We cover transition intensities and transition probabilities, nonparametric estimation and regression models, as well as methods to obtain predictions of future events, given the event history and clinical characteristics of a patient. We show how to perform many types of analyses using standard software, using the same techniques as in the multi-state approach of the competing risks analysis.

More details on competing risks, as well as the analysis in situations with intermediate states, can be found in the *Tutorial in biostatistics: competing risks and multi-state models* [41] and the book *Data analysis with competing risks and intermediate states* [15]. A more recent review on competing risks is given in *Competing Risks: Concepts, Methods and Software* [17].

Contents

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1 Introduction

In the end we all die. More interesting than the death event itself is the time component: at what age does one die and what characteristics make some individuals die earlier than others? Survival analysis provides the set of tools that help answering the question of which factors influence the time until the occurrence of some event (which is not restricted to be death).

In the end we all die, but not all from the same cause. Information on the spectrum in causes of death has added value. A competing risks model extends the classical survival setting by considering a collection of mutually exclusive potential event types. Figure 1 shows the number of individuals that died of different causes in the twentieth century. The smallest subgroups may not be readable (have a look at the website if you want to read them[1]), but it is seen that the most frequent category is made up of the non-communicable diseases, within which cardiovascular diseases form the largest subgroup. The causes of death are competing: if one occurs, the others do not. Again, including the time component has added value. Some causes, like measles, tend to occur at a young age, whereas others, like prostate cancer, almost exclusively occur in older men.

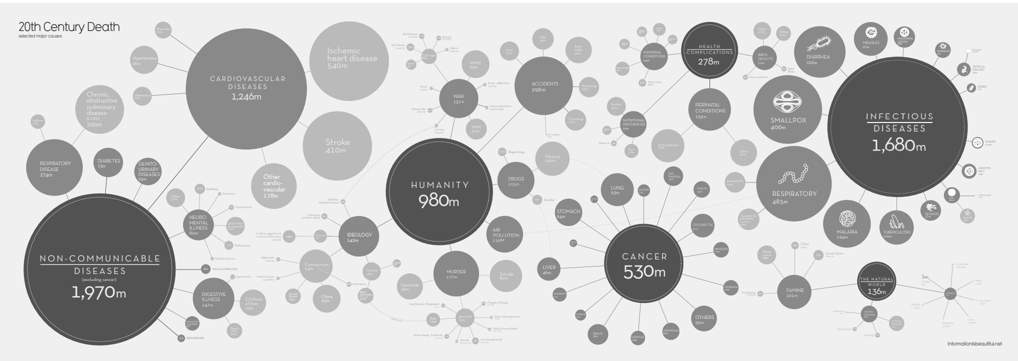

Figure 1: All causes of death in the 20th century.

In the end we all die, but not all for the same reason nor with the same life histories. Even if two individuals die at the same age and of the same cause, their life courses have been different. Between birth and death, intermediate events occur that influence one's life course. The are several ways to model the occurrence of such events and how they impact later events. One approach is via a multi-state model, in which events are seen as transitions from one state to another. Under the frequently assumed Markov property, a multi-state model can be seen as a sequence of competing risks models. For Markov models with exactly observed transitions and right censored data, most of the estimation techniques are a direct extension of those from the competing risks setting and the same software can be used. Some computations are more complex and require software that has been written specifically for such multi-state models.

[1]The figure was downloaded on September 1st, 2013 from the website `http://www.informationisbeautiful.net/visualizations/20th-century-death`. Author: David McCandless.

Part I
Main Concepts and Quantities

2 Examples Competing Risks

Competing risks are event-type outcomes that are mutually exclusive. The event can be biological, such as infection, disease onset or death from a natural cause. It can also be a human intervention such as organ transplantation, start of treatment or death from suicide. In both situations we write that the event "occurs", although an intervention is typically a human decision.

Competing risks can be seen as levels of a binomial or multinomial outcome. The probability of each of the competing risks to occur may be related to covariables via a (multinomial) logistic regression. Usually there is a time component involved in the study question. Special techniques have been developed for competing risks analyses with incomplete time-to-event data. These techniques, which extend the ones from standard survival analysis, are explained in this course.

Sometimes each of the event types are of interest, but more often there is one event type of interest and a competing event may prevent it from occurring. The most classical example is death as a competing risk of some non-fatal event. If the competing risk is a non-fatal event, the event of interest can still occur, but it changes the risk of the event of interest to such extent that it creates a completely new setting

Event-type outcomes that are "competing" (mutually exclusive)

- Each event type is of interest
  - A: Effect of dexamethasone on recovery and death after SARS-CoV-2 infection (binary outcome, complimentary levels)
  - B: Spectrum in causes of death after HIV infection (role of cART)
  - C: AIDS and SI switch after HIV infection
  - D: Death and relapse after bone marrow transplant (Practical)
- One event type of interest, intervention prevents it from occurring
  - E: Intervention (transplantation or treatment) competing risk for natural disease course
  - F: Treatment change or interruption competing risk for treatment failure
  - G: Discharge competing risk for hospital-acquired infection
- One event type of interest, biological event prevents it from occurring
  - H: Death competing risk of non-fatal event (e.g. disease onset)

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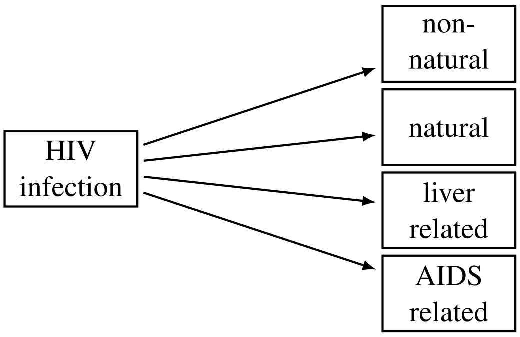
2.1 All event types of interest

A: Treatment for COVID-19. The RECOVERY randomized controlled trial (RCT) has been one of the most influential studies investigating the efficacy of treatment in hospitalized patients with COVID-19. One of the comparisons was between dexamethasone and standard of care [42]. The primary outcome was all-cause mortality within 28 days after randomisation. Time to hospital discharge within 28 days was defined as a separate secondary outcome. Hospital discharge within 28 days could be seen as a complimentary competing risk, representing recovery or cure. They used time-to-event methods (rather than a logistic regression) because of censored data: ten percent of the patients remained in the hospital for more than 28 days. One could make "hospitalized for more than 28 days" into another competing risk. However, it would not make sense as an efficacy outcome, because it combines the beneficial event "discharge" and the unfavourable event "death" and all patients will ultimately die from COVID-19 or be discharged. Also, they may have been interested in the effect of dexamethasone on time-to-discharge rather than discharge within 28 days.

B: Spectrum in causes of death after HIV infection. Combination anti-retroviral therapy (cART) greatly reduced AIDS-related mortality in HIV infected individuals. Its efficacy has been shown in RCTs. An interesting question is what HIV infected individuals will die of if they no longer die of AIDS. Is their spectrum in causes of death and its relation with age similar to that of individuals without HIV infection? Side effects of cART may increase the risk to die of other causes like cardiovascular disease. But even if cART has no direct effect on other causes of death (COD), these other COD may become more common because we all die of some cause: *"in the end we all die, but not all at the same age and from the same cause"*. Using a large data set of HIV infected individuals from different risk groups, we studied how the introduction of cART changed progression to four different causes of death [24]. We chose 1997 as the single cutpoint (which is certainly too simplistic because the efficacy of cART has improved over time). Individuals who became infected before 1997 contributed to both calendar periods. The different causes of death act as competing risks, as depicted in slide 4.

B: Spectrum in causes of death after HIV infection



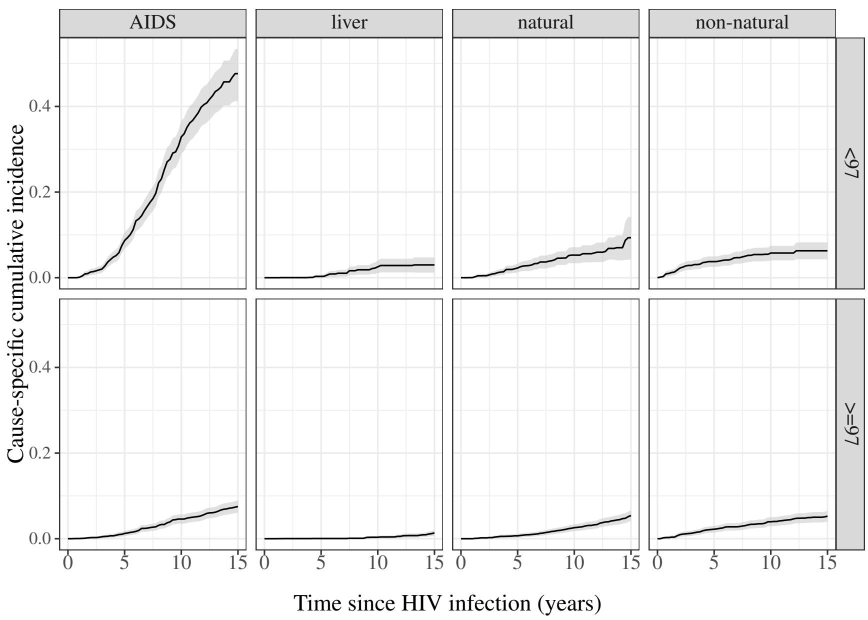
- Has the spectrum in causes of death changed after the introduction of cART (combination anti-retroviral therapy)
- Side effects of cART? *Etiology* (≈ cause-specific hazard)
- In the end we all die. *Prediction* (≈ subdistribution hazard)

.4

The question whether cART use itself increases other types of mortality is etiologic/causal in nature. The clinical/predictive question is to quantify the effect of cART on cumulative cause-specific mortality, taking into account that other causes of death will become more apparent if persons no longer die of AIDS. In the competing risks setting, they are related to two different definitions of hazard.

Estimates of progression to the competing causes of death by calendar period are shown in slide 5. There is no indication that the other causes death had become more frequent. Note however that this may be explained by the presence of risk factors that act as confounders for the relation between calendar period and COD. Risk group is one of them. In the populations from which our data were sampled, HIV infection through injecting drug use went down over time. This may explain the decrease in liver related mortality (hepatitis C infection is very common in HIV infected injecting drug users) as well as in the mortality due to non-natural causes (overdose, suicide). We briefly return to this example in Section 16.2.

Cause-specific mortality by calendar period



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C: First event after HIV infection During the course from HIV infection to death, intermediate events may occur that have an impact on subsequent disease progression. One such event is a switch of the HIV virus to the so-called syncytium inducing (SI) phenotype[2]. Since neither of the events is final, we can also describe what happens after the first event using a multi-state model. The whole process is depicted in Figure 2 by the states that are connected via dashed arrows (for HIV related events) and dotted arrows (for death due to other causes). In this course we focus on the first event from the initial state ("HIV infected"), which is either "AIDS before SI switch" or "SI switch before AIDS". These are depicted via the two solid arrows in Figure 2.

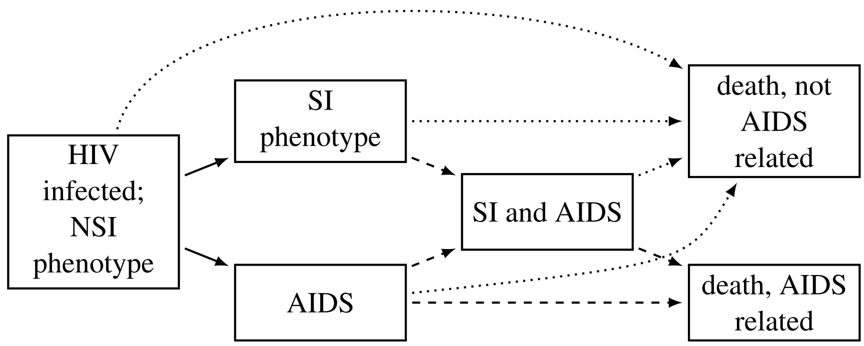


Figure 2: Example: progression from HIV infection to AIDS and death, with an SI switch as possible intermediate event.

A deletion in the gene that codes for the C-C chemokine receptor 5 (CCR5) reduces susceptibility to HIV infection. This deletion is called CCR5-Δ32. Persons with the CCR5-Δ32 deletion on both chromosomes rarely become HIV infected. If the deletion is present on one of the chromosomes, HIV infection does occur but progression to AIDS is slower than for individuals in which both chromosomes do not have the deletion. The reason is that the NSI virus uses the CCR5 receptor for cell entry, and the deletion makes

[2] Upon infection, the virus is almost always of non-SI (NSI) phenotype.

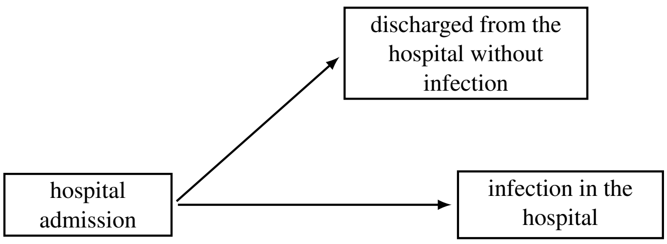
cell entry and multiplication more difficult. However, if the virus in an individual has switched to the SI phenotype, it can use the CXC chemokine receptor 4 (CXCR4) as an alternative for cell entry[3]. Little is known about factors that cause the SI phenotype to appear. Since in individuals with the CCR5-Δ32 deletion the NSI phenotype has more difficulties to replicate, it may be that the SI phenotype appears more rapidly. We use this example in Sections 7, 9, 12, 13, 15, 16 and 17.

2.2 One event type of interest, intervention as competing risk

E: Disease outcome before intervention. For many diseases, treatments have been developed that greatly improve disease course. When studying the natural disease course, treatment initiation act as a competing risk. One example is start of cART in HIV infected individuals [20]. Another intervention that stops the natural disease course is transplantation. For example, liver transplantation is a life saving intervention in patients with primary sclerosing cholangitis [54].

F: cART treatment failure; early versus late starters. In the search for the best moment to start cART, we investigated whether individuals who start cART in the first year after HIV infection ("early starters") have a different risk of treatment failure than individuals who start later, during chronic infection [58]. Treatment change and treatment interruption were interventions that act as competing risks of treatment failure. This example is Exercise IV.19.3.

G: Staphylococcus infection during hospital stay



- Etiology (biological question): infection risk in hospital. What would happen *if* everyone stayed in hospital (no discharge)?
  - *marginal distribution/net risk*
- Predict (clinical question): burden due to infection *while* in hospital; discharge prevents event to occur
  - *cause-specific cumulative incidence /subdistribution/crude risk*
- *Which of two hospitals has higher risk may depend on type of risk*

.6

Infection with *Staphylococcus aureus* is frequent among patients with burn wounds, leading to high morbidity and mortality [37]. "Infection in the hospital" is the event of interest, with "discharge without infection" as competing risk, as shown schematically in slide 6. How we deal with patients that are discharged without infection depends on the study question.

The *etiologic* question is to quantify the *marginal* distribution of infection in the hospital. It describes the distribution of one outcome (time to infection), in the (hypothetical) situation that the other outcome (discharge) does not occur. It depends on

[3] Nowadays, CCR5-tropic and CXCR4-tropic virus are used as names instead of non-SI and SI phenotype.

environmental conditions that lead to infection, such as the prevalence of staphylococcus bacteria. Typically the marginal cumulative probability increases to 100% over time since admission.

The *clinical* objective is to predict the number of infections *while* staying in the hospital. This is important if we want to know the disease burden of infection, and its costs, during hospital stay. The time until the event “infection in the hospital” is described by the *infection-specific* distribution, with “discharge without infection” as competing risk. Some individuals never become infected because they are discharged, and the cumulative probability of infection over time since admission will not reach 100%.

If we want to estimate the marginal distribution, we interpret “discharge without infection” as an event that *censors* infection in the hospital. If we use the Kaplan-Meier estimator of the net risk, we assume that discharged individuals, had they remained in the hospital, would have had the same infection risk as the ones that did remain in follow-up. If the independence assumption does not hold, then estimation of the marginal distribution is difficult. The Kaplan-Meier can still be computed, but it doesn’t estimate anything meaningful (see also Section 8).

If we want to estimate the infection-specific distribution, then we do not need to make this assumption. Discharge is seen as a separate event, and we do not make the assumption that individuals who are discharged can be represented by the ones that remain in the hospital. We explain the difference via an artificial data example.

Estimation

week	0-1	1-2	2-3	3-4	4-5	5-6	6-7	>7
infection	1	2	6	11	9	11	2	18
cumulative	1	3	9	20	29	*40*	42	*60*
discharge	5	9	6	6	9	12	4	35
cumulative	5	14	20	26	35	*47*	51	*86*

- *Subdistribution*. Estimated as frequency of events:
  $\widehat{P}(\text{infection } \leq 6\,\text{weeks}) = 40/146$
  $\widehat{P}(\text{discharge } \leq 6\,\text{weeks}) = 47/146$
  Individuals with competing event remain in denominator, competing event ignored in estimation
- *Marginal distribution*. Discharged individuals treated and interpreted as censored; Kaplan-Meier nonparametric estimator, assumes independent censoring .7

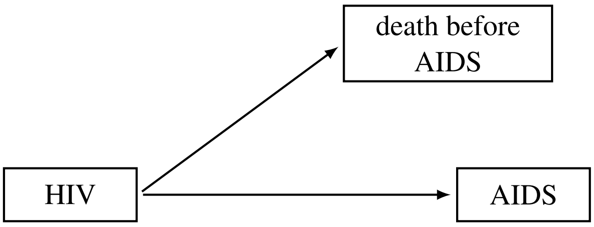
When estimating the infection-specific distribution, we compute the (cumulative) fraction of individuals over time that becomes infected. Those that are discharged without infection will never contribute to the numerator, but they continue to contribute to the denominator; discharge is not treated as a censoring event. In our example there are no individuals that leave the study within the first seven weeks without having experienced either death or discharge (i.e. there are no right censored data). Therefore, the probability to become infected within six weeks can be estimated as the observed frequency $\widehat{P}(\text{infection } \leq 6 \text{ weeks}) = 40/146$.

When comparing two hospitals, results of the marginal analysis may differ from results of the competing risks analysis. If the force of infection is the same in both hospitals, then the marginal distribution of infection is the same. But if the rate of discharge is different, fewer infections will be observed in the hospital with the larger rate of discharge: more individuals have already left the hospital before they could be infected.

2.3 One event type of interest, biological event as competing risk

H-1: Bladder cancer relapse, death other causes (DOC) competing. Elderly people can die from many different causes. Porta-Bleda *et al.* [39, p.32] studied whether the time from diagnosis of bladder cancer to relapse depends on gender. In the data, death from other causes (DOC) was a competing risk for relapse. She observed a strong difference by gender. This example is Exercise II.10.3.

H-2: Death prevents AIDS in HIV infected injecting drug users.

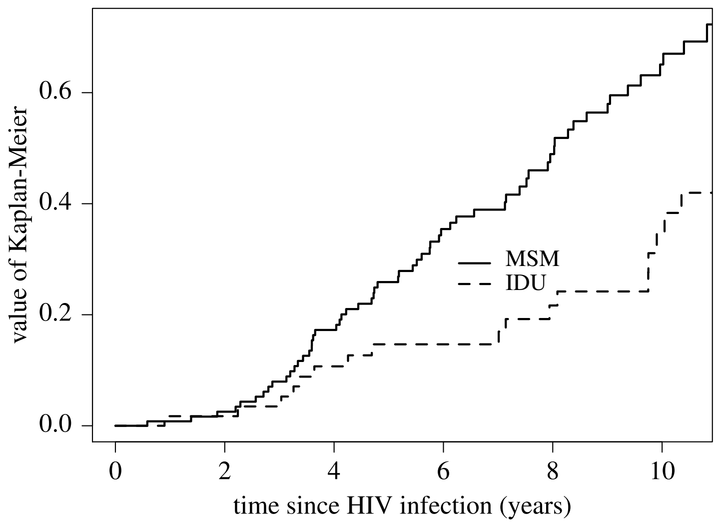


- Interest in time to AIDS if there were no pre-AIDS death.
  Interest in etiology and marginal distribution
- Compare men who have sex with men (MSM) and injecting drug users (IDU) Data from Amsterdam Cohort Studies: 99 IDU; 127 MSM
- Kaplan-Meier: leave risk set at death before AIDS
  Assumption: those that died had equal AIDS risk .8

Biological events like death may hamper the study of the natural disease course. Especially among HIV infected injecting drug users (IDU), studying time from HIV infection to AIDS has been hampered by high mortality from other causes [16, Section 2.3]: death before AIDS is a competing risk. We want to answer the etiologic question whether the natural history from HIV infection to AIDS differs between men who have sex with men (MSM) and IDUs. The natural history is the progression that would be observed if there were no interventions that change natural AIDS progression (like use of therapy) or prevent AIDS from occurring (like pre-AIDS mortality).

The Kaplan-Meier estimator of the marginal distribution suggests that injecting drugs is beneficial in preventing progression to AIDS. The problem is that this estimate is severely biased because pre-AIDS death is not independent from progression to AIDS (see Section 8.5).

Kaplan-Meier: IDU much slower progression ($p = 0.001$)



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3 On Rates and Risks

Beyond classical survival analysis

- Classical: transition between two states, one event type.
  "We all die, but not all at the same age"

$$\boxed{\text{initial state}} \xrightarrow{\text{time}} \boxed{\text{event}}$$

- *Life* and *death* are richer than that
  1. Multiple event types. Competing risks:
     "we all die, but not all at the same age and from the same cause"
  2. Intermediate events. Multi-state model:
     "we all die, but not all at the same age, not from the same cause and with different life histories"
- Two components
  - *Events/Transitions.* Initial, intermediate and final states
  - *Time.* What is the time origin? Multiple time scales?

.10

3.1 Classical survival analysis

We define the main quantities in classical time-to-event ("survival") analysis. The time component allows for two different quantifications: instantaneous and cumulative.

The instantaneous measure is the fraction of individuals that develops an event at some time, amongst the ones that are at risk for experiencing the event at that moment. In epidemiology, it is commonly called the rate, but when the event is a disease, the term incidence is used as well. It is also been called the risk, but we reserve that word for the cumulative measure. The statistical name is the hazard, which is the one that we will use most of the time.

The cumulative measure is the fraction of individuals that develops an event within some time span, amongst the ones that are at risk for experiencing the event at the start of that time period. In epidemiology it is commonly called cumulative incidence or (actuarial) risk. The statistical name is the cumulative distribution function. It is the complement of what is called the "survival function".

Due to the presence of right censored and left truncated data, the analysis of time-to-event data is easier via the hazard. Most models and estimation methods are based on the hazard. Examples are the Kaplan-Meier estimate of the survival function and the Cox proportional hazards regression model.

The formal definition differs slightly depending on whether the distribution of T is discrete or continuous. The distribution of T is discrete if events can only occur at a limited number of time points, say at $\mathscr{T} = \{t_1, \ldots, t_L\}$. The hazard at t_l is defined as $\mathrm{P}(T = t_l \mid T \geq t_l)$. The distribution of T is continuous if T can take any value over some time range. The hazard at time t is defined as the conditional probability of an event in a tiny interval from t to $t + \Delta t$, given that the individual is still at risk at the beginning of that interval, divided by the length of the interval.

Main quantities in classical survival analysis

- T: time span from origin $\rightarrow$ end.
  Time origin determined by entry into "at risk" state.
- *Rate/incidence/hazard* $h(t)$: instantaneous event probability, conditionally on being at risk
- *Risk/cumulative incidence* $F(t) = \mathrm{P}(T \leq t)$
  survival function $\overline{F}(t) = 1 - F(t)$: probability (not) to have experienced the event by time t
- Event probability $p(t_l) = \mathrm{P}(T = t_l)$ (discrete distributions)
  Event density $f(t) = dF(t)/dt$ (continuous distributions)

.11

In classical survival analysis with one type of event, hazard and cumulative incidence are unambiguously related via a one-to-one relation: if we know one, we also know the other. An important consequence of this relation is that the effect of a covariable as quantified via a regression model for the hazard translates to a similar effect on the cumulative incidence.

One-to-one relation

Each of the three quantities completely describes distribution:

discrete Event times: $t_1, \ldots, t_L$

$$\begin{aligned} h(t_l) &= \mathrm{P}(T = t_l \mid T \geq t_l) = \mathrm{P}(T = t_l \mid T > t_l-) \\ &= \mathrm{P}(T = t_l \mid T > t_{l-1}) = \frac{F(t_l) - F(t_{l-1})}{\overline{F}(t_{l-1})} \\ \overline{F}(t) &= \mathrm{P}(T > t) = \prod_{t_l \leq t} \{1 - h(t_l)\} \end{aligned} \tag{1}$$

continuous

$$\begin{aligned} h(t) &= \lim_{\Delta t \downarrow 0} \frac{\mathrm{P}(t \leq T < t + \Delta t \,|\, T \geq t)}{\Delta t} \\ &= \frac{f(t)}{\overline{F}(t)} = -\frac{d \log\{\overline{F}(t)\}}{dt} \\ \overline{F}(t) &= \exp\left\{-\int_0^t h(s)ds\right\} = \exp\{-H(t)\} \end{aligned} \tag{2}$$

.12

3.2 Competing risks analysis

In a competing risks analysis we want to quantify the progression to a specific event type, taking into account that individuals can also progress to other end points. Let the random variable T_k represent the time until an event of a specific type k occurs. The cumulative progression over time to the event of interest, $\mathrm{P}(T_k \leq t) = \mathrm{P}(T \leq t, E = k)$, is called the cause-specific cumulative incidence. Two hazards can be defined. One is the subdistribution hazard, which is defined in Section 3.4. The other hazard is close to the standard rate concept: it is the fraction of individuals that develops the event of interest at some time point amongst those that are still at risk of experiencing the event (see definition on slide 14). It is usually called the cause-specific hazard. Its definition has a direct generalisation to the multi-state setting. Therefore, we call this the multi-state approach to competing risks.

Notation and definitions

- Competing risks: type $E \in \mathscr{K}$. We can assume $\mathscr{K} = \{1, \dots, K\}$
- $T \sim F$ time to event (of any type); $F(t) = \mathrm{P}(T \leq t)$
- Overall hazard h: $\overline{F}(t) = 1 - F(t) = \mathrm{P}(T > t) = \exp\{-\int_0^t h(s)ds\}$
- Cause-specific cumulative incidence: $F_k(t) = \mathrm{P}(T \leq t, E = k)$; $E \in \mathscr{K}$
  Property: $\sum_{e=1}^K F_e(t) = \sum_{e=1}^K \mathrm{P}(T \leq t, E = e) = F(t)$
- Subdistribution random variable $T_k \sim F_k$:
  $T_k = T \times I\{E = k\} + \infty \times I\{E \neq k\}$ (event of interest)
- $\mathrm{P}(T_k \leq t) = \mathrm{P}(T \leq t, E = k) = F_k(t)$
- Subdistribution hazard (continuous distribution):

$$h_k = \lim_{\Delta t \downarrow 0} \frac{1}{\Delta t} \times \mathrm{P}\{t \leq T_k < t + \Delta t \,|\, T_k \geq t\}$$

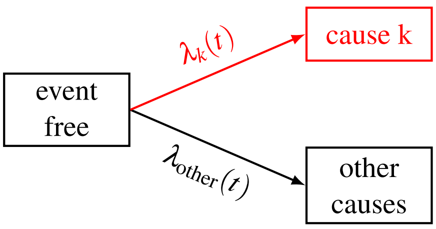
$$\overline{F_k}(t) = \mathrm{P}(T_k > t) = \exp\left\{ -\int_0^t h_k(s)ds \right\}$$

.13

3.3 Multi-state approach

The cause-specific hazardλ_k describes the rate of occurrence of the event of interest. Individuals who experience a competing event leave the risk set. It is a special case of the transition hazard in multi-state models [15, 41]. Therefore, using the cause-specific hazard as basis can be seen as the multi-state approach to competing risks.

The multi-state approach: cause-specific hazard



- Transition rate to cause k. For continuous distribution:

$$\lambda_k(t) = \lim_{\Delta t \downarrow 0} \frac{\mathrm{P}(t \leq T < t + \Delta t, E = k \,|\, T \geq t)}{\Delta t}$$

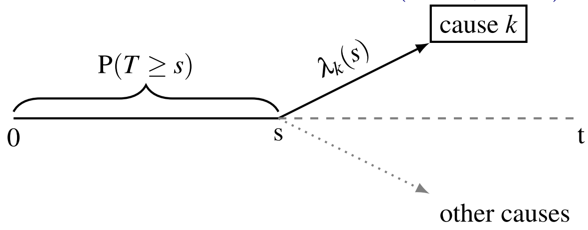
- Sum over causes is overall hazard: $\sum_{e=1}^K \lambda_e(t) = h(t)$
- Cause-specific hazard directly generalizes to multi-state setting (called transition hazard) .14

Using the laws of probability, it is easy to derive that the overall hazard is equal to the sum of the cause-specific hazards

$$\begin{aligned}
\sum_{e=1}^K \lambda_e(t) &= \sum_{e=1}^K \lim_{\Delta t \downarrow 0} \frac{\mathrm{P}(t \leq T < t + \Delta t, E = e \,|\, T \geq t)}{\Delta t} \\
&= \lim_{\Delta t \downarrow 0} \frac{\sum_{e=1}^K \mathrm{P}(t \leq T < t + \Delta t, E = e \,|\, T \geq t)}{\Delta t} \\
&= \lim_{\Delta t \downarrow 0} \frac{\mathrm{P}(t \leq T < t + \Delta t \,|\, T \geq t)}{\Delta t} = h(t) \,.
\end{aligned}$$

The cause-specific cumulative incidence combines the cause-specific hazard and the probability to remain free of any event type, see slide 15. Contrary to the classical survival setting, it does not have a one-to-one relation with the cause-specific hazard. As a consequence, the way in which covariables are associated with a cause-specific hazard may not coincide with the way in which these covariables are associated with the cause-specific cumulative incidence (see Section 9.2 for an example).

From hazard to cumulative scale $\mathrm{P}(T \le t, E = k)$



- $F_k(t) = \mathrm{P}(T \le t, E = k) = \int_0^t \overline{F}(s-)\lambda_k(s)ds$
- Depends on all cause-specific hazards via overall "survival"

$$\overline{F}(s) = \exp\left\{-\int_0^s h(u)du\right\} = \exp\left\{-\sum_{e=1}^{K}\int_0^s \lambda_e(u)du\right\}$$

$\Longrightarrow \lambda_k$ does not uniquely specify $\mathrm{P}(T \le t, E = k)$ .15

3.4 Subdistribution approach

There is a hazard that does have the one-to-one relation with the cause-specific cumulative incidence. Since the event of interest never happens for part of the individuals, the cause-specific cumulative incidence doesn't go up to the value one. Therefore, it is most often called the *subdistribution hazard* or subhazard. Definitions and properties in the subdistribution approach are completely similar to the standard setting with a single event type, but with T and F replaced by T_k and F_k. There is a hazard function h_k, called the subdistribution hazard, that uniquely describes the distribution. For a discrete distribution it is defined as

$$h_k(\tau_j) = \mathrm{P}(T_k = \tau_j \mid T_k > \tau_{j-1}) = \mathrm{P}(T_k = \tau_j \mid T_k \ge \tau_j)\,.$$

Using the law of conditional probabilities, it directly relates to F_k via the product formula

$$\begin{aligned}\overline{F_k}(t) = 1 - F_k(t) &= \mathrm{P}(T_k > t) \\ &= \prod_{\tau_j \le t} \mathrm{P}(T_k > \tau_j \mid T_k > \tau_{j-1}) = \prod_{\tau_j \le t}[1 - h_k(\tau_j)]\,. \qquad (3)\end{aligned}$$

In this approach, the competing event is basically ignored. An individual that experiences a competing event is assumed to remain at risk for the event type of interest. When we rewrite the subdistribution hazard in terms of the multi-state variable (T, E), we see what is different from the cause-specific hazard: the term in the denominator not only contains the probability to be event-free just before s, but also the probability to have experienced a competing event at an earlier time point.

Subdistribution hazard

$$\begin{aligned}h_k(s) &= \lim_{\Delta s \downarrow 0} \frac{1}{\Delta s} \times \mathrm{P}\{s \le T_k < s + \Delta s \,|\, T_k \ge s\} \\ &= \lim_{\Delta s \downarrow 0} \frac{\frac{1}{\Delta s}\mathrm{P}\{s \le T < s + \Delta s, E = k\}}{\mathrm{P}\{T \ge s \text{ or } (T < s, E \ne k)\}}\end{aligned}$$

- Denominator: event free *or with earlier competing event*
- Interpretation controversial
  - Not a rate in epidemiological sense, unless we can include immune individuals in the risk set (e.g. vaccination; unobserved cure)
- One-to-one relation with crude risk

$$\overline{F_k}(t) = \prod_{t_l \le t}\left\{1 - h_k(t_l)\right\} \quad \text{or} \quad \overline{F_k}(t) = \exp\{-\int_0^t h_k(s)ds\}$$

.16

3.5 Type of estimands in the presence of competing risks

We summarize the main quantities in the table in slide 17.

Rates and risks in competing risks setting

	hazard		cumulative	
competing risks	marginal	λ^m	net risk marginal survival function marginal cumulative incidence	$F_k^m(t)$
	cause-specific	λ_k	no corresponding quantity	
	subdistribution	h_k	*crude risk* *cause-specific cumulative incidence*	$F_k(t)$
combined	overall	h	overall risk overall survival function overall cumulative incidence	$F(t)$

.17

In the presence of competing risks, we distinguish between a marginal analysis and a competing risks analysis. In a marginal analysis we want to quantify the hazard and cumulative incidence that would be observed if the competing events did not occur. So we want to perform a classical survival analysis with a single event type. For example, we quantify the rate and risk of staphylococcus infection that would be observed if everyone would stay in hospital. We use the names *marginal hazard* for the rate and *marginal cumulative incidence* or *net risk* for the cumulative measure.

In a competing risks analysis we want to quantify the progression to a specific event type, taking into account that individuals can also progress to other events instead. The cumulative progression over time to the event of interest has been called absolute risk, crude risk, cause-specific cumulative incidence (function), crude cumulative incidence (function), crude incidence curve, crude probability function, actual probability, cause-specific risk, cause-specific failure probability, subdistribution function and even just cumulative incidence. We will use the names *cause-specific cumulative incidence*, *crude risk* and *subdistribution function*. It is called a subdistribution function because it is not a proper distribution: its value does not increase from zero to one over time because competing events prevent the event of interest from happening.

We can also combine all event types into a single composite event type. Then we are back in the standard setting with a single event type and a random variable T that takes values according to this time-to-event distribution. For example, instead of discriminating between causes of death, we look at overall mortality. We will use the names overall hazard and overall cumulative incidence for this quantity. If death and some other event are competing, the cumulative probability to remain free of both events is often called the *event-free survival*. The combined event plays a role in estimation of the cause-specific cumulative incidence based on the cause-specific hazard.

4 Data Structure

4.1 Time scales

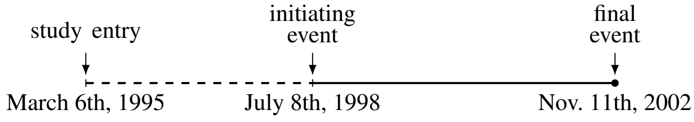
Data is collected in calendar time. Calendar time is also the time scale of interest in the analysis if we want to investigate the development of an epidemic like HIV or SARS-CoV-2 at the population level. Time scales have a reference value. The standard reference value of calendar time in large part of the world is the start of the Christian calendar.

Statistical packages by default choose another origin when representing calendar time[4].

More often, the time scale of interest is initiated by the occurrence of some event. Examples of initiating events are birth, hospital admission, HIV infection, AIDS diagnosis, start of cART and receiving a bone marrow transplant.

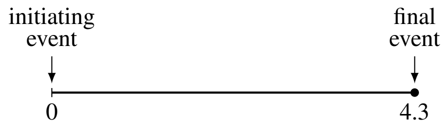
Time scales

- Data collection in calendar time



calendar time scale

- Often personal/patient time scale more relevant



personal time scale (years)

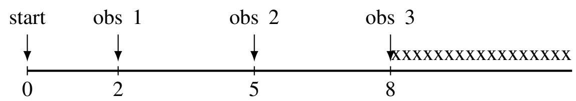
- Every time scale has a reference value: origin → end
- Classical survival analysis/competing risks: often single principal time scale .18

4.2 Right censored data

Usually part of the observed time-to-event data is incomplete. The most common type is right censored data, which is the type that we will consider in this course. An individual is right censored with respect to an event if he was observed to be at risk for experiencing the event during some time but the event itself was not observed. Data is left censored if the the event happened before the first observation time. Sometimes, information on the event status is only known at certain observation times. We may know that the event happened in-between two observation times, but the exact timing is unknown. This is called interval censored data.

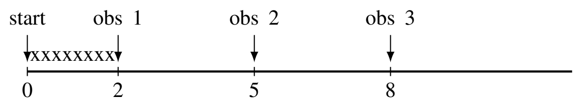
The observation process: censored event time

- Right censored: event free until last observation time



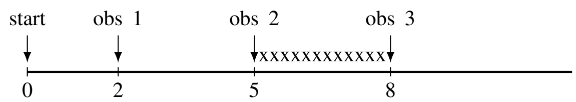
event happened after observation time 8

- Left censored: event before first observation time



event happened before observation time 2

- Interval censored: event between two observation times



event happened between observation times 5 and 8

.19

[4] SPSS uses October 14, 1582 (the start of the Gregorian calendar), Stata uses January 1st, 1960 and R uses January 1st, 1970.

We distinguish between three reasons for right censored data. Each of the three reasons differs with respect to the occurrence and possible observation of the event.

Right censored event time: three reasons

Administrative censoring At date of analysis. Still in follow-up and event may be observed in the future

Loss to follow-up Left the study before date of analysis. May happen for different reasons (move to another country, too ill to participate). Event may have happened, but is not observed.

Competing risks Two types:

- **Absorbing event:** other event (e.g. death) prevents occurrence of event of interest. No further follow-up information
- **Non-absorbing event:** "artificial censoring". Still in study, but conditions have changed too much to stay included.
  Another example: first event analysis

.20

We comment on each of them in more detail.

Administrative censoring This occurs if individuals are still in follow-up without having experienced the event yet. Usually the administrative censoring time is based on a fixed calendar date that is chosen as the end of follow-up for the analysis, aptly named "date of analysis". All information after this date is not taken into consideration. On a personal time scale the administrative censoring time will differ per individual, unless all individuals entered the initial state at the same calendar date. In the statistical literature administrative censoring has been called progressive type I censoring [3].

Loss to follow-up Individuals are lost to follow-up if they stopped being under observation before the date of analysis and before the event occurred. This can happen for several reasons. They may have moved to another country, they may have lost motivation to continue in the study or they may have become too ill to continue participation.

Competing risk A competing risk is another event that prevents the event of interest from happening. It may be death, but it may also be that an individual can still experience the event, but another event has happened first that changes the conditions under which the event of interest can occur. This has been called artificial censoring [26]. Examples of artificial censoring are discharge from hospital if infection during the hospital stay is the event of interest (example G), or AIDS before SI switch if SI switch before AIDS is the event of interest (Example C).

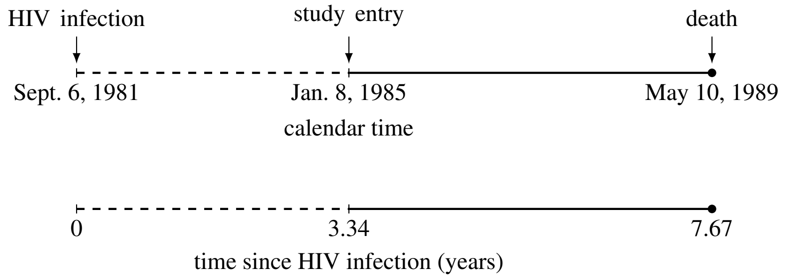
4.3 Late entry/left truncation

A typical example of left truncated data is found in cohort studies with late entry: some individuals enter the study after the event that defines the origin of the personal time scale. As an example, consider the data from the Amsterdam Cohort Studies on HIV Infection and AIDS (ACS). These studies were started in October 1984, shortly after the first test to detect HIV infection had become available. Suppose that an individual became infected in September 1981[5]. If he entered the ACS in January 1985 and died as a consequence of HIV infection in May 1989, his time and event information is as illustrated in slide 21. In his personal time scale, he entered the study 3.34 years after HIV infection.

[5] This may be known through information from stored blood samples.

Late entry

Individuals enter risk set after time origin



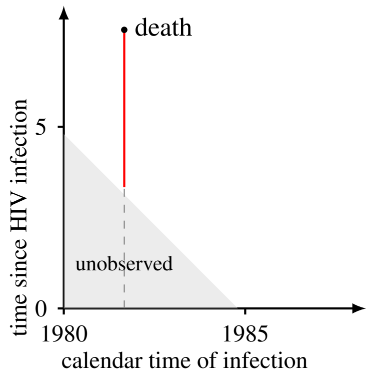
- Late entry usually induces length biased sampling (left truncation)

.21

Left truncation is present when the data is subject to a form of length biased sampling: individuals that became HIV infected at the beginning of the epidemic and had a short time to AIDS and death are missed. The information with respect to both the calendar time of HIV infection and the personal time from HIV infection to death can be shown graphically as in slide 22. A fast progressor who became infected in 1980 and died before the study started, for example in July 1984, did not have the possibility to enter the study. The grey area represents the "unobserved" period: all the combinations of the infection date and the subsequent time to death that make it impossible to enter the study because death occurred before the study started.

Left truncation: fast progressors underrepresented

- Example: time from HIV infection to death
- study started in 1985
- individual missed if died before 1985
- individual infected in 1982 only at risk from 1985
- not same as left censoring



.22

Not only HIV infected individuals that died before the start of the study will be missed. The same holds for any HIV infected individual that died before he would enter the study. For example, the individual from slide 21 entered the study on January 8, 1985 and would have been missed as well if he had died between the start of the ACS at the end of October 1984 and his time of entry. The shorter the time from HIV infection to death, the more likely it is that an individual is missed. This is the mechanism behind length-biased sampling.

In general, there is left truncation with respect to an event if both of the following conditions hold i) there are individuals that came under observation for the event some time after they became at risk for experiencing the event and ii) they would be missed if the event had happened before they came under observation for the event.

Another mechanism that generates a form of left truncated data is when some characteristic of an individual that changes over time is included in the analysis. Examples are the inclusion of a time-varying covariable in a regression model (see Section 21.1) or an intermediate state in a multi-state model. This type of truncation has been called ??
internal left truncation [3]. Another data structure that is represented and analysed in a similar way is when we have time-varying weights in the estimate the subdistribution hazard. We explain how to represent and analyse such data in Section 21. ??

4.4 Setup data analysis

Probabilistic setup data

- Event time T_i; may be right censored (administratively, loss to follow-up)
  Censoring time $C_i \sim \Gamma$, $\Gamma(t) = \mathrm{P}(C_i \leq t)$
  $X_i = \min\{T_i, C_i\}$, $\Delta_i = I(T_i \leq C_i)$
- Left truncation (late entry). *Entry time* $V_i \sim \Phi$, $\Phi(t) = \mathrm{P}(V \leq t)$
- Sample: $\{V_i, X_i, E_i\Delta_i\}, i = 1, \ldots, N$
- Assume (V_i, C_i) independent of/non-informative for (T_i, E_i) (possibly conditional on covariables in regression model) .23

If there is late entry that induces left truncation, the time at which an individual comes under observation is represented by V_i. X_i is observed only if $X_i > V_i$. If there is no late entry, V_i is zero for all individuals. It will turn out to be useful to attach a probability distribution to the censoring and entry times, even though we are not primarily interested in estimating them. We write $C_i \sim \Gamma$, hence $\Gamma(t) = \mathrm{P}(C_i \leq t)$ and $V_i \sim \Phi$, hence $\Phi(t) = \mathrm{P}(V_i \leq t)$.

Observed data

$\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$
Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number "in risk set" (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \ldots, Z_{iq})$ .24

The observed data is represented as $\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$, where N is the sample size. We use $t_{(1)} < t_{(2)} < \ldots < t_{(n)}$ to denote the ordered distinct observed event times. Note that often $n < N$ due to right censoring and tied event times. Similarly we let $c_{(1)} < \ldots < c_{(n_c)}$ and $v_{(1)} < \ldots < v_{(n_l)}$ denote the ordered distinct observed censoring and entry times respectively. Let m_i be the number of censorings at $c_{(i)}$, and let w_i be the number of entries at $v_{(i)}$. Entries, censorings and events may occur at the same time point. In that case we assume the ordering $t_i < c_j < v_{j'}$.

For each $t_{(i)}$, define $d_k(t_{(i)})$ as the number of observed events of type k at $t_{(i)}$ and $d(t_{(i)})$ as the number of observed events of any type at $t_{(i)}$. We make a distinction between individuals that are "observed to be at risk" and those that are "in the risk set" as well as between "censoring" and "leaving the risk set". This distinction is necessary for estimation of the subdistribution hazard: individuals that are censored due to a competing event remain in the risk set, even though they are no longer at risk for experiencing the event of interest (see Section 11.1). Although conceptually a different mechanism, the correction for left truncation is similar to the standard way right censored data are incorporated: an individual is included while he is observed to be at risk. Let $r(t)$ be the number of individuals that is observed and event free at time t. For estimation of the subdistribution hazard we need an extended risk set, to which individuals who experienced a competing event contribute as well. We use the notation $r^*(t)$ for the size of this risk set.

5 Exercises Part I

1. **Human life course.** This is a simple exercise to train in the concepts of hazard, probability and cumulative incidence. We estimate the distribution of time T from

birth to death for individuals in Sweden. Data have been downloaded from the human mortality database (`http://www.mortality.org`). For every age a (rounded to years), the estimates of probability $\mathrm{P}(T = a)$, hazard $\mathrm{P}(T = a|T \geq a)$ and survival $\mathrm{P}(T > a)$ are shown in Figure 3. Which line represents which quantity?

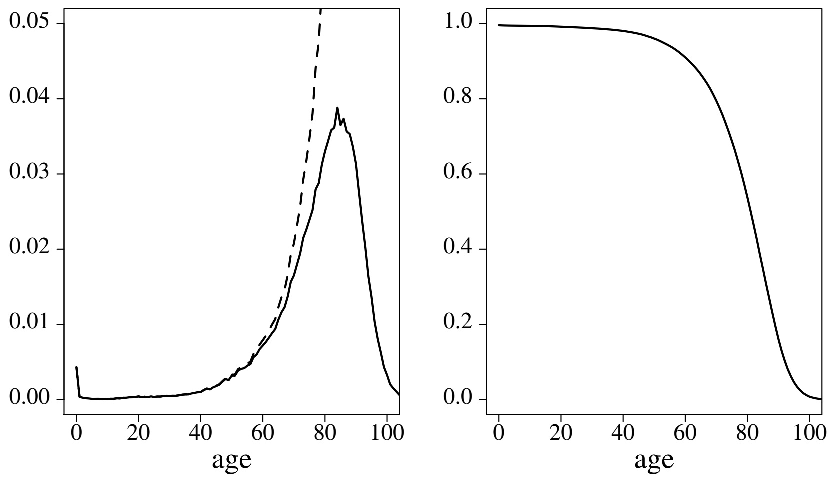


Figure 3: Mortality in Dutch population.

2. **Censoring due to discharge.** Psychiatric patients are treated in two different hospitals, A and B. We want to investigate whether the effect of a treatment on the development of side effects is the same for both hospitals. Hospital A collects information on side effects after discharge, and hospital B does not. In our analysis for hospital A we also use information after discharge, while for hospital B we censor individuals when they are discharged. What type of distribution do we want to quantify? Under which conditions can this be a fair comparison between both hospitals?
3. **Definition of cause-specific hazard.** Comment on this interpretation of cause-specific hazard

   > We want to compare the cancer event rates in the virtual situation when the competing risks did not exist. The analysis of the cause-specific hazard models the event of interest in the absence of competing risk events and thus is the appropriate method.

4. **Definition of competing risk.** Comment on the following definition of a competing risk [19]:

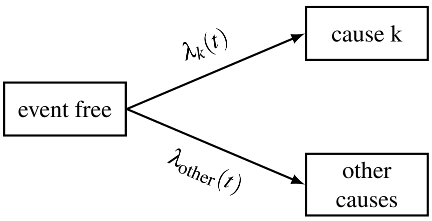
   > A competing risk is an event whose occurrence either precludes the occurrence of the event under examination or fundamentally alters the probability of occurrence of that event.

Part II
Estimation: Multi-State Approach

6 Aalen-Johansen Estimator

The Aalen-Johansen estimator is the basic estimator of the cause-specific cumulative incidence under right censored and/or left truncated data. It is based on an estimator of the cause-specific hazard. The estimator of the cause-specific hazard treats a censoring due to competing risks similar to the other reasons of censoring (administrative and loss to follow-up), but it is not interpreted as a censoring.

Cause-specific hazard



- Individuals with a competing event are no longer at risk $\Longrightarrow$ leave the risk set

$$\widehat{\lambda}_k(t_{(i)}) = \frac{d_k(t_{(i)})}{r(t_{(i)})}$$

- *Standard rate estimate*: same estimator as for marginal hazard (assuming independence), but different interpretation
- Cumulative hazard (Nelson-Aalen estimator):

$$\widehat{\Lambda}_k(t) = \sum_{t_{(i)} \leq t} \widehat{\lambda}_k(t_{(i)})$$

.25

If we can assume that the occurrence of the competing events is noninformative for the event of interest, the marginal hazard is estimated in the same way. However, if the occurrence of another event is informative, we no longer estimate the marginal hazard but only the cause-specific hazard. Interpreted as estimate of the cause-specific hazard, we do not have to assume that individuals who experience a competing event can be represented by the individuals that remain in follow-up. See also Section 18.

Aalen-Johansen estimator of F_k

- Plug-in estimator based on $F_k(t) = \int_0^t \mathrm{P}\{T \geq s\} \lambda_k(s) ds$:

$$\widehat{F}_k^{\mathrm{AJ}}(t) = \sum_{i: t_{(i)} \leq t} \widehat{\overline{F}}^{\mathrm{PL}}(t_{(i)}-) \times \widehat{\lambda}_k(t_{(i)}) \text{ with}$$

$$\widehat{\lambda}_k(t_{(i)}) = \frac{d_k(t_{(i)})}{r(t_{(i)})} \qquad \text{cause specific hazard}$$

$$\widehat{\overline{F}}^{\mathrm{PL}}(t_{(i)}-) = \prod_{j: t_{(j)} < t_{(i)}} \left(1 - \frac{d(t_{(j)})}{r(t_{(j)})}\right) \text{ Kaplan-Meier}$$

- Single event type: equal to Kaplan-Meier
- Competing risks: same hazard estimate as for marginal distribution, but cumulative quantity different

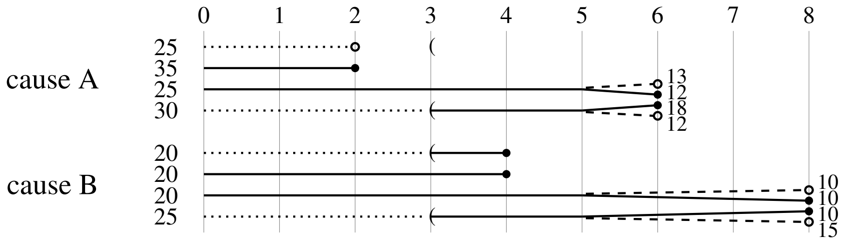
.26

We use the superscript AJ to denote this form. Detailed explanations by example of how the Aalen-Johansen estimator is calculated can be found in several papers (e.g. [28, 45]). It is a special case of the Aalen-Johansen estimator of the transition probability in a multi-state model. We use an example with artificial data [15, Chapter 2], which is also used as example for the calculations with the subdistribution approach.

The data are shown graphically in slide 27. We have 200 individuals, competing causes of death A and B. Interest is in cause A. Events are denoted by circles. Events that end in an open circle are not observed, due to right censoring at the start of the dashed lines. The unobserved period before late entry is denoted by a dotted line and the time of entry by "(". The 25 individuals in the first row, who would enter at 3 years, are not observed at all since they had event A before they entered the study. Hence the late entry leads to left truncation.

Example calculations

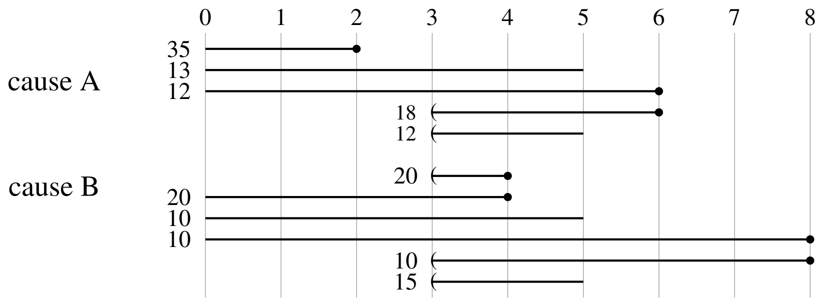
- 115 individuals die of A (60 at 2 years, 55 at 6 years)
- 85 individuals die of B (40 at 4 years, 45 at 8 years)



- 50 individuals censored after 5 years
- 100 individuals enter 3 years after the time origin (if they are still event free) .27

In slide 28 we show the observed data and the calculation of the Aalen-Johansen estimate for cause A. The estimate $\widehat{\lambda_A}$ of the cause-specific hazard for cause A has a value unequal to zero at $t = 2$ and $t = 6$ only. The estimate of the overall hazard $\widehat{h}$ additionally has a value unequal to zero at $t = 4$, when the competing event occurs. Next, we calculate the Kaplan-Meier $\widehat{F}^{\mathrm{PL}}$ for both end points combined. In the last line we combine both. Note that $\widehat{\mathrm{P}}(T \geq t) = \widehat{F}^{\mathrm{PL}}(t-)$ is based on the event times before t.

Detailed calculation of $\widehat{F_A}^{\mathrm{AJ}}(2)$ and $\widehat{F_A}^{\mathrm{AJ}}(6)$



	2 years	4 years	6 years
$\widehat{\lambda}_{\mathrm{A}}(t)$	$\frac{35}{100}$	0	$\frac{30}{50}$
$\widehat{h}(t)$	$\frac{35}{100}$	$\frac{40}{140}$	$\frac{30}{50}$
$\widehat{\mathrm{P}}(T \geq t)$	1	$(1 - \frac{35}{100}) = \frac{65}{100}$	$\frac{65}{100} \times (1 - \frac{40}{140}) = \frac{65}{140}$
$\widehat{F_A}^{\mathrm{AJ}}(t)$	$1 \times \frac{35}{100} = \frac{35}{100}$	$\frac{35}{100} + \frac{65}{100} \times 0 = \frac{35}{100}$	$\frac{35}{100} + \frac{65}{140} \times \frac{30}{50} = \frac{22}{35}$

.28

6.1 Confidence intervals

We first discuss confidence intervals around the Kaplan-Meier. Let $\widehat{\text{var}}$ denote the estimate of the variance and let $\widehat{\text{s.e.}} = \sqrt{\widehat{\text{var}}}$ denote the estimate of the standard error. Let k_α be the $1-\alpha/2$ quantile of the standard normal distribution. We leave out the explicit reference to the Kaplan-Meier via the subscript PL and write $\widehat{\widehat{F}}$.

The confidence interval is best calculated based on the variance of the Nelson-Aalen estimator $\widehat{H}(t) = \sum_{t_{(j)} \leq t} \frac{d(t_{(j)})}{r(t_{(j)})}$ of the cumulative hazard.

Confidence intervals, Kaplan-Meier

Start with variance of Nelson-Aalen estimator $\widehat{H}(t)$

$$\text{Aalen:} \qquad \widehat{\text{var}}\left\{\widehat{H}(t)\right\} = \sum_{t_{(j)} \leq t} \frac{d(t_{(j)})}{r^2(t_{(j)})} \tag{4}$$

$$\text{Greenwood:} \qquad \widehat{\text{var}}\left\{\widehat{H}(t)\right\} = \sum_{t_{(j)} \leq t} \frac{d(t_{(j)})}{r(t_{(j)})\{r(t_{(j)}) - d(t_{(j)})\}} \tag{5}$$

Use the delta method and $\widehat{\widehat{F}}(t) \approx \exp\{-\widehat{H}(t)\}$ to obtain

$$\widehat{\text{var}}\left\{\widehat{\widehat{F}}(t)\right\} = \left\{\widehat{\widehat{F}}(t)\right\}^2 \times \widehat{\text{var}}\left\{\widehat{H}(t)\right\}. \tag{6}$$

.29

The Greenwood form is preferred when estimating the variance of the survival function, whereas the Aalen estimate has been shown to perform better when estimating the variance of the cumulative hazard [29].

Confidence intervals, choice in scale

- On "plain" or "linear" scale:

$$\widehat{\widehat{F}}(t) \pm k_\alpha \widehat{\text{s.e.}}\left\{\widehat{\widehat{F}}(t)\right\} = \widehat{\widehat{F}}(t) \pm k_\alpha \widehat{\widehat{F}}(t) \times \widehat{\text{s.e.}}\left\{\widehat{H}(t)\right\}$$

- On "log scale" [log survival; use $\log\{\widehat{\widehat{F}}(t)\} \approx -\widehat{H}(t)$]:

$$\exp\left[-\widehat{H}(t) \pm k_\alpha \widehat{\text{s.e.}}\{\widehat{H}(t)\}\right] = \widehat{\widehat{F}}(t) \times \exp\left\{\frac{\pm k_\alpha \widehat{\text{s.e.}}\{\widehat{\widehat{F}}(t)\}}{\widehat{\widehat{F}}(t)}\right\} \tag{7}$$

- "log-log" or "complementary log-log" scale (use $\log \widehat{H}(t) \approx \log[-\log\{\widehat{\widehat{F}}(t)\}]$)

The "log" scale has best overall performance, but "log-log" scale best for small sample sizes (≈ 25, $\leq 50\%$ censoring)[7] .30

The linear scale can give values below zero and above one and is not recommended [49, p.17]. The recommended approach is to first calculate the $(1-\alpha) \times 100\%$ confidence interval around the estimate of the cumulative hazard as is done in the "log" scale [49, p.17]. It can give values above one, but never below zero. An alternative form, which is guaranteed to give confidence intervals between zero and one, first calculates the confidence interval around the logarithm of the cumulative hazard

$$\log \widehat{H}(t) \pm k_\alpha \widehat{\text{s.e.}}\left\{\log \widehat{H}(t)\right\}.$$

If we transform it back to a confidence interval around the estimated survival function, it obtains the form a^b, with

$$
\begin{aligned}
a &= \widehat{\overline{F}}(t) \\
b &= \exp\left[\pm k_\alpha \,\widehat{\text{s.e.}}\left\{\log \widehat{H}(t)\right\}\right]
\end{aligned}
\tag{8}
$$

Using the delta method, $\widehat{H}(t) \approx -\log\{\widehat{\overline{F}}(t)\}$ and (6), the estimate of the standard error can be rewritten as

$$
\widehat{\text{s.e.}}\left\{\log \widehat{H}(t)\right\} = \frac{\widehat{\text{s.e.}}\left\{\widehat{H}(t)\right\}}{\widehat{H}(t)} = \frac{\widehat{\text{s.e.}}\left\{\widehat{\overline{F}}(t)\right\}}{-\widehat{\overline{F}}(t)\log\left\{\widehat{\overline{F}}(t)\right\}}. \tag{9}
$$

Because of the relation $\log\{\widehat{H}(t)\} \approx \log[-\log\{\widehat{\overline{F}}(t)\}]$, it is called the "log-log" or "complementary log-log" scale, abbreviated as "cloglog".

For the Aalen-Johansen estimator of the cause-specific cumulative incidence, which is based on the cause-specific hazard, there is no one-to-one relation like $H(t) = -\log \overline{F}(t)$ and the first expression in (7) cannot be used. Instead, the second expression in (7) and the last one in (9) are used, which do not make any direct reference to a hazard. Estimation of the standard error of the Aalen-Johansen estimate is more involved than for the Kaplan-Meier estimate and is discussed in [2].

Confidence intervals, competing risks

- Based on estimator of $\text{var}\{\widehat{\Lambda_k}(t)\}$, same as in classical setting
- Transformation to $\widehat{\text{var}}\left\{\widehat{F_k}(t)\right\}$. Complicated formula (see [2])
- Same scales as in classical setting, obtained directly via expression in $\widehat{\overline{F_k}}$; e.g. "log" scale based on

$$
\widehat{\overline{F_k}}(t) \times \exp\left\{\frac{\pm k_\alpha \,\widehat{\text{s.e.}}\{\widehat{\overline{F_k}}(t)\}}{\widehat{\overline{F_k}}(t)}\right\},
$$

- CI around $\widehat{\overline{F_k}}$ can be transformed to CI around $\widehat{F_k}$.
  There is also a direct approach, in which $\overline{F_k}$ in above formula is replaced by F_k.
  Naming of scales inconsistent

.31

We gave expressions for the confidence interval around the estimate of the complement of the (cause-specific) cumulative incidence (the "survival function"). If we want to calculate the confidence intervals around the estimate of the cumulative incidence itself, we can first calculate the confidence interval around $\widehat{\overline{F}}$ and use $y = 1-(1-y)$. However, we can also construct confidence intervals directly around $\widehat{F}$, applying the delta method for the log and log-log transformations. They do not give the same formulas. Note that $\widehat{\text{s.e.}}\{\widehat{\overline{F}}(t)\} = \widehat{\text{s.e.}}\{\widehat{F}(t)\}$. We obtain the following expressions

$$\text{linear scale: } \widehat{F}(t) \pm k_\alpha \widehat{\text{s.e.}}\left\{\widehat{F}(t)\right\}$$

$$\text{log scale: } \widehat{F}(t) \times \exp\left\{\frac{\pm k_\alpha \widehat{\text{s.e.}}\{\widehat{F}(t)\}}{\widehat{F}(t)}\right\}$$

$$\begin{aligned} \text{log-log scale: } & a^b \text{ with} \\ & a = \widehat{F}(t) \\ & b = \exp\left[\frac{\pm k_\alpha \widehat{\text{s.e.}}\left\{\widehat{F}(t)\right\}}{\widehat{F}(t)\log\left\{\widehat{F}(t)\right\}}\right] \end{aligned} \tag{10}$$

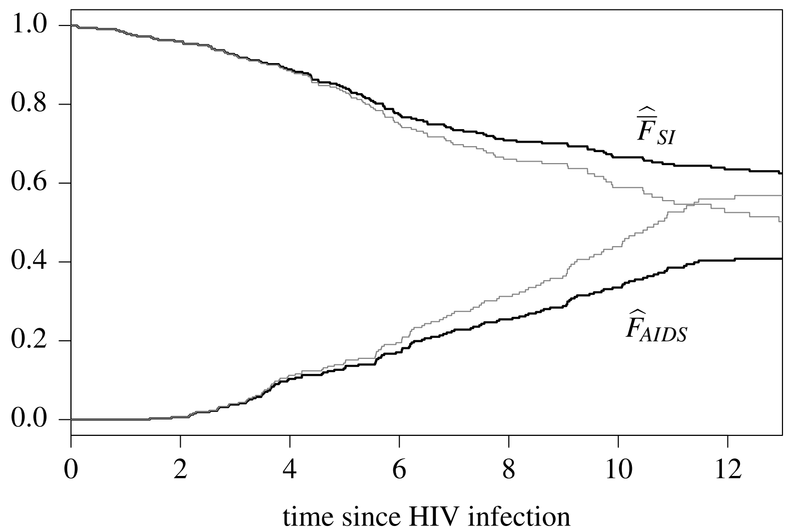
These expressions have mainly been used in the multi-state setting. When the software refers to the log or log-log scale, it is important to know whether it refers to (7) and (8) or the ones in (10).

7 Display Formats

There are several ways to show the estimates graphically. We illustrate two ways with the data from example C.

In slide 32 we use the ***alternate*** display format. One outcome, SI, is plotted as the "survival" curve $\overline{F}_{SI}$, starting at one and decreasing, and the other one, AIDS, as the cumulative incidence curve F_{AIDS}, starting at zero and going up. This format can be informative if there are two event types and we want to compare the cause-specific cumulative incidence by subgroup for both competing risks. In the figure we also show the marginal Kaplan-Meier curves that are based on the "estimate" of the marginal hazard[6]. These curves are steeper than the cause-specific curves. We also see that the curves cross, which can never happen with cause-specific cumulative incidence curves; they always add up to a number ≤ 1.

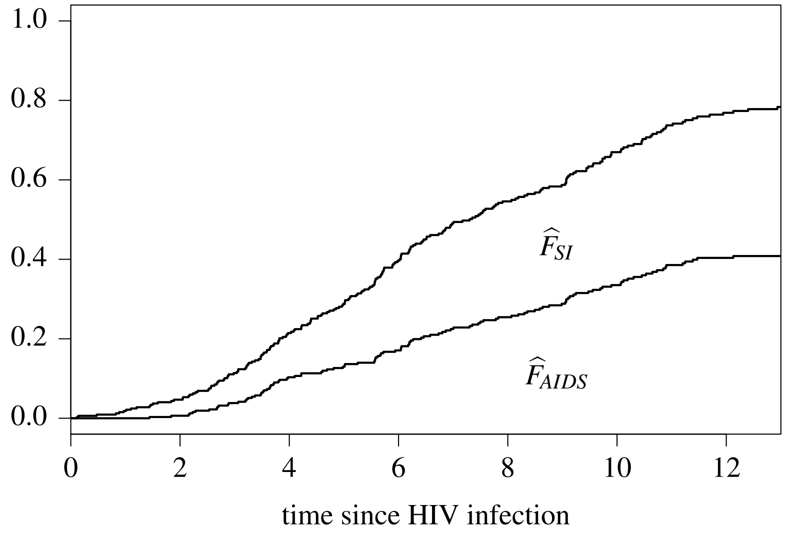
Alternate display format



.32

[6]We write estimate in quotes because it probably does not estimate the marginal distribution, due to informative censoring.

Stacked display format



.33

Another option is the ***stacked*** format. In slide 33, the AIDS-specific curve F_{AIDS} is the same as in slide 32. The SI-specific curve is plotted on top of it. The difference between both curves quantifies the probability to have progressed to SI. Since $F_{AIDS} + F_{SI} = F$, the upper curve estimates the overall cumulative incidence. Therefore the upper curve can be obtained by calculating the overall Kaplan-Meier in which both end points are combined. This format is informative if we want to emphasize how all event types add up to the overall risk. It can be used with any number of competing event types.

In the ***overlaid*** format we plot the estimates for the different end points using the same base. This format is informative when comparing progression to the different competing risks. An example is Figure 5 on Page 37, in which crude risk of symptomatic cardiac events and crude risk of death from other causes are plotted. A fourth option is to plot the cumulative incidence for each end point in a separate figure or panel. This ***separate*** format is useful if we are interested in the difference by subgroup and we have more than one competing risk. One example is Figure 6 on page 38, in which the main interest is in the relation between gender and each event type separately.

8 Marginal versus Cause-Specific

8.1 All quantities and their estimators

We summarized the main quantities in slide 17. In slide 34, we give an overview of their estimators. The subdistribution hazard is covered in Section 11.2.

Estimators in competing risks setting

hazard	**estimate**	**cumulative**
marginal	$d_k(t)/r(t)\{=\widehat{\lambda}_k(t)\}$	$\widehat{F_k^m}^{\text{PL}} = \prod_{t_{(j)}\leq t}\left\{1-\widehat{\lambda}_k(t_{(j)})\right\}$
cause-specific	$\widehat{\lambda}_k(t) = d_k(t)/r(t)$	$\widehat{F}_k^{\text{AJ}}(t) = \sum_{t_{(j)}\leq t}\widehat{F}^{\text{PL}}(t_{(j)}-)\widehat{\lambda}_k(t_{(j)})$
subdistribution	$\widehat{h}_k(t) = d_k(t)/r^*(t)$	$\widehat{F}_k^{\text{PL}}(t) = \prod_{t_{(j)}\leq t}\left\{1-\widehat{h}_k(t_{(j)})\right\}$
overall	$\widehat{h}(t) = d(t)/r(t)$	$\widehat{F}^{\text{PL}}(t) = \prod_{t_{(j)}\leq t}\left\{1-\widehat{h}(t_{(j)})\right\}$

.34

If we are interested in the marginal distribution, then we typically compute the Kaplan-Meier estimate and censor individuals when they experience a competing event. However, this is valid only under the assumption of independent censoring, because only then do the individuals who remain in the risk set r represent the ones who are censored. Unfortunately, independent censoring cannot be tested for (see Section 8.3). Extra information may allow to show dependent censoring (Section 8.5), but independence can never be tested for.

Marginal distribution

- Estimated via (marginal) hazard, basis for Kaplan-Meier estimate of cumulative incidence/net risk
- Assumption: censoring (including from competing risk) is independent; censoring should give no information on residual time-to-event. Then censored individuals can be represented by the ones that remain at risk
- Otherwise Kaplan-Meier has no interpretation. Does not describe survival in (hypothetical) world with competing event removed
- Extra information may allow to show dependent censoring (Example H-2, IDU and pre-AIDS death), but *independence* can never be tested for .35

In the estimator of the cause-specific hazard, the occurrence of a competing event is treated as a censored observation. However, it is not interpreted as a censored observation. It is only interpreted as a censored observation when we use it to estimate the marginal hazard. Interpreted as estimate of the cause-specific hazard, we do not have to assume that individuals who experience a competing event can be represented by the individuals that remain in follow-up. If the competing events are independent and the estimate represents both the cause-specific hazard and the marginal hazard, the cumulative estimates are different. When estimating the marginal distribution, we use the Kaplan-Meier estimator $\widehat{F_k^m}^{\mathrm{PL}}$ instead of the Aalen-Johansen estimator $\widehat{F}_k^{\mathrm{AJ}}$.

Competing risks

- Competing risk is treated as a separate event
  - Individuals censored by competing event don't have to be represented by the ones that remain at risk.
    Other censoring (administrative/loss to follow-up) must be independent
- Cause-specific hazard
  - Basis for Aalen-Johansen estimator of cause-specific cumulative incidence/crude risk
  - If censoring due to competing event is independent, then marginal and cause-specific hazard are equal. Cumulative quantities different: Kaplan-Meier versus Aalen-Johansen
- Subdistribution hazard: one-to-one relation with crude risk.
  Product-limit estimator equal to Aalen-Johansen estimator with our choice of weights .36

8.2 Assumptions on censoring and delayed entry

Unbiased estimation requires some assumptions with respect to the right censoring and left truncation mechanisms. When right censoring is caused by a competing risk, the assumptions differ between a marginal analysis and a competing risks analysis. For left truncation due to late entry, the necessary assumptions are the same for both types of analyses.

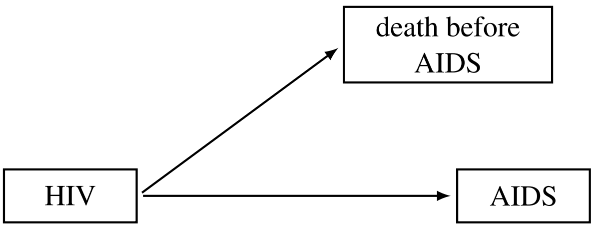
Assumptions for data analysis

- Competing risks: administrative right censoring and loss to follow-up non-informative
- Marginal analysis: censoring due to competing risks non-informative as well
- If competing risks are independent (may hold in Example G: staphylococcus infection)
  - Hazard estimate both marginal and cause-specific interpretation
  - Cumulative quantity different (Kaplan-Meier versus Aalen-Johansen)
- *If competing risks are dependent, marginal and cause-specific hazard may be different* (Example H-2)
- Late entry non-informative

.37

In Example G, staphylococcus infection in hospital and discharge may be unrelated. In that case the hazard that we estimate by censoring individuals at discharge can be interpreted as both the cause-specific and the marginal hazard of infection. If used in the Kaplan-Meier, it estimates the marginal distribution; if used in the Aalen-Johansen estimator, it estimates the infection-specific cumulative incidence. If progression to the competing events is not unrelated, then the Kaplan-Meier does not represent an interpretable quantity and the marginal distribution cannot be estimated without incorporating extra information or making additional assumptions. This is an issue in example H-2. Based on the Kaplan-Meier estimate (slide 9) we would conclude that injecting drugs prevents progression to AIDS. However, the estimate is only valid under the assumption that individuals that die can be represented by the ones that remain in follow-up. If IDUs die from causes that are partially related to AIDS progression, the assumption is violated.

Time from HIV infection to AIDS



- Compare risk groups MSM and IDU
- IDUs expected to have faster progression to AIDS
- Interest in time to AIDS if there were no pre-AIDS death.
  Interest in etiology and marginal distribution
- *Kaplan-Meier: those who die before AIDS leave the risk set*
  Assumption: can be represented by those that do not die
- *Can we test for independence?*

.38

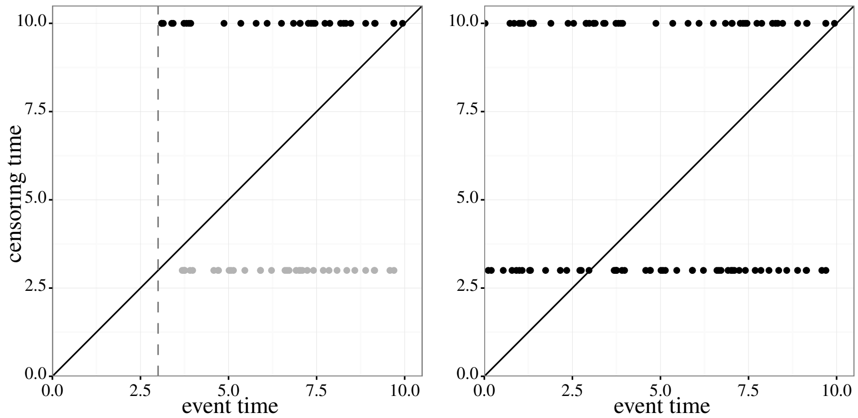
8.3 Independent right censoring?

We first concentrate on the assumptions with respect to right censoring in a setting without competing risks. Since there is only one event type, there is only one type of hazard. Since there are no competing risks, all censoring is administrative or due to loss to follow-up. Every censored individual has an event time, which is larger than his censoring time. One way to model independent, non-informative censoring is by assuming that every individual who has the event observed also has a censoring time. In this way, we can define a bivariate distribution for the event and censoring times. Non-informative censoring now means that the time-to-censoring distribution and the time-to-event distribution are independent.

We can visualize independence via a scatter plot. We start with a simple example. Suppose that we follow individuals for a maximum of ten months. Death is the event of interest. After three months it is decided that the study will continue with half of the participants that are still alive. They are selected at random. Hence, censoring is non-informative. In the left panel of slide 39, the event times and censoring times are shown for the individuals that were still alive after three months. The diagonal line is the line for which x and y have the same value. The event and censoring times of the individuals that leave the study after three months are shown by the grey points along the horizontal line at $y = 3$. Since censoring is non-informative, their (unobserved) death times have approximately the same distribution as the (observed) death times of individuals that continue to be in follow-up until year ten (the black points). Hence, in order to quantify mortality after three months, the censored individuals can be represented by the individuals that remain under observation.

Since the decision on censoring is not made for individuals that died within three months, strictly speaking they did not have a censoring time. But an equivalent situation would arise if the decision on censoring at three months were made at random at the start of the study. Then also the individuals that died in the first three months would have a censoring time. The right panel of slide 39 shows how the complete data on censoring and event time for the first ten months could have been. Independence between censoring time and event time is reflected by the absence of a correlation in the scatterplot of the complete data.

Observed and complete data



.39

Since the assumption of independent censoring is essential for unbiased estimation, we may try to use the data to test for it. Unfortunately, it has been shown that this is not possible [50]. The reason is that, if an individual is right censored, we do not know whether the event would have happened the day after or many years later. We give a graphical argument by showing that the same observed data with respect to event time and censoring time can be generated by a mechanism in which they are independent as well as by one in which they are dependent. Hence, the observed data does not allow us to discriminate between both.

Again, every individual is assumed to have both an event time and a censoring time, which follows some bivariate distribution. This is similar to having data on weight and height from individuals. Data on weight and height from a sample of individuals could be as in the scatterplot A of Figure 4. In subfigure B, exactly the same points as in subfigure A are plotted, but now they refer to event times and censoring times. The difference in the survival setting is that we only observe one, the earliest, of the event times and censoring times. For the other time, we only know that it occurs later. Because the data is only partially observed, it is plotted in grey. For example, the arrow in subfigure B points to an event time that is not observed because censoring occurred first. We do know the censoring time, which is read from the y-axis. For the event time we only

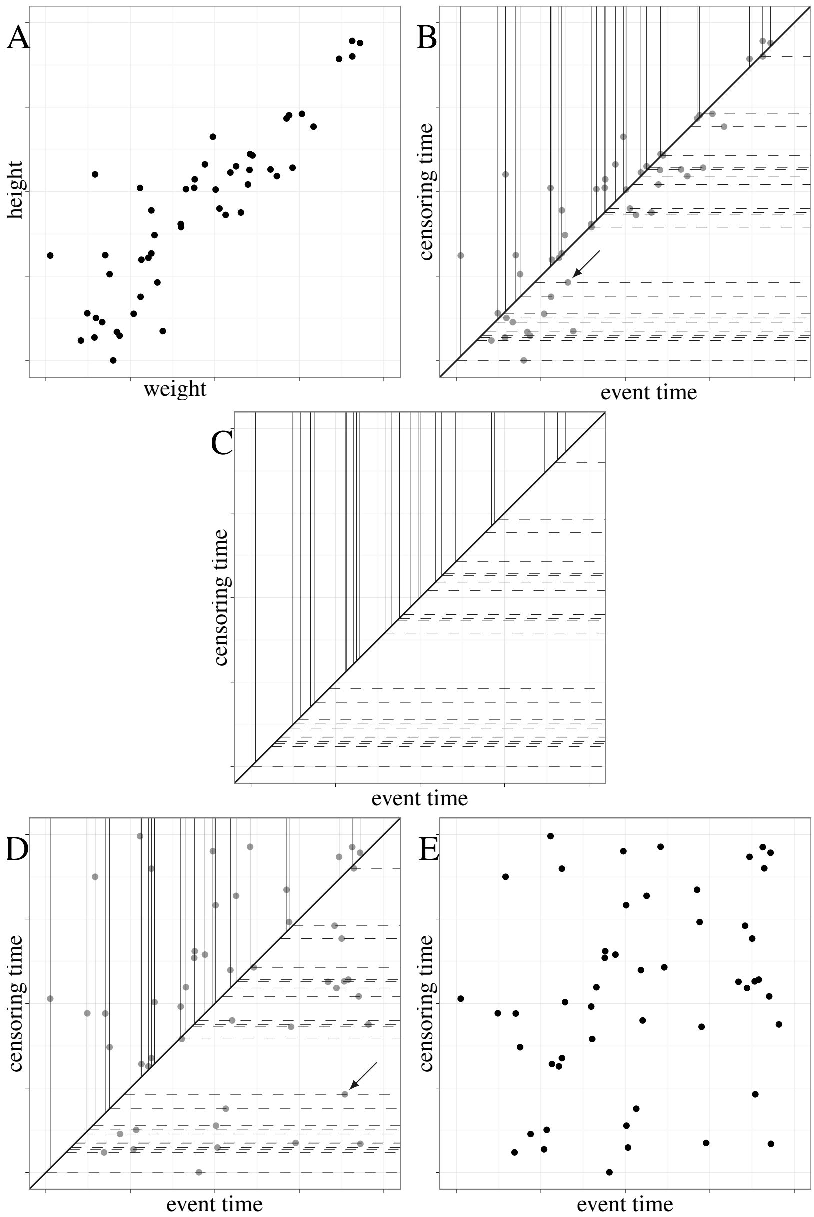


Figure 4: Example showing that we cannot decide from the time-to-event data whether censoring is independent.

know that it has some value along the horizontal dashed line that starts at the diagonal and crosses the unobserved true event time. Hence for every individual, the observed information consists of either a horizontal line (right censored event) or a vertical line (event observed, censoring time is larger). The information that we observe is shown in subfigure C. In subfigure D, we add points again in such a way that they are still in correspondence with the observed information in subfigure C. For example, the point with the arrow has moved to the right along the horizontal line. In subfigure E, we remove the lines and the scatterplot is seen to correspond to a situation in which event time and censoring time are independent.

In summary, if individuals experienced the event shortly after they were censored (as in A/B), such that censoring time and event time were dependent, this would lead to the same observed data (as in C) as in a situation of independence between the time-to-event and time-to-censoring distributions (which could generate a pattern as in D/E). Hence, the observed data does not allow us to decide which of the two (or any other degree of dependence) is the real one. If censoring is due to a competing risk, the event of interest will never occur but the same property holds.

8.4 Late entry

With left truncation due to late entry we need to make a similar assumption in order to obtain unbiased estimates. At every time point, individuals that are in follow-up and event-free should have the same future event risk as individuals that have not entered the study (yet) at that time point. And the fact that an individual enters the study event-free should not provide any information with respect to his future event time. Then, the unobserved individuals can be represented by the ones that are in follow-up.

Similar to right censoring, one way to model non-informative left truncation is by defining a distribution for the entry times that is independent from the time-to-entry distribution. This refers to the distribution of the whole sample, including the individuals that are not observed because they experienced the event before entry.

8.5 What can we do?

Conclusion

- Same data can represent independent as well as dependent censoring
  - We cannot test for independence based on observed event/censoring data
  - No correction for dependence possible based on observed event/censoring data
- *Extra information may allow to show dependence, but independence can never be tested for* .40

If we cannot determine whether right censoring is non-informative based on the data of the event times and censoring times, what else can we do? We can argue why we think a non-informative observation scheme is a reasonable assumption, possibly after correction for variables that are known to induce dependence. For right censored data, it helps to have another look at the three general reasons for censoring and argue whether censoring is likely to be independent.

Reasons for right censored data

Cutoff date of analysis (administrative censoring) Censoring usually non-informative

Loss to follow-up May be problematic

- Sicker individuals discontinue participation in study (lack of energy, too ill, return to home country)
- Healthier individuals discontinue participation (don't feel the need to continue, start new life in other country)

Competing risks Often there is some background mechanism that influences progression to the different end points, which makes censoring informative. In competing risks analysis, independence is not required. .41

In the IDU/MSM comparison, the data on time to AIDS and death does not allow us to see whether censoring due to death is informative or not. However, additional information was available on the causes of death. We see that 17 out of the 99 IDUs died and some of the causes of death were related to AIDS progression. They were not far from developing AIDS and therefore they cannot be represented by the IDUs that did not die: pre-AIDS death is informative for AIDS. Therefore, the Kaplan-Meier curve for the IDUs does not reflect the marginal distribution of time from HIV infection to AIDS. In fact, it has no interpretation. Among the MSM, only 5 out of 127 died, and the causes of death were less clearly related to AIDS progression.

Explanation difference between IDU and MSM

- Extra information on cause of death before AIDS

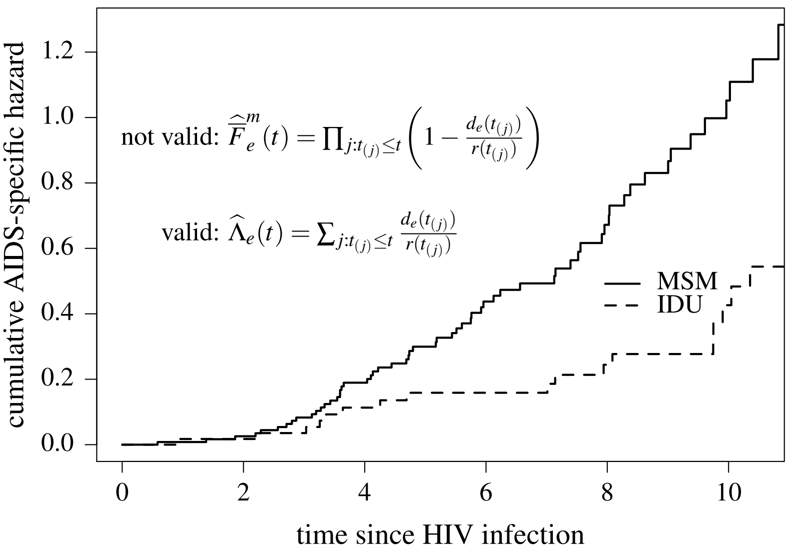
	IDU	MSM
Reason of death	Number	
HIV related infections	3	0
overdose/suicide	6	0
violence/accident	2	0
liver cirrhosis	2	0
cancer	0	1
heart attack	0	1
unknown	4	3

- Some pre-AIDS death causes in IDU related to AIDS progression. Censoring close to AIDS, hence net risk for IDU biased downwards

.42

In slide 9 we saw that the Kaplan-Meier was strikingly different between IDU and MSM. Because of independent censoring, it is invalid as estimate of the marginal cumulative incidence. However, the Kaplan-Meier monotonically relates to the estimate of the cumulative hazard that censors individuals when they experience a competing event, which is the cause-specific hazard. Therefore, a similar difference is observed in the estimates of the AIDS-specific cumulative hazards, which are valid:

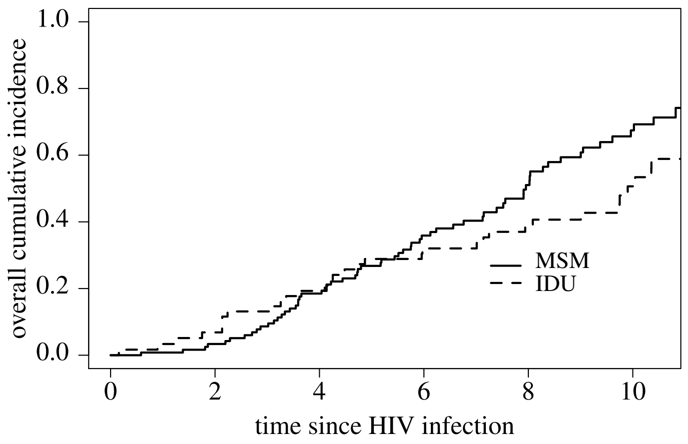
AIDS-specific cumulative hazard



.43

Whether the estimate of the cause-specific hazard is close to the marginal hazard depends on when the event of interest would have happened if a competing event happened first. We can look at the most extreme situations. On one extreme, if it would have happened right after the competing event, then the marginal hazard is equal to the overall hazard that combines all event types. In our example, we would obtain an estimate of the marginal cumulative incidence via the Kaplan-Meier based on the overall hazard, as in slide 44.

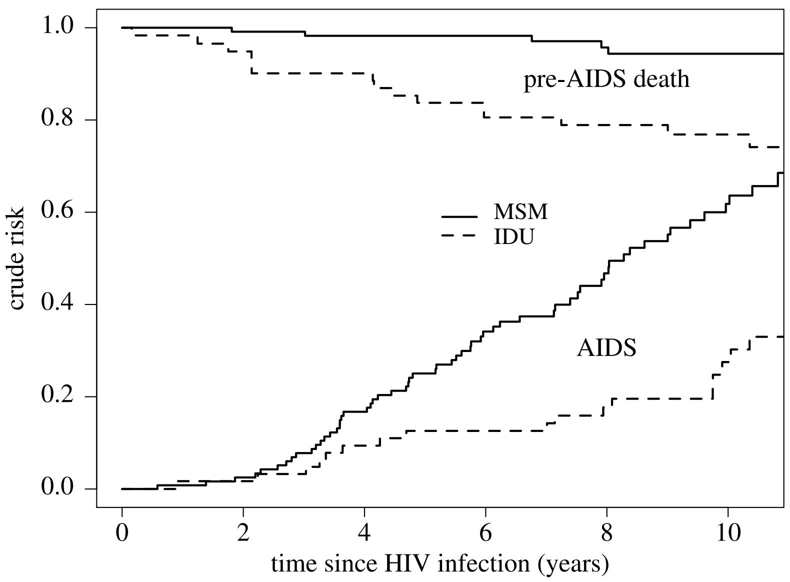
AIDS at time of death: combine AIDS and pre-AIDS death



Log-rank test: $p = 0.14$. .44

On the other extreme, if AIDS would never have happened among those that died first, then the marginal hazard is equal to the subdistribution hazard. Then the estimates in slide 45 would reflect the marginal distributions.

Never AIDS: death before AIDS treated as competing risk



.45

Probably, neither of the two situations approximate reality. The curves in slide 44 only reflect the AIDS-free survival, whereas those in 45 only quantify the cumulative AIDS risk in a situation in which individuals can also experience death as a competing event.

9 Regression; proportional cause-specific hazards

Often, we want to quantify how hazard and cumulative incidence depend on characteristics that vary within the population. If we are interested in one single categorical characteristic like gender or CCR5 genotype, we can compute nonparametric estimates per value and compare them via the log-rank test (see Section 12). Beyond this situation, we need to make further assumptions and specify some type of regression model in which effects are quantified through parameters.

The most popular approach is to use a model in which parameters quantify effects on a hazard with the extra condition that effect sizes don't change over follow-up time: the proportional hazards model. We first explain proportional cause-specific hazards models, which immediately extend to transition hazards in the multi-state setting. In Section 13 we quantify effects via a proportional subdistribution hazards model.

9.1 Standard Cox model

Basic structure Cox model

$$h(t\,|\,\mathbf{Z}_i) = h_0(t)\,\exp(\beta^\top \mathbf{Z}_i)$$

$h_0(t)$ baseline hazard, same for all individuals

$$\beta^\top \mathbf{Z}_i = \beta_1 Z_{i1} + \ldots + \beta_q Z_{iq}$$

Relative hazards constant over time:

$$\frac{h(t\,|\,\mathbf{Z}_i)}{h(t\,|\,\mathbf{Z}_j)} = \frac{\exp(\beta^\top \mathbf{Z}_i)}{\exp(\beta^\top \mathbf{Z}_j)}$$

.46

The parameter vector β is estimated by maximizing the partial likelihood. We assume that all event times are distinct[7]. Given that there is an event observed at $t_{(i)}$, the conditional probability that it is from individual l is equal to

$$\frac{h_0(t_{(i)})\exp(\beta^\top \mathbf{Z}_l)}{\sum_{j\in R(t_{(i)})} h_0(t_{(i)})\exp(\beta^\top \mathbf{Z}_j)} = \frac{\exp(\beta^\top \mathbf{Z}_l)}{\sum_{j\in R(t_{(i)})} \exp(\beta^\top \mathbf{Z}_j)}.$$

The partial likelihood is the product of this expression over the observed event times. We use a slightly different, but equivalent notation via an "at-risk" function $\psi(t)$.

Estimation

Estimate of β through maximisation of partial likelihood.

$$L(\beta) = \prod_{i=1,\delta_i=1}^{N} \left\{ \frac{\exp(\beta^\top \mathbf{Z}_i)}{\sum_{j=1}^{N} \psi_j(t_i)\exp(\beta^\top \mathbf{Z}_j)} \right\}.$$

with

$$\psi_j(t) = \begin{cases} 0 & \text{if person } j \text{ not at risk at } t \\ 1 & \text{if person } j \text{ at risk at } t \end{cases}$$

- Only values at observed event times matter
- Numerator: $\mathbf{Z}_i$ from individual with event
- Denominator: all individuals in follow up at event time
- Directly extends to time-varying covariables $Z_i(t)$

.47

We use the data set from Example C, which is available in the `mstate` R package and in Stata. We show the information from the individuals in rows 14, 3, 15 and 8 of the data set.

`aiddsi` data set, four example individuals

patnr	time	status	cause	ccr5
14	5.05	0	event-free	WW
3	2.23	1	AIDS	WW
15	10.20	1	AIDS	WM
8	8.61	2	SI	WW

.48

[7] A number of methods exist to deal with tied event times; we do not explain them.

We first fit a Cox model for the combined end point AIDS/SI switch. In R we write

Results Cox model for combined end point

```
coxph(formula = Surv(time, status != 0) ~ ccr5,
                                    data = aidssi)
```

Result:

```
         coef exp(coef) se(coef)      z       p
ccr5WM -0.701     0.496    0.186 -3.77 0.00016

Likelihood ratio test=16.5  on 1 df, p=4.95e-05  n= 324,
                        number of events= 220
   (5 observations deleted due to missingness)
```

.49

On slide 49 we observed that, relative to the reference category `WW`, individuals with the CCR5-Δ32 deletion have a much slower progression to the combined event AIDS/SI. We want to disentangle this effect with respect to both end points, which will be done via a competing risks analysis.

9.2 Proportional cause-specific hazards

In the competing risks setting we can specify a similar type of model for the cause-specific hazard and the subdistribution hazard. The difference is that everything refers to a specific type of end point via the subscript k. We start with a proportional cause-specific hazards model. The proportional subdistribution hazards (or Fine and Gray) model is covered in Section 13.

Basic structure

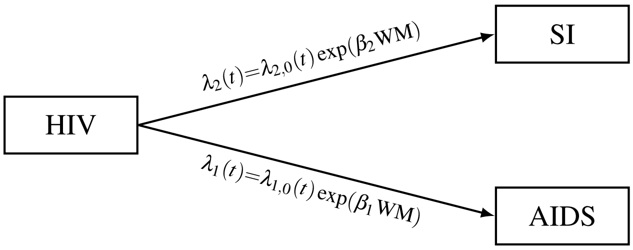
- $\lambda_k(t \mid \mathbf{Z}_i)$: cause-specific hazard of cause k for a subject with covariable vector $\mathbf{Z}_i$ given by
$$\lambda_k(t \,|\, \mathbf{Z}_i) = \lambda_{k,0}(t) \exp(\beta_k^\top \mathbf{Z}_i) \; ,$$
  with
  - $\lambda_{k,0}(t)$ is the baseline cause-specific hazard for cause k
  - $\beta_k^\top = (\beta_{k1}, \ldots, \beta_{kq})$ represent the covariable effects on cause k
- Estimation via standard Cox model: individuals with a competing event leave the risk set

.50

The parameters β_k quantify the effect of the covariables $\mathbf{Z}_i$ on the progression rate to cause k and $\lambda_{k,0}(t)$ is the baseline cause-specific hazard for cause k. The cause-specific hazard is estimated by removing individuals from the risk set when they experience a competing event. As a consequence, the estimation procedure is the same as in a setting with only one type of event. When fitting the model, we can use all the techniques that are used when fitting a standard Cox proportional hazards model. The only difference is in the interpretation: we quantify effects on the cause-specific hazard, not on the marginal hazard. Only if the censoring due to the competing risks is non-informative, then the estimates can also be interpreted as effects on the marginal hazard. Since a different quantity is estimated, we call it a proportional cause-specific hazards model and not a Cox model.

As example, we quantify the effect of the CCR5-Δ32 deletion on progression to AIDS and to SI switch as first event. A schematic representation of the situation that we want to investigate is given in slide 51.

Example: effect CCR5Δ32 on AIDS and SI



.51

The baseline hazards $\lambda_{1,0}$ and $\lambda_{2,0}$ describe the AIDS-specific and SI-specific hazards for the reference group, those with value `WW`. For the individuals with the deletion on one of the chromosomes, the AIDS-specific and SI-specific hazard are multiplied by a factor $\exp(\beta_1)$ and $\exp(\beta_2)$ respectively.

If we quantify the effect of CCR5 on the AIDS-specific hazard, only the individuals that were observed to develop AIDS before an SI switch have an event. These are the individuals that have the value 1 in the **status** column or the value `AIDS` in the **cause** column. In R we write and obtain:

Analysis per risk type (in R)

```
coxph(Surv(time, status==1) ~ ccr5, data = aidssi)
------------------------
         coef exp(coef) se(coef)      z       p
ccr5WM  -1.24     0.291    0.307  -4.02 5.7e-05

Likelihood ratio test=22  on 1 df, p=2.76e-06   n= 324,
                                   number of events= 113
   (5 observations deleted due to missingness)
```

```
coxph(Surv(time, status==2) ~ ccr5, data = aidssi)
------------------------
         coef exp(coef) se(coef)     z       p
ccr5WM -0.254     0.776    0.238 -1.07    0.29

Likelihood ratio test=1.19  on 1 df, p=0.275  n= 324,
                                   number of events= 107
   (5 observations deleted due to missingness)
```

.52

Individuals with the deletion have a slower progression rate to the combined AIDS/SI end point (as shown in slide 49) mainly because of their lower AIDS-specific hazard (relative hazard $\exp(\beta_{\mathrm{AIDS}}) = 0.291$). The estimated SI-specific hazard for individuals with the deletion is only slightly lower than the hazard for those without the deletion (relative hazard $\exp(\beta_{\mathrm{SI}}) = 0.776$).

`status==1` is a logical statement with outcomes `FALSE` or `TRUE`, which are interpreted as 0 and 1. For the switch to SI phenotype as first event, we use `status==2`, which again is translated to 0 and 1.

10 Exercises part II

1. **Ignore the competing event?** In a study on long-term effects of treatment in childhood cancer survivors [38], the end point of interest was symptomatic cardiac event (CE). Comment on the following statement:

    > Some individuals died without CE (DOC), but they had already left the study before they died. Therefore, for CE as end point, no competing risk of DOC is present in our data and the Kaplan-Meier curve is a valid estimate of the marginal distribution of time to CE.

2. **What if Kaplan-Meier and Aalen-Johansen are almost equal?** We continue with the same example, but no longer assume that all individuals that died from other causes had left the study before they died. Hence, death from other causes (DOC) was an observed competing risk. In Figure 5, the estimate of the CE-specific cumulative incidence is compared with the Kaplan-Meier curve for CE as end point. The cause-specific cumulative incidence curve for DOC is shown as well. (Note that analysis was restricted to persons that survived the first five years after diagnosis.)

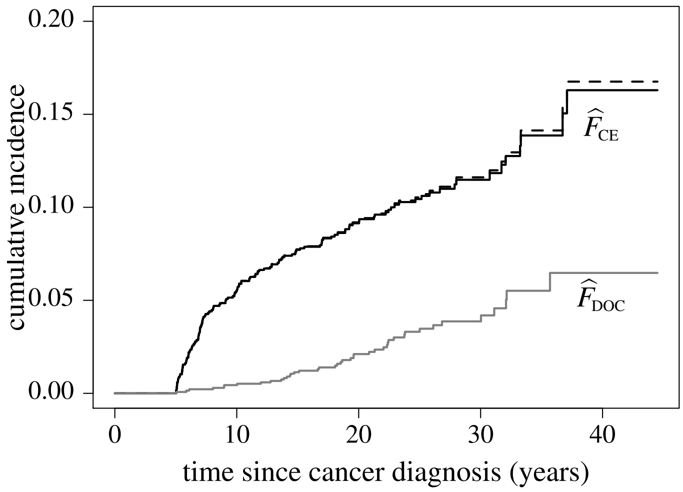


Figure 5: Estimate of CE-specific cumulative incidence (black solid line), Kaplan-Meier (dashed line) and DOC-specific cumulative incidence (gray line).

We see that the Kaplan-Meier (on the complementary scale of the cumulative incidence) and the estimate of the CE-specific cumulative incidence are almost equal. Comment on the following statement

> Death that is not caused by symptomatic cardiac events may not be related to having symptomatic cardiac events. In this case, ignoring the informative censoring mechanism does not substantially influence the estimates of symptomatic cardiac events.

3. **Interpretation of gender difference.** In a study on the effect of gender on time from diagnosis of bladder cancer to relapse, death from other causes (DOC) was a competing risk. In Figure 6, we observe a strong effect of gender on relapse to both end points.
One may wonder whether the lower risk of relapse for males may be explained by the higher mortality from other causes: the larger number of deaths among men prevents relapse to occur. In Figure 7, we plot both the Kaplan-Meier for relapse, in which individuals that die from other causes leave the risk set, and the estimate of relapse-specifc cumulative incidence estimate.
Comment on the following statement:

> The gender difference is also present in the Kaplan-Meier curves. Moreover, for both genders the cause-specific cumulative incidence function and the Kaplan-Meier are almost the same. Hence, the difference in relapse by gender cannot be explained by the larger competing death rates for males.

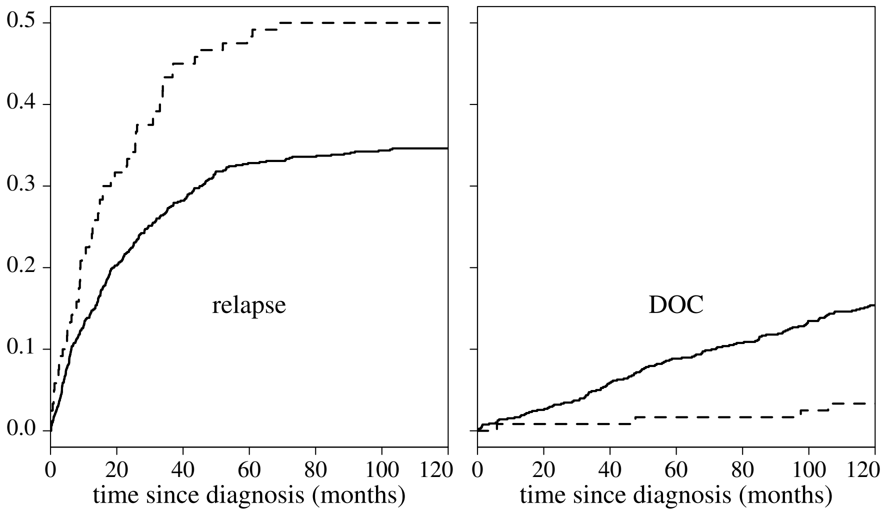


Figure 6: Estimate of cause-specific cumulative incidence for relapse (left panel) and death (right panel) for females and males. Solid lines: males. Dashed lines: females.

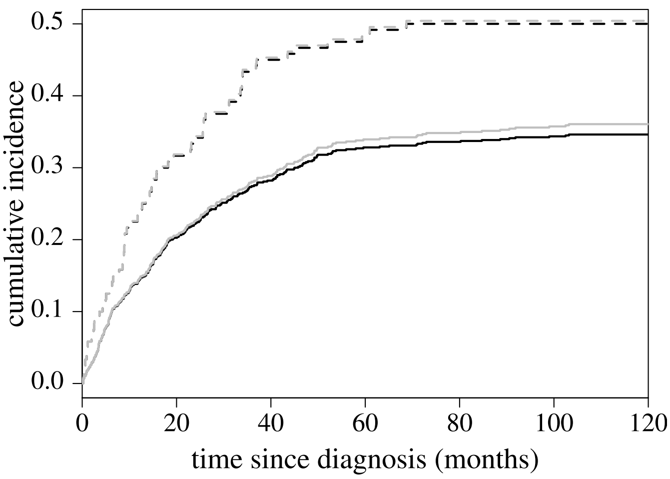


Figure 7: Estimate of relapse-specific cumulative incidence (black lines) and Kaplan-Meier (grey lines) for males (solid lines) and females (dashed lines).

4. **Cause-specific hazard and independent censoring.** In patients that receive allogeneic hematopoietic stem cell transplantation, relapse and treatment related mortality (TRM) are two competing events. Comment on the following statements:

   I. The effect of a covariable on an event from either a cause-specific (e.g. relapse) model or overall (e.g. relapse and TRM combined) model may be very different from the effect of the covariable on the event (e.g. relapse) in the presence of competing risks.

   II. Using a cause-specific Cox regression model is incorrect because it ignores competing risks and treats them as censored.

Part III
Estimation: Subdistribution Approach

11 Subdistribution

In the previous part we considered estimation of the cause-specific hazard. Since the cause-specific hazard is a standard rate, fitting a proportional cause-specific hazards regression model boils down to fitting a standard Cox proportional hazards model. The cause-specific hazard forms the basis of the Aalen-Johansen estimator of the cause-specific cumulative incidence, but they do not have a one-to-one relation because the cause-specific cumulative incidence is determined by the cause-specific hazards for all competing risks.

11.1 Subdistribution hazard

Since the occurrence of a competing event is ignored, an individual that experiences a competing event remains in the risk set until the time at which he would have been censored (either administratively or due to loss to follow-up). In the rare situation that we have complete information, individuals who experience a competing event remain in the risk set forever. In actual data analysis, they should be given a follow-up that is at least as long as the longest observed follow-up time in the data. If all censoring is administrative, we know for every individual when he would have been censored, even if he experienced the competing event first. When computing the estimate based on a data set, individuals that experienced a competing event should remain in the risk set until this administrative censoring time. The most common situation is that we don't know the censoring time because the competing event comes first. Then we estimate the time-to-censoring distribution $\overline{\Gamma}$. One approach is to use this estimate to multiply impute a censoring time [43]. An alternative is to use weights [14, 15]: an individual that experiences a competing event remains in the risk set with a weight that changes over time according to the estimated probability to remain uncensored. A similar correction can be made in the presence of left truncated data, based on an estimate of the time-to-entry distribution Φ.

Estimation of subdistribution hazard

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Complete data (no censoring/truncation): individuals with competing event remain in risk set forever. Small change in data setup, standard software

Administrative censoring only: individuals with a competing event leave risk set at their date of censoring. Standard software can be used with small change in follow-up

Censoring time unknown in some individuals: *Estimate time-to-censoring distribution*. Then for those with competing event

- multiply impute censoring times
- *reweight* by probability to remain uncensored

Left truncation Weights determined by time-to-entry distribution

.53

We explain how these weights to correct for unobserved right censoring and left truncation are created. For censoring, we estimate the time-to-censoring distribution by reversing the role of event (of any type) and censoring. Then the censoring times are either observed or right censored by event. For left truncation, we reverse the role of observed entry time and the observed event or censoring $T \wedge C$.

Estimation of $\overline{\Gamma}$ and Φ

Simplest: via standard PL-form

- $\widehat{\overline{\Gamma}}$: reverse role of event time T_i and censoring C_i:

$$\widehat{\overline{\Gamma}}(t) = \widehat{\mathrm{P}}(C > t) = \prod_{i:c_{(i)} \leq t} \left\{ 1 - \frac{m_i}{r(c_{(i)})} \right\}$$

  m_i: number of censorings at $c_{(i)}$

- $\widehat{\Phi}$: reverse role of $X_j = T_j \wedge C_j$ and truncation time V_j, V_j ("event") is right truncated by X_j:

$$\widehat{\Phi}(t) = \widehat{\mathrm{P}}(V \leq t) = \widehat{\mathrm{P}}(-V \geq -t) = \prod_{i:-v_{(i)} < -t} \left\{ 1 - \frac{w_i}{r(v_{(i)})} \right\}$$
$$= \prod_{i:v_{(i)} > t} \left\{ 1 - \frac{w_i}{r(v_{(i)})} \right\}.$$

  w_i: number of entries at $v_{(i)}$

.54

11.2 Product-limit estimator

We are now ready to formulate an alternative estimator of the cause-specific cumulative incidence based on the subdistribution hazard. It has the same form as the classical Kaplan-Meier. This is not surprising, because it follows from the same relation between the hazard and the corresponding cumulative quantity for discrete distributions. For the Kaplan-Meier we have

Kaplan-Meier: product-limit formula

- Based on law of conditional probability

$$\mathrm{P}\{T > t\} = \prod_{i:t_{(i)} \leq t} \mathrm{P}\{T > t_{(i)} | T > t_{(i)}-\}$$
$$= \prod_{i:t_{(i)} \leq t} \left\{ 1 - h(t_{(i)}) \right\}$$

- and estimate of hazard

$$\widehat{h}(t_{(i)}) = \frac{d(t_{(i)})}{r(t_{(i)})}$$

- Same for subdistribution

$$\mathrm{P}\{T_k > t\} = \prod_{i:t_{(i)} \leq t} \mathrm{P}\{T_k > t_{(i)} | T_k > t_{(i)}-\}$$
$$= \prod_{i:t_{(i)} \leq t} \left\{ 1 - h_k(t_{(i)}) \right\}$$

.55

General product-limit estimator of F_k

$$\widehat{F_k}^{\mathrm{PL}}(t) = \prod_{i: t_{(i)} \le t} \left\{ 1 - \widehat{h}_k(t_{(i)}) \right\}$$

$$\widehat{h}_k(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Weights: contribution $\omega_l(t_{(i)})$ of individual l to $r^*(t_{(i)})$ is

- censored or event of type k before $t_{(i)}$: 0
- still at risk at $t_{(i)}$: 1
- competing event at $t_{(j)}$ before $t_{(i)}$: weight

$$\frac{\widehat{\widehat{\Gamma}}(t_{(i)}-)}{\widehat{\widehat{\Gamma}}(t_{(j)}-)} \times \frac{\widehat{\Phi}(t_{(i)}-)}{\widehat{\Phi}(t_{(j)}-)}$$

$$\approx \widehat{\mathrm{P}}\{C > t_{(i)} | C > t_{(j)}\} \times 1/\widehat{\mathrm{P}}\{V < t_{(j)} | V < t_{(i)}\}$$

.56

If we assume censoring and entry time to be independent, the censoring weight $\widehat{\widehat{\Gamma}}(t_{(i)}-)/\widehat{\widehat{\Gamma}}(t_{(j)}-)$ can be seen as an estimate of $\mathrm{P}\{C \ge t_{(i)} | C \ge t_{(j)}\}$, i.e. the probability that an individual remains free of censoring until the event time $t_{(i)}$, given that he was not censored before he experienced the competing event at $t_{(j)}$[8]. Phrased otherwise: if we don't know when an individual that has a competing event at $t_{(j)}$ would have been censored, what we can do is to distribute his censoring time over the later observed censoring times according to the conditional distribution $\widehat{\widehat{\Gamma}}(t-)/\widehat{\widehat{\Gamma}}(t_{(j)}-)$.

The truncation weights $\widehat{\Phi}(t_{(i)}-)/\widehat{\Phi}(t_{(j)}-)$ can be seen to correct for individuals that experienced a competing event before they would have entered the study. They are not observed, but ought to be included in the risk set because individuals remain in the risk set when a competing event occurs. They are represented by the individuals that had their competing event observed. At each event time $t_{(i)}$ of the type of interest, an individual that had a competing event observed at an earlier time point $t_{(j)}$ is still at risk. He represents individuals that had their competing event missed with a weight $1/\widehat{\mathrm{P}}\{V < t_{(j)} | V < t_{(i)}\}$. The condition $V < t_{(i)}$ reflects that we calculate the risk set at $t_{(i)}$; we do not correct for late entries after $t_{(i)}$. The event $V < t_{(j)}$ reflects that this individual with the competing event at $t_{(j)}$ represents individuals that had a competing event at the same $t_{(j)}$ but were missed. This weight is estimated by $\{\widehat{\Phi}(t_{(j)}-)/\widehat{\Phi}(t_{(i)}-)\}^{-1} = \widehat{\Phi}(t_{(i)}-)/\widehat{\Phi}(t_{(j)}-)$. Note that individuals that experienced the event of interest before they would enter the study are represented by the individuals that are observed to be at risk, as reflected in $r(t)$. For them, no further correction is needed.

If we choose the weights as described on slide 56, the resulting product-limit estimator is algebraically equivalent to the Aalen-Johansen estimator $\widehat{F_k}^{\mathrm{AJ}}$ [14]:

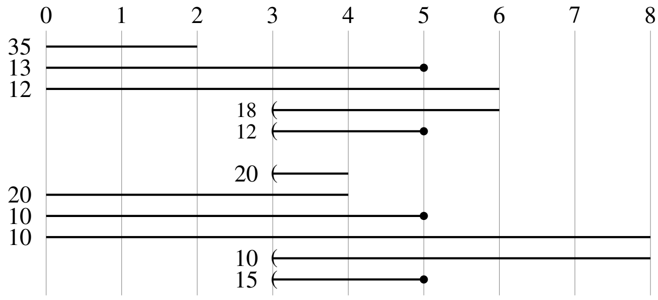
$$\widehat{F_k}^{\mathrm{AJ}} \equiv \widehat{F_k}^{\mathrm{PL}} \tag{11}$$

Note that this equivalence is an empirical relation, i.e. based on the data. It holds whatever the dependence between censoring time and entry time. $\widehat{\widehat{\Gamma}}$ and $\widehat{\Phi}$ can be calculated irrespective of whether they represent the marginal distributions of Γ and Φ (which only holds if C and V are independent). Similarly, the equivalence holds irrespective of whether Γ and Φ depend on covariables, which are not included in the calculation.

We show the calculations by example, using the data as in slide 27. We first calculate the product-limit forms $\widehat{\Gamma}^{\mathrm{PL}}$ and $\widehat{\Phi}^{\mathrm{PL}}$. In slide 57 we show the same data as in slide 28, but with the role of events and censorings interchanged.

[8] Since we assumed events to occur before censorings in case of ties, we use $\ge$ instead of $>$.

Calculation of $\widehat{\Gamma}^{\mathrm{PL}}$



$\widehat{\Gamma}^{\mathrm{PL}}(t) = 1 - \frac{50}{100} = 0.5$ if $t \geq 5$; $\widehat{\Gamma}^{\mathrm{PL}}(t) = 1$ if $t < 5$ .57

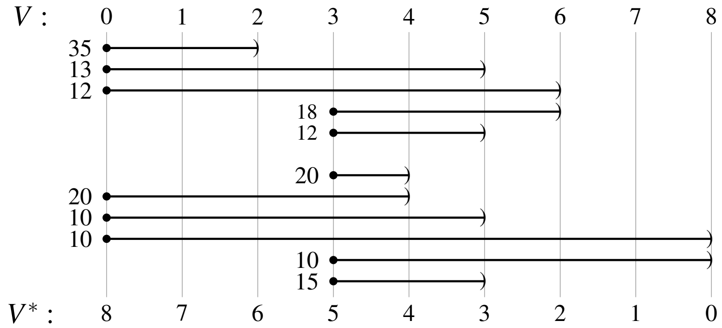
The only censoring, which is treated as event, occurs at $c = 5$. The estimate of the censoring hazard at that time is

$$\frac{13+12+10+15}{13+12+18+12+10+10+10+15} = \frac{50}{100} = 0.5\,.$$

Hence we have $\widehat{\Gamma}^{\mathrm{PL}}(t) = 1$ for $t < 5$ and $\widehat{\Gamma}^{\mathrm{PL}}(t) = 1 - 0.5 = 0.5$ thereafter.

The data needed for the calculation of $\widehat{\Phi}^{\mathrm{PL}}$ in our example data set is shown in slide 58. Right truncation as caused by the events and censorings is denoted by "$)$". The entries, which play the role of events, occur at $v = 0$ and $v = 3$.

Calculation of $\widehat{\Phi}^{\mathrm{PL}}$



$\widehat{\Phi}(t) = 1 - \frac{75}{140} = \frac{65}{140}$ if $0 \leq t < 3$; $\widehat{\Phi}(t) = 1$ if $t \geq 3$ .58

Using the notation on slide 54, at $v = 3$ we have

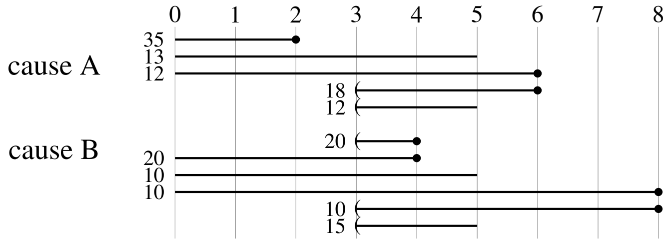
$$\frac{w_2}{r(v_{(2)})} = \frac{18+12+20+10+15}{13+12+18+12+20+20+10+10+10+15} = \frac{75}{140}\,,$$

hence $\widehat{\Phi}(t) = 1 - 75/140 = 65/140$ for $0 \leq t < 3$ and $\widehat{\Phi}(t) = 1$ thereafter. Hence, the probability to be observed from $t = 0$ onwards, $\mathrm{P}(V \leq 0)$, is estimated to be $65/140$ and the estimate of the probability to enter at $t = 3$ is $75/140$.

When using software that computes product-limit estimators, we may create the variable $V^* = -V + 8$ as at the bottom on slide 58. It transforms right truncated into left truncated event time data that don't have a negative value. Events or censorings occur at $v^* = 0, 2, 3, 4$ and 6. The only entries occur at $v^* = 5$ and $v^* = 8$. Hence $\widehat{\mathrm{P}}(V^* > v^*) = 1$ for $v^* < 5$, $\widehat{\mathrm{P}}(V^* > v^*) = \frac{65}{140}$ for $5 \leq v^* < 8$ and $\widehat{\mathrm{P}}(V^* > v^*) = 0$ for $v^* \geq 8$, translating into $\widehat{\mathrm{P}}(V < v) = 1$ for $v > 3$, $\widehat{\mathrm{P}}(V < v) = \frac{65}{140}$ for $0 < v \leq 3$ and $\widehat{\mathrm{P}}(V < v) = 0$ for $v \leq 0$.

Now we are ready to calculate the product-limit form.

Calculation of $\widehat{F_A}^{\mathrm{PL}}(2)$ and $\widehat{F_A}^{\mathrm{PL}}(6)$



	2 years	4 years	6 years
$\widehat{\widehat{\Gamma}}(t-)$	1	1	0.5
$\widehat{\Phi}(t-)$	$\frac{65}{140}$	1	1
$d(t)$	35 (A)	40 (B)	30 (A)
$r^*(t)$	100		$50+40\times\frac{0.5}{1}\times\frac{1}{1}=70$
$\widehat{h_A}(t)$	$\frac{35}{100}$	0	$\frac{30}{70}$
$\widehat{F_A}^{\mathrm{PL}}(t)$	$(1-\frac{35}{100})=\frac{65}{100}$	$\frac{65}{100}$	$(1-\frac{35}{100})\times(1-\frac{30}{70})=\frac{13}{35}$

.59

11.3 Confidence intervals based on PL-estimator

The PL form has the same structure as the Kaplan-Meier and we can extend the approach for the Kaplan-Meier estimator to the competing risks setting, replacing $r(t_{(j)})$ by $r^*(t_{(j)})$ in (4) or (5). Using the delta method, the variance of $\widehat{F_k}^{\mathrm{PL}}$ is estimated as

$$\widehat{\mathrm{var}}\left\{\widehat{F_k}^{\mathrm{PL}}(t)\right\}=\{\widehat{F_k}^{\mathrm{PL}}(t)\}^2\times\widehat{\mathrm{var}}\left\{\widehat{H_k}(t)\right\}.$$

Using this estimate of the variance, we can use the formulas (7) and (8)/(9) from section 6.1 and apply them to $\widehat{\mathrm{s.e.}}\{\widehat{\widehat{F_k}}(t)\}$. Or we can use the ones from (10) that are based on $\widehat{F}_k(t)$ instead of $\widehat{\widehat{F_k}}(t)$.

Standard error/confidence intervals

Based on formulas from Kaplan-Meier

- Based on estimator of subdistribution hazard instead of cause-specific hazard.

$$\widehat{\mathrm{var}}\left\{\widehat{H_k}(t)\right\}=\widehat{\mathrm{var}}\left\{\sum_{t_{(j)}\le t}\frac{d_k(t_{(j)})}{r^*(t_{(j)})}\right\}$$

  Two forms: Aalen and Greenwood.
- Confidence intervals: e.g. on log-scale (via $\log\{\widehat{\widehat{F_k}}(t)\approx-\widehat{H_k}(t)\}$):

$$\exp\left[\log\{\widehat{\widehat{F_k}}(t)\}\pm k_\alpha\,\widehat{\mathrm{s.e.}}\{\widehat{H_k}(t)\}\right]$$

- PL-form seems to perform as well as approach based on AJ-form in most settings .60

In a simulation study, the performance of this approach was similar or even better than the one based on the AJ-form [15, Section 2.4], except when there was heavy left truncation.
We also give some relations and properties. They are formulated for the estimators, but they equally hold for the theoretical quantities.

Some relations and properties

- Estimate of overall hazard is sum of cause-specific hazard estimates
$$\frac{\sum_{e=1}^{K} d_e(t_{(i)})}{r(t_{(i)})} = \frac{d(t_{(i)})}{r(t_{(i)})}$$
- Sum of Aalen-Johansen (or weighted PL) estimates is Kaplan-Meier of overall event risk
- Marginal Kaplan-Meier always steeper than weighted PL (or Aalen-Johansen) estimate, because
$$r(t_{(i)}) \leq r^*(t_{(i)})$$
- Relation between both hazards
$$\widehat{\lambda}_k(t) \times \widehat{\overline{F}}^{\mathrm{PL}}(t-) = \widehat{h}_k(t) \times \widehat{\overline{F_k}}^{\mathrm{PL}}(t-)$$

.61

12 Log-rank tests

In the situation without competing risks, the log-rank test for the null hypothesis that the time-to-event distributions for two or more groups are the same is in essence a nonparametric test for equality of the corresponding hazards. At every event time, the estimates of the hazards in the groups are compared.

The log-rank test for equality of the cause-specific hazards is just the log-rank test from the classical survival setting. The reason is that the nonparametric estimator of the cause-specific hazard is the standard hazard estimator. If all competing events are independent, it additionally tests for equivalence of the marginal hazards and marginal cumulative distributions. However, since it has no one-to-one relation with the cumulative scale, it does not test whether the cause-specific cumulative incidences are the same.

If we want to use a nonparametric test for equality of cause-specific cumulative incidence curves, we can use the log-rank test based on subdistribution hazards, which do have the one-to-one correspondence with the cumulative scale. This test was first introduced by Gray [21] for right censored data [9], but can be extended to left truncated data. It is based on the extended at-risk set r^*. Another issue is the estimation of the variance of the test statistic. We can use the formula from the standard log-rank test, replacing $r(t)$ by $r^*(t)$. This approach is equivalent to the one suggested in [5], but differs from the one used by Gray. In a simulation study (unpublished), the difference between the approaches was small.

Testing for differences between curves

- Log-rank test is a nonparametric test for equality of hazards over groups. One of family of tests. E.g. with two groups in classical setting:
$$Z = \sum_{j=1}^{n} W(t_{(j)}) \frac{r^{(1)}(t_{(j)}) r^{(2)}(t_{(j)})}{r^{(1)}(t_{(j)}) + r^{(2)}(t_{(j)})} \left[\frac{d^{(1)}(t_{(j)})}{r^{(1)}(t_{(j)})} - \frac{d^{(2)}(t_{(j)})}{r^{(2)}(t_{(j)})} \right]$$
Log-rank: $W \equiv 1$
- Standard log-rank test (one event type): nonparametric comparison of marginal hazards $\rightarrow$ compares marginal time-to-event distributions (1-1 relation)
- Standard log-rank test (competing risks): nonparametric comparison of cause-specific hazards
- Gray's log-rank test: replace r by r^*. Nonparametric test to compare subdistribution hazards $\rightarrow$ compares cause-specific cumulative incidences (1-1 relation)

.62

[9] It's often called Gray's test

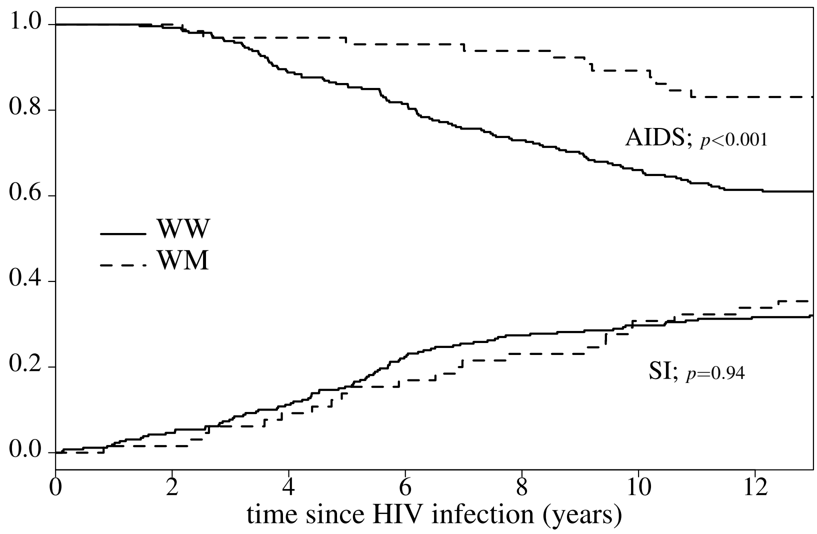
Remember the three types of hazards that we defined in a setting with competing risks:

Three types of hazards

1. marginal, only estimable under independence, conditionally on the covariables in the model
2. cause-specific; estimable (standard censoring)
   No 1-1 relation with cause specific cumulative incidence
3. subdistribution; estimable (weighted risk set)
   1-1 relation with cause specific cumulative incidence .63

As an example, we compare the cause-specific cumulative incidence by CCR5 genotype for both event types. Individuals that have the CCR5-Δ32 deletion on one of their chromosomes have the value `WM`. The value `WW` stands for the wild type, i.e. without the deletion, on both chromosomes. The estimates for both event types are shown in slide 64, together with the p-values based on Gray's log-rank test.

Estimates by value of CCR5, with Gray's log-rank test



.64

13 Proportional subdistribution hazards

Since the subdistribution hazard defines the cumulative incidence, parameter estimates from a proportional subdistribution hazards model have a qualitatively similar interpretation on the cumulative scale. The model that assumes proportionality in the subdistribution hazard is usually called the Fine and Gray model.

Fine and Gray model

- *Regression on subdistribution hazard*

$$h_k(t \,|\, \mathbf{Z}_i) = h_{k,0}(t)\, \exp(\beta_k^\top \mathbf{Z}_i)$$

- Original idea by Fine and Gray (1999)
  Sometimes called "competing risks proportional hazards"

- Estimation based on weighted partial likelihood:

$$L(\beta_k) = \prod_{\substack{i=1 \\ e_i\delta_i=k}}^{N} \left\{ \frac{\exp(\beta_k^\top \mathbf{Z}_i)}{\sum_{j=1}^{N} \omega_j(t_i)\exp(\beta_k^\top \mathbf{Z}_j)} \right\}. \tag{12}$$

.65

Parameters can be estimated by maximizing a partial likelihood in which the risk set is adapted to estimation of the subdistribution hazard. If there is no left truncation and if all censoring is administrative, the only change compared to the partial likelihood from the Cox model is in the definition of the risk set: individuals that experience a competing event remain in the risk set until their administrative censoring time. The general approach with left truncation due to late entry and/or right censoring is to use weighted at-risk sets that correct for this missed information. If the truncation and censoring weights $\widehat{\Phi}$ and $\widehat{\widehat{\Gamma}}$ are chosen to be the same for every individual, i.e. if they do not depend on covariables, this score function can be rewritten in a form based on the weights ω_l that were defined in slide 56. Whether and how to make the weights depend on covariables is an open question [12].

P-values from the proportional subdistribution hazards model are more in correspondence with the log-rank test on subdistribution hazards:

Test for effect of CCR5-Δ32 deletion

- Cause-specific hazards (WM versus WW)

	HR	p-value (Cox)	log-rank	df
AIDS	0.291	<0.001	<0.001	1
SI	0.776	0.29	0.28	1

- Subdistribution:

	HR	p-value (F&G)	log-rank	df
AIDS	0.366	<0.001	<0.001	1
SI	1.024	0.92	0.94	1

.66

13.1 Standard errors

Fine and Gray derived the asymptotic distribution of the parameter estimates [13]. They came up with a sandwich-type estimator of the standard error. However, in a simulation study the regular estimator of the standard error performed better in small samples [14]. The sandwich estimator of the variance tended to be too small, leading to confidence intervals that are too narrow.

Standard errors

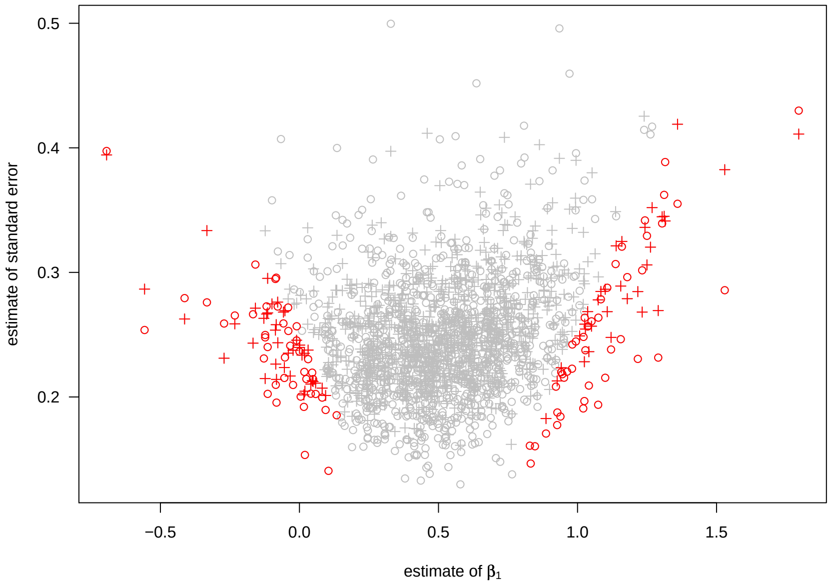
- Fine & Gray use sandwich type estimator
  Also default in `coxph` in recent versions of `survival`
- In simulation study, standard estimator performed better:
  - Two covariables, Z_1 and $Z_2 \sim N(0,1)$, effect $\beta_1 = \beta_2 = 0.5$
  - T° unit exponential with mass 0.7 at ∞ if $Z_1 = Z_2 = 0$
  - Censoring uniform on $[0.5, 1]$

Results for β_1 (5000 runs):

	N	bias	p-value	$\mathrm{var}(\widehat{\beta}_1)$	$\overline{\mathrm{var}(\beta_1)}$	$P(0.5 \in \mathrm{CI})$	p-value
`coxph` with	100	0.018	<0.001	0.069	0.065	0.946	0.21
`robust`	200	0.005	0.043	0.032	0.030	0.944	0.074
`=FALSE`	500	0.001	0.38	0.012	0.011	0.950	1.00
`crr` or	100	0.018	<0.001	0.069	0.062	*0.928*	<0.001
`coxph`	200	0.005	0.043	0.032	0.029	*0.934*	<0.001
	500	0.001	0.38	0.012	0.011	0.945	0.14

.67

Standard errors



.68

Why Does Sandwich Perform Worse?

- Sandwich asymptotically unbiased
- Censoring characteristics in sample is what matters, not the theoretical distribution of the time-to-censoring distribution
- Only individuals without event are reweighted in ω_i-weights.

.69

14 Exercises part III

1. Comment on the following statement

   When there are no competing risks, a log-rank test is used to compare Kaplan-Meier cumulative incidence curves. In competing risks settings, this test is inappropriate.

2. **Cause-specific and subdistribution as surrogate of marginal.** Comment on the following statement

   When the ratio of the marginal hazards is the desired estimand and a regression model on the cause-specific hazard or subdistribution hazard is used, they can be considered ad hoc methods for estimating the ratio of the marginal hazards.

Part IV

Interpretation; Summary

15 Combined analysis

In sections 9.2 and 13 we fitted a separate model for each event type. This resembles a stratified regression analysis. A stratified analysis is performed if the effect of covariables is quantified separately for each value of some categorical variable such as gender. The main reason to perform a stratified analysis is simplicity; the parameters are easy to interpret. However, exactly the same results are obtained by fitting a single regression model that includes gender as extra covariable and an interaction term between gender and all other covariables. Using one single model with interaction terms has several advantages. We can *test* whether the effect of the covariables differs by gender. And if we want to *assume* the effect of a covariable to be the same for both genders, we can remove the interaction term from the model. This increases power if the assumption is correct.

Similarly, in separate competing risks analyses per event type, we cannot test the hypothesis whether the effect of the CCR5-Δ32 deletion is the same for both event types, i.e. whether $\beta_{\text{AIDS}} = \beta_{\text{SI}}$. And we cannot perform an analysis in which β_{AIDS} and β_{SI} are assumed to be equal. Also, we cannot test whether hazards for different causes are proportional or fit models in which we assume baseline hazards to be proportional.

Analysis per event type

We cannot

- test whether effect of covariable differs per event type
- assume equality of effect for some (or all) event types
- test for proportional baseline hazards or assume them proportional

Similar issues as with stratified regression analysis .70

However, we can do all these things if we create an augmented data set using a duplication method [36]. This is explained in the next section.

15.1 Stacked data set

The regression analyses per end point all use the same time and covariable information. Only the definition of the event variable is different. If we want to perform the analysis for both end points at once, we basically stack the data sets that are used in the separate analyses. We can write code to create such data set, but we can also use the `crprep` of `finegray` function. Below we use the data set created via the `crprep` function.

For the cause-specific hazard analysis, the data of the four example individuals in the stacked format is shown in slide 71. The **time** and **ccr5** variables from the original data set are duplicated. The data has an extra column **failcode**, denoting which of the two separate analyses that row refers to. Hence, per individual one row has the value `AIDS` and one row has the value `SI`. Every individual is at risk to progress to either of the two end points and thus has both values of **failcode**; unlike **ccr5**, it is not a characteristic of the individual. Just like in the separate analyses, the event variable should have the value 0 if censored and 1 if the event of interest occurred. The event of interest occurred if the values in **cause** and **failcode** coincide. If there were K competing events, each individual would have K rows in the stacked format, one for every possible cause of failure. The last three columns are explained below.

Stacked data set for combined analysis on cause-specific hazard

- Combine data sets for each event type *and adapt covariables*:

patnr	time	status	status2	cause	failcode	ccr5	ccr.comb	ccrAIDS	ccrSI
14	5.05	0	0	event-free	AIDS	WW	WW	0	0
3	2.23	1	1	AIDS	AIDS	WW	WW	0	0
15	10.20	1	1	AIDS	AIDS	WM	WM.AIDS	1	0
8	8.61	2	0	SI	AIDS	WW	WW	0	0
14	5.05	0	0	event-free	SI	WW	WW	0	0
3	2.23	1	0	AIDS	SI	WW	WW	0	0
15	10.20	1	0	AIDS	SI	WM	WM.SI	0	1
8	8.61	2	1	SI	SI	WW	WW	0	0

- Baseline hazard differs by event type: stratified Cox model, with `failcode` as stratum .71

To replicate the separate analyses, we need to use a stratified Cox model.

Stratified Cox model

Used if baseline hazard may differ per subgroup (nonproportional hazards)

$$h_g(t \,|\, \mathbf{Z}_i) = h_{g,0}(t)\, \exp(\beta^\top \mathbf{Z}_i)\ .$$

with g subgroup ("stratum") to which person i belongs.
Maximise product of partial likelihoods per stratum

$$L(\beta) = \prod_{g \in \mathscr{G}} L^g(\beta)$$

with $L^g(\beta)$ partial likelihood over individuals in stratum g

- If all parameters differ by stratum, same as fitting separate Cox model per stratum
- In R:

```
coxph(Surv(time,status)~strata(var) + covar,
                                  data=...)
```

.72

15.2 Analyses

If we perform the analysis per event type, each analysis yields its own estimates of the baseline hazard and of the effect of the deletion. In the combined analysis, we estimate a separate baseline hazard per event type by including the type of event as stratum variable in the model. The effect of the CCR5-Δ32 deletion per event type can be modeled in several ways. The most straightforward way is by including an interaction between `ccr5` and `failcode`. In R, if we call the stacked data set `aidssi.stack`, we use[10]

Combined analysis I: interaction term

```
coxph(Surv(time, status2) ~ ccr5 * strata(failcode),
                                 data = aidssi.stack)
```

which yields

```
                   coef exp(coef) se(coef)     z       p
ccr5WM           -1.236     0.291    0.307 -4.02 5.7e-05
ccr5WM:strata(fa  0.982     2.669    0.389  2.53 1.2e-02

Likelihood ratio test=23.2  on 2 df, p=9.29e-06  n= 648,
                                   number of events= 220
   (10 observations deleted due to missingness)
```

Compare with -1.236 and -0.254 from analysis per event type. .73

[10] In R, the asterisk in the regression formula is used to denote both the main effects and their interaction.

Since `failcode` is a stratum variable, it does not have a parameter estimate. The value -1.236 is, as before, the effect of the deletion on the AIDS-specific hazard. The value 0.982 is different from the earlier analyses. It represents the *additional* effect of the deletion on the SI-specific hazard. The value -0.254 that we found in the separate analysis with SI as the event type is the effect on AIDS plus the additional effect that is represented by the interaction term: $-1.236+0.982=-0.254$ (or equivalently the product of 0.291 and 2.669 is equal to 0.776 on the hazard scale). Note that the value of the likelihood ratio test statistic is the sum of the two values from the separate analyses.

We see the first advantage of the use of the stacked data format. We can *test* whether the CCR5-Δ32 deletion has the same association with the AIDS-specific and the SI-specific hazard. The p-value 0.012 suggests that the association of the CCR5-Δ32 deletion with the cause-specific hazard differs by event type.

Instead of including an interaction term, we can also create a compound variable that combines information on CCR5 and event type. We create a categorical variable with three levels: one for the baseline `WW` level and two for the deletion in order to allow the effect of `WM` to depend on type of event. This is the column **ccr.comb** in slide 71. Contrary to the standard regression setting we have three instead of four levels, because the baseline hazards are estimated nonparametrically.

Combined analysis II: compound variable

```
coxph(Surv(time, status2) ~ ccr.comb + strata(failcode),
                                     data = aidssi.stack)
```

Same effect estimates and p-values as in separate analyses

```
                    coef exp(coef) se(coef)     z       p
ccr.combWM.AIDS   -1.236     0.291    0.307 -4.02 5.7e-05
ccr.combWM.SI     -0.254     0.776    0.238 -1.07 2.9e-01

Likelihood ratio test=23.2  on 2 df, p=9.29e-06  n= 648,
                                    number of events= 220
   (10 observations deleted due to missingness)
```

.74

Since they cannot compute on text representations, statistical programs transform the categorical variable **ccr.comb** into two dummy variables that represent the non-reference values `WM.AIDS` and `WM.SI`. They are given in the columns **ccrAIDS** and **ccrSI** in slide 71. The variable **ccrAIDS** has the value 1 when the genotype is not the reference category and the event type in **failcode** has the value `AIDS`, and it has the value 0 otherwise (see person 15); similarly for **ccrSI** and `SI`.

Combined analysis III: type-specific covariables

- The variables `ccrSI` and `ccrAIDS` are examples of *type-specific* covariables
- Instead of specifying (in separate models k)

  $$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{k,0}(t)\,\exp(\beta_k^\top \mathbf{Z}_i)\,,$$

  we can equivalently specify (in one model)

  $$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{k,0}(t)\,\exp(\beta^\top \mathbf{Z}_{k,i})\,,$$

- $\mathbf{Z}_{k,i}$ equals $\mathbf{Z}_i$ for cause k, and has value 0 otherwise
- Eg. Z_i = CCR5 genotype coding (`WM`=1, `WW`=0); AIDS: $k=1$; SI: $k=2$:

  $$\texttt{ccrAIDS}_i = Z_{1,i} = \begin{cases} Z_i, & \text{if } \texttt{failcode=AIDS} \\ 0, & \text{if } \texttt{failcode=SI} \end{cases}$$

  and

  $$\texttt{ccrSI}_i = Z_{2,i} = \begin{cases} 0, & \text{if } \texttt{failcode=AIDS} \\ Z_i, & \text{if } \texttt{failcode=SI} \end{cases}$$

.75

The variables `ccrSI` and `ccrAIDS` are examples of so-called *type-specific covariables* [3, p.478]. The general idea is that parameters β_k that are modeled for each separate analysis are replaced by one set of parameters β that model the effect of all type-specific covariables. In the CCR5 example we have one variable, $Z_i = \texttt{ccr5}_i$. Continuous type-specific variables can be constructed in a similar way. For example, if we want to investigate the effect of age on both cause-specific hazards, we create two columns **ageAIDS** and **ageSI**. The column **ageAIDS** has the value of age if the event type in **failcode** has the value `AIDS` and has value zero otherwise; **ageSI** has the value of age if the event type in **failcode** is `SI` and has value zero otherwise.

Combined analysis III: results

```
coxph(Surv(time, status2) ~ ccrAIDS + ccrSI +
                    strata(failcode), data = aidssi.stack)
```

Same effect estimates and p-values as before

```
            coef exp(coef) se(coef)     z       p
ccrAIDS   -1.236     0.291    0.307 -4.02 5.7e-05
ccrSI     -0.254     0.776    0.238 -1.07 2.9e-01
```

.76

If the effects of all covariables are allowed to differ by stratum, the partial likelihood of the combined model using the stacked data set is the product of the partial likelihoods of the separate analyses. In slide 77, we show this for the effect of the CCR5 genotype on progression to AIDS and SI switch as competing first events. We use the model specification via the type-specific variables **ccrAIDS** and **ccrSI**.

The first term in the stratified partial likelihood is a product of terms over the first 324 rows in `aidssi.stack`, for which **failcode** is equal to `AIDS`. In these rows, **ccrSI** is always zero. $R_{\text{AIDS}}(t)$ is the set of individuals that are still at risk at time t over the rows with **failcode** equal to `AIDS`. It is the same as the risk set $R(t)$ of individuals in `aidssi` that are still at risk at time t, because the event times have been duplicated in the stacked data set. Furthermore, **ccrAIDS** and **ccr5** have the same value in these rows[11]. Therefore, the expression can be rewritten as the partial likelihood based on the original data set `aidssi`, with AIDS as the event type. The same argument holds for SI as event type.

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\text{AIDS}}, \beta_{\text{SI}})$ with `failcode` as stratum: $L(\beta) = L^{\text{AIDS}}(\beta) \times L^{\text{SI}}(\beta)$ ($R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i)$: risk set under `failcode=AIDS/SI`)

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1\ldots\\ \texttt{status2}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta_1 \times \texttt{ccrAIDS}_j + \beta_2 \times 0)} \right\} \qquad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1\\ \texttt{status}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_1 \times \texttt{ccr5}_j)} \right\} \qquad \texttt{aidssi}$$

$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325\ldots\\ \texttt{status2}_i=1}}^{648} \left\{ \frac{\exp(\beta_1 \times 0 + \beta_2 \times \texttt{ccrSI}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta_1 \times 0 + \beta_2 \times \texttt{ccrSI}_j)} \right\} \qquad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=325\\ \texttt{status}_i=2}}^{648} \left\{ \frac{\exp(\beta_2 \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_2 \times \texttt{ccr5}_j)} \right\} \qquad \texttt{aidssi}$$

- `ccrAIDS` only used for AIDS stratum; `ccrSI` only for SI stratum

.77

[11] Apart from the coding as categorical variable with levels `WW` and `WM` or as dummy 0-1 variable.

Note that a stratified proportional hazards model is not the same as a stratified analysis. A stratified analysis is performed when separate models are fitted for each value of some categorical variable. A stratified proportional hazards model is a single proportional hazards model that includes a categorical variable defining the “strata”, such that a separate baseline hazard is estimated for each level. A stratified proportional hazards model is a combined analysis. If we include an interaction between the stratum variable and every other variable in the model, as we did in the CCR5 example, all effects are allowed to differ over all levels of the stratum variable. Fitting such a stratified proportional hazards model is in essence the same model as when separate proportional hazards models are fitted for each value of the stratum variable. Depending on how we define the contrasts, all parameter estimates are exactly the same (as in the combined analyses II and III), or some parameters reflect the difference in effect over the event types (as in the combined analysis I).

15.3 Standard error

There has been some controversy surrounding the best computation of the standard error. It has been suggested to use a sandwich-type robust estimate for the standard errors in order to correct for the fact that individuals are replicated in the stacked data set [36].This is not correct. The non-robust standard error replicates the results from the separate analyses because the parameter effects of the covariables are maximized separately per end point and individuals can experience one event at most. If we use a robust estimate, the standard errors and p-values are no longer equal to the ones from the separate analyses:

Robust estimate of standard error?

```
coxph(Surv(time, status) ~ ccr.comb + strata(failcode) +
                    cluster(patnr), data = aidssi.stack)
```

gives

```
                  coef  se(coef) robust se      z        p
ccr.combWM.AIDS -1.236     0.307     0.296 -4.18 0.00003
ccr.combWM.SI   -0.254     0.238     0.233 -1.09 0.27000
```

Robust estimate differs from separate analyses .78

16 Advantages combined analysis

Performing the analysis for all event types at once allows for great flexibility in model specification. The major issue is to include the appropriate covariables. We can assume that a covariable has no effect on some event type by leaving out that type-specific covariable. For example, if we include **ageAIDS** but leave out **ageSI** then we assume that there is no effect of the CCR5 genotype on the progression to SI as the first event. But we can do much more.

16.1 Test for equality

We can test whether the effects of **ccrAIDS** and **ccrSI** are equal. One way is to test the null hypothesis $\beta_1 = \beta_2$. An alternative is to use a different contrast and replace one of the type-specific covariables by the original `ccr5` variable.

Advantage I: test whether effects are equal

```
coxph(Surv(time,status) ~ ccr5 + ccrSI + strata(failcode),
                              aidssi.stack)
```

gives

```
          coef exp(coef) se(coef)     z       p
ccr5WM -1.236     0.291    0.307 -4.02 5.7e-05
ccrSI   0.982     2.669    0.389  2.53 1.2e-02
```

Note: same result via interaction term:

```
coxph(Surv(time, status) ~ ccr5 * strata(failcode),
                                  data = aidssi.stack)
```

.79

This is the same output as when using the CCR5-by-stratum interaction. Equivalently, we can use `ccrAIDS` instead of `ccrSI`. Then we obtain the value $\beta_1 = -0.254$ for the effect on the SI-specific hazard, and the value $\beta_2 = -0.982$ for the additional effect on the AIDS-specific hazard.

When creating the type-specific variables and specifying the contrasts in the model, close attention should be given to the interpretation of the parameters. It is recommended to write down the model and investigate, for each combination of covariable and end point, which parameters contribute to the effect size. In this simplest of all models we have

Explanation model

$$\lambda_k(t) = \lambda_{k,0}(t)\exp(\beta_1 \times \texttt{ccr5} + \beta_2 \times \texttt{ccrSI}).$$

AIDS; WW	AIDS; WM	SI; WW	SI; WM
$\lambda_{1,0}(t)$	$\lambda_{1,0}(t)\exp(\beta_1)$	$\lambda_{2,0}(t)$	$\lambda_{2,0}(t)\exp(\beta_1+\beta_2)$

.80

The AIDS-specific hazard for `WM` relative to `WW` is $\exp(\beta_1)$; the effect of `WM` on the SI-specific hazard is $\exp(\beta_1+\beta_2)$. These effects are relative to the reference value `WW` *within* the specific event type. The effect of `WM` on the SI-specific hazard, relative to its effect on the AIDS-specific hazard is

$$\frac{\lambda_2(t|WM)}{\lambda_1(t|WM)} = \frac{\lambda_{2,0}(t)}{\lambda_{1,0}(t)}\exp(\beta_2).$$

16.2 Assume equality

Advantage II: assume equality

Same effect of CCR5 mutation on relative hazards for both transitions

```
coxph(Surv(time, status) ~ ccr5  + strata(failcode),
                                 data = aidssi.stack)
```

	coef	exp(coef)	se(coef)	z	p
ccr5	-0.701	0.496	0.186	-3.77	0.00016

This is similar to

```
coxph(Surv(time, status!=0) ~ ccr5, data = aidssi)
```

with original data set

patnr	time	status	ccr5
14	5.05	0	WW
3	2.23	1	WW
15	10.20	1	WM
8	8.61	2	WW

.81

If we assume that the effect of every covariable is the same for all event types, the analysis based on the stacked data has an equivalent analysis that uses the original data. We show this for the CCR5 example by comparing the partial likelihoods.

Why?

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1\ldots \\ \texttt{status2}_i=1}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=1}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi}$$

$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325\ldots \\ \texttt{status2}_i=1}}^{648} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=2}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi}$$

Same as partial likelihood based on `aidssi` data:

$$\prod_{\substack{i=1 \\ \texttt{status}_i>0}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right)$$ .82

Sometimes we need to assume equality in order to analyse the effect of covariables on event types that are rare. In the example B from Section 2, we investigated the how the different causes of death changed after the introduction of cART [24]. However, we wanted to correct for the potential confounders “risk group”, “sex” and “age”. When correcting for possible confounders, we needed to estimate at least 10 parameters per cause of death. However, liver related death was very rare in our data set[12]. The only way to analyse this end point is to *assume* that the effect of the other covariables on progression to liver related mortality is equal to their effects on some other more common cause of death. A detailed explanation of the model that we used can be found in Section 4.6 of my book [15].

Assuming equality necessary when some event types are rare

$N = 9164$, correct for age, sex, risk group, calendar period

	sex between men	idu	haemophiliac	sex between men and women	total
alive	5052	1310	73	2252	8446
aids	158	148	65	24	397
liver	3	15	22	1	39
non natural	24	98	2	6	128
natural	68	47	14	23	154

Too few events for proper analysis per event type .83

16.3 Proportional baseline hazards

We can also assume cause-specific baseline hazards to be proportional. If this is a reasonable assumption, it has the added value that we obtain a parametric quantification of the relative rate of progression to the different event types.

[12] Note that the main aim was to investigate the effect of hepatitis C coinfection on mortality. For that reason liver-related death was a separate endpoint.

Advantage III: Proportional baseline hazards

$$\lambda_{\mathrm{SI},0}(t) = \lambda_{\mathrm{AIDS},0}(t) \times \exp(\alpha)$$

Total model now reads (AIDS: $k=1$; SI: $k=2$):

$$\lambda_k(t\,|\,\mathbf{Z}_i) = \lambda_{1,0}(t)\exp\{\alpha \times (k-1) + \beta_1 \times \texttt{ccrAIDS}_i + \beta_2 \times \texttt{ccrSI}_i\} \tag{13}$$

In R:

```
coxph(Surv(time, status) ~ ccrAIDS + ccrSI +
                    failcode, data = aidssi.stack)
```

Results:

```
              coef exp(coef) se(coef)     z       p
ccrAIDS     -1.166     0.311    0.306 -3.81 0.00014
ccrSI       -0.332     0.718    0.237 -1.40 0.16000
failcodeSI  -0.184     0.832    0.148 -1.25 0.21000

Likelihood ratio test=21.5  on 3 df, p=8.12e-05  n= 648,
                                  number of events= 220
   (10 observations deleted due to missingness)
```

.84

Although `(k-1)` in (13) is not a covariable but a failcode specification, a model with proportional baseline hazards can be fitted by letting "event type" play the role of a covariable. Again a robust standard error is not needed because every individual has at most one row with an event.

The general model can be formulated as

$$\lambda_k(t\,|\,\mathbf{Z}_i) = \lambda_0(t)\exp\{\alpha_k + \beta_k^\top \mathbf{Z}_i\}\,. \tag{14}$$

The reference event type k_0 has $\alpha_{k_0} = 0$. It translates to a model for the overall probability to experience an event type of the form (see [40])

$$\frac{\mathrm{P}(E=k\,|\,\mathbf{Z})}{\mathrm{P}(E=k_0\,|\,\mathbf{Z})} = \exp\left\{\alpha_k + \beta_k^\top \mathbf{Z} - \beta_{k_0}^\top \mathbf{Z}\right\}. \tag{15}$$

17 From regression on hazard to cumulative scale

We can use results from a regression model on the hazard to estimate the cumulative incidence curve for any combination of values of the included covariables. Assume that all covariables are time-fixed; they do not change value over time. In the classical survival setting, if we use a proportional hazards model and an individual has covariable values $\mathbf{Z}$, then his probability to remain event-free can be estimated via:

Standard survival setting: one event type

- Cox regression, $\mathbf{Z}$ time-fixed: $h(t) = h_0(t)\exp(\beta^\top \mathbf{Z}_i)$
- One-to-one relation between effects on hazard and survival

  $$\overline{F}(t|\mathbf{Z}_i) = \exp\{-\int_0^t h_0(s)\exp(\beta^\top \mathbf{Z}_i)ds\} = \overline{F}_0(t)^{\exp(\beta^\top \mathbf{Z}_i)},$$

  with $\overline{F}_0(t) = \exp\{-\int_0^t h_0(s)ds\} = \exp\{-H_0(t)\}$

- Estimate becomes

  $$\widehat{\overline{F}}(t|\mathbf{Z}_i) = \exp\left\{-\widehat{H}_0(t)e^{\widehat{\beta}^\top \mathbf{Z}_i}\right\} = \widehat{\overline{F}}_0(t)^{\exp(\widehat{\beta}^\top \mathbf{Z}_i)}$$

  with $\widehat{\overline{F}}_0(t) = \exp\{-\widehat{H}_0(t)\}$ estimated "baseline" survival curve
- Curves cannot cross
- *Similar for regression model on subdistribution hazard*

.85

This one-to-one relation between hazard and cumulative incidence implies that the effect of a variable on the hazard translates to a qualitatively similar effect on the cumulative probability—the direction of an effect on the cumulative probability can already be learned from its effect on the hazard.

17.1 Competing risks

In a model for the cause-specific hazard, this no longer holds. The cause-specific cumulative incidence not only depends on the effect of the covariable on that cause, but also on the effects of the covariable on all other causes and on the baseline hazards of all other causes.

Cause-specific hazard

- Cumulative progression to one end point also depends on progression to other end points
- "No one-to-one relation with cumulative scale"
- Example C: CCR5-Δ32 deletion and SI switch

$$F_{SI}(t) = \mathrm{P}\{T \leq t, E = SI\} = \int_0^t \mathrm{P}\{T \geq s\}\lambda_{SI}(s)ds$$

with $\mathrm{P}\{T \geq s\} = \exp[-\int_0^s \{\lambda_{SI}(u) + \lambda_{AIDS}(u)\}du]$ .86

As an example, we quantify the SI-specific cumulative incidence, with AIDS as competing risk. We use a proportional cause-specific hazards model for both end points with the CCR5-Δ32 deletion as the only covariable. This model provides the parameter estimates for the effect of the deletion on the SI-specific hazard $\widehat{\beta}_{SI}$, the AIDS-specific hazard $\widehat{\beta}_{AIDS}$ as well as the estimates of the cumulative baseline hazards $\widehat{\Lambda}_{0,SI}$ and $\widehat{\Lambda}_{0,AIDS}$. The relation between the value of CCR5 and the estimate of the SI-specific cumulative incidence becomes

Estimate based on proportional cause-specific hazards

$$\begin{aligned}
\widehat{F}_{\mathrm{SI}}(t \mid \mathtt{ccr5}) &= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{\mathrm{SI}}(t_{(i)} \mid \mathtt{ccr5}) \times \widehat{\widehat{F}}(t_{(i)}- \mid \mathtt{ccr5}) \\
&= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{0,\mathrm{SI}}(t_{(i)}) \exp\big(\widehat{\beta}_{\mathrm{SI}}\mathtt{ccr5}\big) \times \\
&\quad \exp\Big\{-\big[\widehat{\Lambda}_{\mathrm{AIDS}}(t_{(i)}- \mid \mathtt{ccr5}) + \widehat{\Lambda}_{\mathrm{SI}}(t_{(i)}- \mid \mathtt{ccr5})\big]\Big\}
\end{aligned}$$

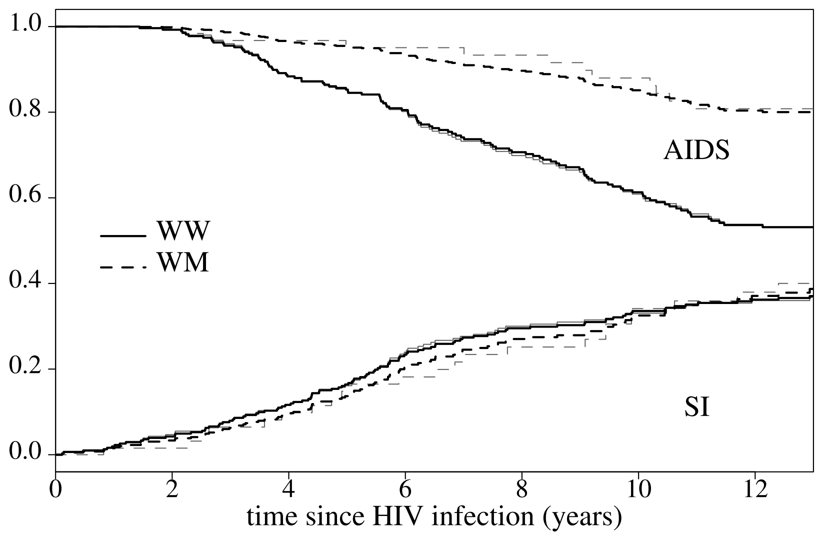
and use

$$\begin{aligned}
\widehat{\Lambda}_{\mathrm{SI}}(t \mid \mathtt{ccr5}) &= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{\mathrm{SI}}(t_{(i)} \mid \mathtt{ccr5}) \\
&= \sum_{t_{(i)} \leq t} \widehat{\lambda}_{0,\mathrm{SI}}(t_{(i)}) \exp(\widehat{\beta}_{\mathrm{SI}}\mathtt{ccr5}) \\
&= \widehat{\Lambda}_{0,\mathrm{SI}}(t) \exp(\widehat{\beta}_{\mathrm{SI}}\mathtt{ccr5}).
\end{aligned}$$

and similarly $\widehat{\Lambda}_{\mathrm{AIDS}}(t \mid \mathtt{ccr5}) = \widehat{\Lambda}_{0,\mathrm{AIDS}}(t) \exp(\widehat{\beta}_{\mathrm{AIDS}}\mathtt{ccr5})$ .87

$\widehat{\lambda}_{0,\mathrm{SI}}$ is the jump in the cumulative baseline hazard. A similar formula holds for $\widehat{F}_{\mathrm{AIDS}}(t)$. In slide 88, we compare these estimates with the nonparametric estimates (thin grey lines).

Cumulative incidence based on proportional cause-specific hazards model



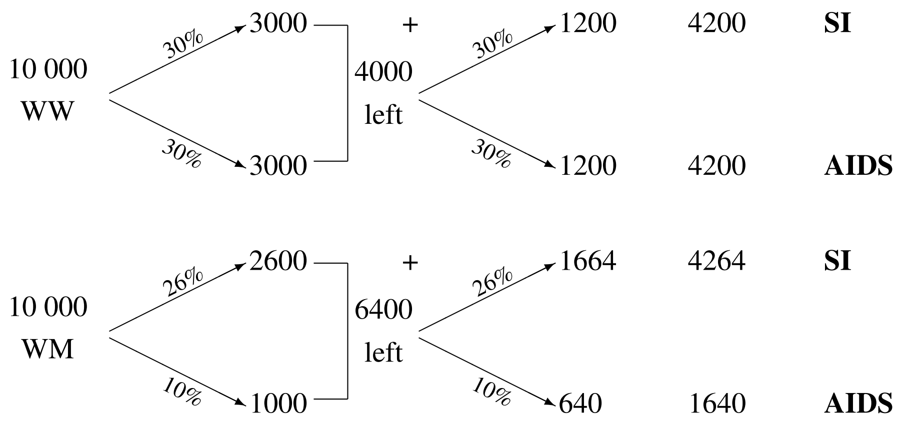
.88

For the WW group, the curves practically overlap. Curves differ slightly more in the WM group, but note that the WM group is much smaller. In the estimate based on the regression model, the cause-specific estimates for the two CCR5 values jump at the same time points. Since the WM curve uses information from the WW group, the jumps are smaller than for the nonparametric estimator.

For SI as end point, the cause-specific hazard ratio was 0.776. In a classical survival setting, a hazard ratio below one implies that the estimated cumulative incidence for the indivduals with the deletion is consistently below the other one. In slide 88, although initially the probability of SI appearance is indeed lower for the persons with the deletion WM, after approximately 9 years the difference decreases rather than increases, and after 11 years the cause-specific cumulative incidence functions even cross. In a model for the cause-specific hazard, the cause-specific cumulative incidence not only depends on the effect of the covariable on that cause, but also on the effects of the covariable on all other causes and on the baseline hazards of all other causes. Although the SI-specific hazard is lower for WM, the AIDS-specific hazard is also lower for WM, and even more so. This affects the impact of CCR5 on the SI-specific cumulative incidence. We can have cumulative incidence curves that cross, even when the cause-specific hazards are proportional. We explain this with a simplified example.

Suppose we have a population of 10000 individuals with the wild type WW and 10000 individuals with the deletion WM. We assume two time points at which events of both types can occur. We assume that WW individuals have a constant cause-specific hazard of 0.3 at both time points, for both SI as well as AIDS. The deletion WM is protective for the SI-specific hazard (hazard ratio $26/30 = 0.87$). However, it is even more protective for AIDS as first event (hazard ratio $10/30 = 0.33$). This latter aspect causes more individuals to remain at risk after the first event time for WM. As a consequence, at the next time point SI appears in more individuals with WM than in individuals with WW (1664 versus 1200). The total effect after the second event time is that the curves for SI have crossed and that the value is higher for individuals with WM than for individuals with WW genotype.

“Reproducing” this result: CS-RH $\neq$ cumulative effect

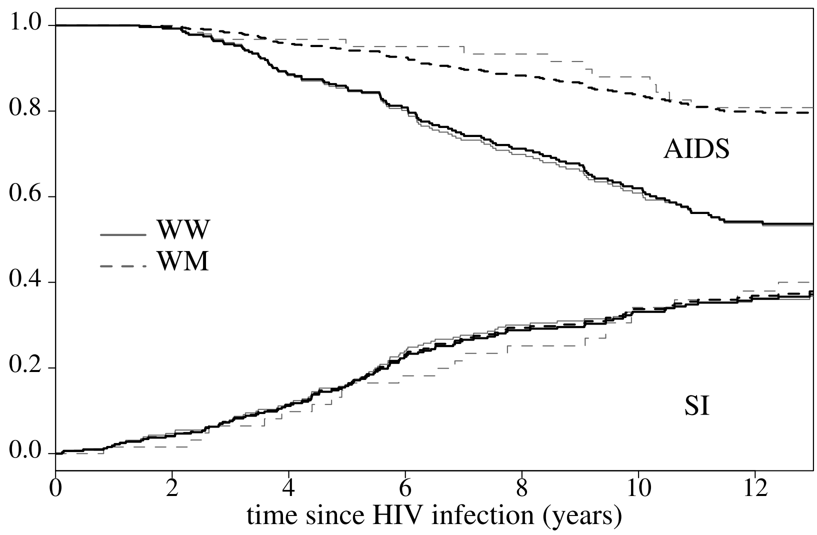


$\mathrm{CS-RH(AIDS)} = 10/30 = 0.33$
$\mathrm{CS-RH(SI)} = 26/30 = 0.87$ .89

In conclusion, if we want to translate effects based on a cause-specific hazards model to the cumulative probability, we need to quantify the effects for every possible combination of covariables. Effects can no longer be quantified via a set of parameters. This makes this approach a bit awkward for investigating effects on the cumulative scale.

In slide 90, we compare the estimates from the Fine and Gray model with the non-parametric estimates (thin grey lines). Contrary to the estimates based on a cause-specific proportional hazards model, the curves cannot cross.

Cumulative incidence based on proportional subdistribution hazards model



.90

18 Which Type of Analysis to Choose?

18.1 Choices...

If there are competing event types, several instantaneous and cumulative quantities can be defined (see slide 17). Also, we have seen that results on the hazard do not always translate to similar results on the cumulative scale. With choice, there is confusion with respect to (i) which quantity do I use in my analysis, (ii) what assumptions do I need to make in order to estimate this quantity and (iii) the interpretation of the results.

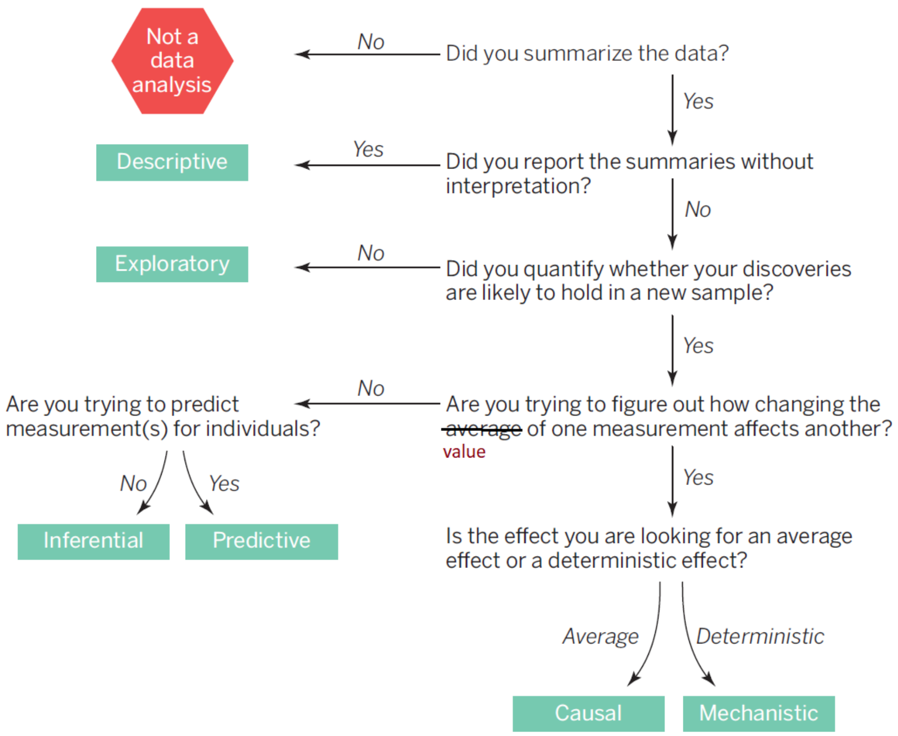
Choices…

- *Interest in the outcome in the presence of competing risks or marginally?*
- *For competing risks analysis: cause-specific hazard or subdistribution hazard?*
- *For competing risks analysis: hazard or cumulative probability?* .91

18.2 Three types of study questions

The appropriate choice relates to the type of study question as well as to the character of the competing risk (biological or human intervention). Typically the study question involves the relation between covariables and outcome, which is translated into the formulation of a regression model. Li *et al.* [35] give an informative flowchart that may help in the decision. Leek & Peng [34] distinguish between six types of study questions. We focus on three of them: inferential, predictive and causal study questions.

Data analysis flowchart



.92

Type of study question

- Inferential: association/correlation; risk factor analysis
- Predictive
  - Relationship at individual level
  - Cumulative event probability most relevant
  - Usually refers to real world, in which competing risks are present
  - Interventions complicate predictions
  - Validation same as in standard time-to-event setting
- Causal (see also Gillies, 2019)
  - Effect at population level, possibly by subgroup
  - Actions, interventions and manipulations
  - Biological mechanisms
  - May refer to counterfactual world without competing risks .93

Inferential questions are about association or correlation. The purpose of the analysis is to determine the existence and strength of a relationship, without formalizing the underlying mechanisms. It may be the most commonly applied type of analysis [34, 46].

Predictive questions are about individual outcomes. The aim is to predict the outcome at the individual level as accurately as possible, preferably using information from few and readily available variables. What makes the competing risks different from the setting with a single event type is that the actual predictions, the cause-specific cumulative incidences, not only reflect the relation between the predictors and the outcome of interest, but also depend on the relation between the predictors and the other outcomes ("in the end we all die, but not all at the same age and from the same cause").

Interventions play a special role in prediction models. They are typically not seen as competing risks. Interventions can be included as variables in the model, but they are decided by humans and do not reflect biological mechanisms in a natural setting. If all individuals are treated according to the same general guidelines, then current practice can be seen as the natural setting. If the decision of when to intervene after baseline varies by setting and location, the model becomes less generalizable and special methods for causal inference are often needed [52].

Validation of prognostic prediction models is based on performance on the cumulative scale. For prediction of a specific event type, methods are essentially the same as in the standard time-to-event setting. Van Geloven *et al.*[53] give an overview of performance measures and modern validation methods.

Gillies [18] describes different approaches to causality in medicine. One philosophical school characterizes causality completely in terms of actions, interventions and manipulations. A randomized controlled trial quantifies the effect of an intervention on the outcome. Its main purpose is to see whether a new intervention improves disease outcome, not to increase understanding of disease mechanisms, and it often does not make the mechanism explicit. Another philosophical school of causality postulates that causality can only be concluded if we can specify a mechanism that explains the relation. According to the Russo-Williamson thesis [44], we need both statistical evidence (often through interventions on humans) and evidence of a mechanism (often through laboratory experiments on animals) to infer causality. For competing risks, it is useful to make a similar distinction between competing risks that are interventions and competing risks that are part of a biological mechanism. We need to decide whether the competing outcomes form part of the causal pathways under study or whether there is only a single outcome that is relevant.

18.3 Etiology: marginal or competing risks analysis

Competing risk intervention or biological?

- Competing risks an intervention: marginal setting can be imagined
  - Example G: hospital discharge
  - Example H-2: death competing risk for AIDS (partly: suicide, overdose)
  - When to start intervention: what happens without intervention?

  Independence of competing risks often problematic
- Competing risk biological but unrelated mechanism
  - Example H-2: death competing risk for AIDS (partly: lung cancer or CVD unrelated diseases)

  Treated and interpreted as censored data
- Competing risk part of same mechanism: competing risks

.94

If the competing risk is an intervention or a direct consequence thereof, it is usually possible to imagine a setting without the competing risk and the marginal distribution may be the one of etiological interest. What is quantified in such analysis has been called a controlled direct effect because it sets the competing events as absent [57]. When comparing two hospitals in Example G, the hospital is a proxy for the exposure (the real exposure is harder to define but it comprises the conditions that impact infection risk in a hospital). The competing risk discharge is an intervention. The marginal distribution describes what would be observed if everyone would stay in hospital. It reflects the etiology of infection as a consequence of food hygienic conditions. [51] discuss a similar type of contrast between marginal and competing risks analysis for patients on dialysis.

In the comparison of time to AIDS for MSM and IDU in Example H-2, the competing risk pre-AIDS mortality covers both interventions (such as suicide) as well as biological mechanisms that cannot be seen separately from the disease process leading to AIDS (such as HIV related infections). The comparison of natural histories refers to a setting without the interventions, but an event like death from HIV related infections should not be ignored. In fact, it had better been included in the definition of AIDS.

When estimating the marginal distribution, the censoring due to the competing risks has to be independent. Then, individuals that are censored because they experience a competing event can be represented by the ones that remain at risk. Censoring due to discharge can be assumed to be independent if the infection rate is the same for every individual. Although the hazard estimate can be interpreted as cause-specific as well as marginal, the resulting estimates of the cumulative probabilities are different: for the net risk or marginal cumulative distribution we use the Kaplan-Meier, for the crude risk or cause-specific cumulative incidence we use the Aalen-Johansen estimator[13]. Often the competing risks are not independent. Then, by censoring individuals when they experience a competing event, we only estimate the cause-specific hazard. The Kaplan-Meier does not represent an interpretable quantity and the marginal distribution cannot be estimated without incorporating extra information or making additional assumptions.

If we want to perform a competing risks analysis, censoring due to a competing risk is not required to be non-informative. The reason is that such censored individuals are not supposed to be represented by the ones that remain in follow-up. We do not want to quantify progression to the event of interest in a situation in which the competing events would not exist. We quantify the rates and risks of the event, taking into account that individuals can also experience some other event instead.

If the competing risk is a biological event, a setting without the competing risk is often irrelevant or difficult to imagine. It is irrelevant for the inferential study question about the impact of the introduction of cART on the spectrum in causes of death in HIV infected individuals (Example B). All competing events are of interest and only a competing risks analysis makes sense. It is also irrelevant for a predictive study question if the competing risks cannot be prevented. A setting without competing risks is difficult to imagine if the question is causal and the competing event is a possible outcome of the same biological mechanism that leads to the event of interest. If the competing risks are different end points of the same process, etiology is best described by the cause-specific hazards. One example is the spectrum of causes of death in HIV infected individuals. Another example is the process of ageing and its effect on the spectrum of causes of mortality. We quantify the cause-specific death rates amongst the individuals that are still alive, i.e. at risk. Since the different causes of death are, at least partly, consequences of the same biological process, individuals that die of one cause are also more likely to die of other causes: time to the competing causes of death is dependent. As a consequence, individuals that die of a competing event are not the same as individuals that remain alive, and the cause-specific hazard is different from the marginal hazard.

[13] Or its equivalent product-limit form based on the subdistribution hazard.

18.4 Competing risks: type of quantity

Cause-specific or subdistribution hazard?

- Both describe competing risks setting
- Cause-specific often closer to causal effects, but any hazard is problematic
- Subdistribution directly relates to cumulative scale
  - Combination of etiology and effect on other event types.
    "Impact" instead of "effect"
  - Regression: parameters directly represent cumulative scale
- Some suggest to report both, especially for inferential study questions (e.g. Latouche *et al.*, 2013)

.95

The Fine and Gray proportional subdistribution hazards model has sometimes been called a competing risks regression model (see e.g. the paper that is referred to in Exercise IV.19.2). The name may give the incorrect suggestion that the regression model for the cause-specific hazard ignores the competing risks issue and therefore does not apply in the presence of competing risks. The misconception that a Fine and Gray model is the only possible approach may be one of the reasons why its results have often been interpreted incorrectly in the medical literature [4]. A regression model for the cause-specific hazard is also a competing risks analysis, even though it uses completely standard approaches with respect to right censoring and left truncation. A regression model on the cause-specific hazard may be better suited for studies on etiology, while a regression model on the subdistribution hazard quantifies what is actually observed over time [33, 55].

Latouche *et al.* [32] suggest to always report results from both hazard regression models, because each may give relevant information that is not revealed by the other. The cause-specific hazards model quantifies the relation with the instantaneous event rate, while the subdistribution hazard directly relates to the cumulative scale. However, we should not blindly restrict to those two models. The choice of model should primarily depend on the type of study question, what we want to report as well as on the structure of the relation between exposure and outcome.

Even though the regression model may be based on the hazard, the cumulative quantity is often more relevant. This is clear when the question is predictive. An individual wants to know the probability of dying within a certain time span, not the probability of dying on a specific day if still alive. It is also the more interpretable quantity when the study question is causal.

Hazard or cumulative probability?

- Prediction: cumulative probability, but may be based on regression model for the hazard
  depends on data characteristics and ease of presentation
- Etiology: value of hazard ratio does not reflect causal effect if there is unexplained variation in distribution (frailty)
- Cumulative probability: regression
  - Via cause-specific hazards, but no one-to-one relation
  - Via subdistribution hazards, one-to-one relation with cumulative scale
  - Directly via cause-specific cumulative incidence or function thereof
    In competing risks setting extra complication because cause-specific cumulative incidence neither reflects causal mechanism

.96

In a standard time-to-event analysis, the strength of the effect is often quantified via the hazard, for example by using a Cox proportional hazards model. The hazard is a probability, conditional on being event free. The value of the hazard ratio may not reflect the size of the causal effect because removing those with an event can generate a selective risk set over time [1, 25]. This even holds in a randomized controlled trial. At baseline the arms are (on average) equal in all characteristics because of the randomisation. Problems occur during the follow-up, unless each individual has exactly the same event risk. This is very unlikely to occur. There are always individuals with higher and with lower event risk that we cannot correct for.

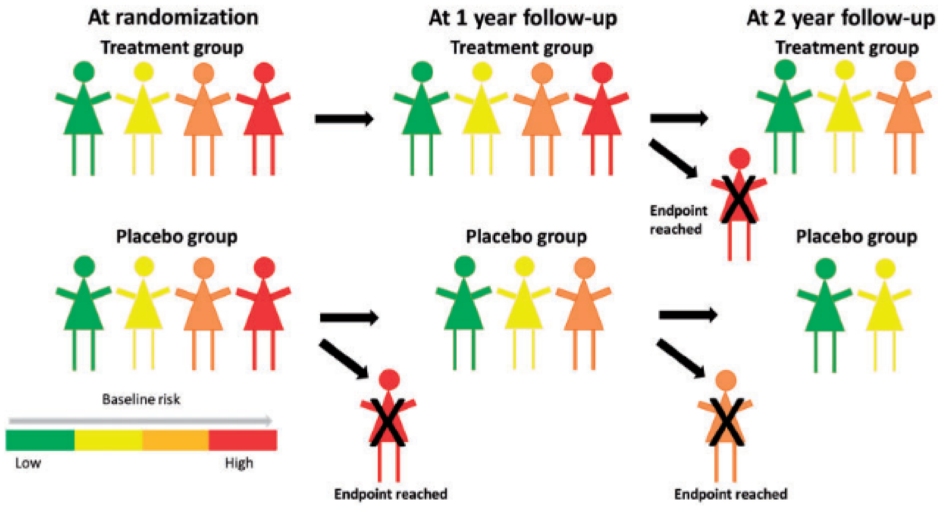
The hazard of hazard ratios

- Individuals in both arms on average equal at randomisation
- Over time
  - high risk individuals more likely to experience event
  - low risk individuals overrepresented
  - stronger selection of low risk individuals in placebo group if treatment reduces event rate
- Observed hazard ratio biased towards "no effect" (may even go in opposite direction)

.97

This selection is illustrated in the figure of slide 98 (Stensrud *et al.* [47]). After 2 years of follow-up, individuals in the placebo group are of lower event risk than in the treatment group. Therefore, the placebo group has a lower hazard than the treatment group.

Built-in selection bias



.98

The cumulative incidence, or functions thereof like the restricted mean survival time/restricted mean time lost, do not suffer from this selection mechanism. A regression model on a cumulative quantity may therefore a better alternative, although there are ongoing discussions on this question..

An effect measure derived from cumulative risk/survival

- *Ratio of cumulative risk*
  Logistic regression at fixed τ:

$$\text{logit}[\text{P}(T \leq \tau \mid \mathbf{Z}_i)] = \beta_0 + \beta_k^\top \mathbf{Z}_i$$

  Proportional odds logistic regression over time range:

$$\text{logit}[\text{P}(T \leq t \mid \mathbf{Z}_i)] = \log[\pi_{k,0}(t)] + \beta_k^\top \mathbf{Z}_i$$

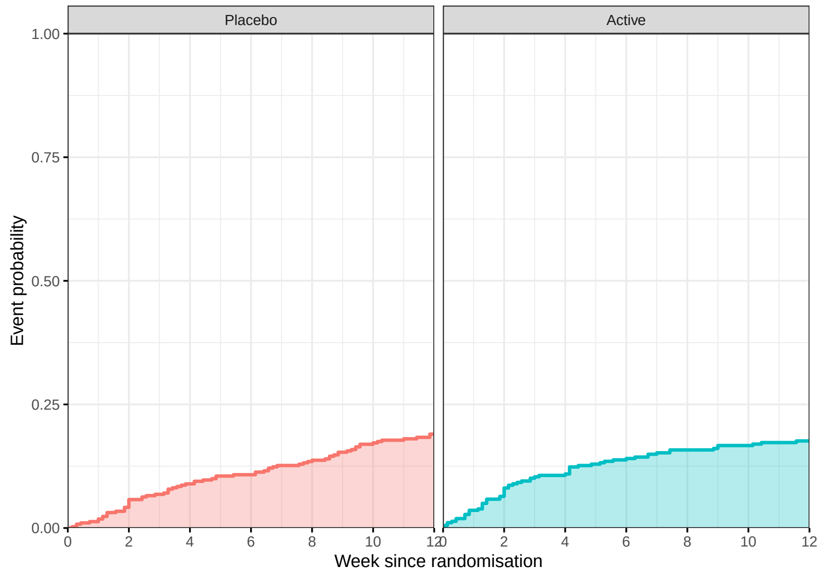
  We can also use difference in cumulative risk as effect measure
- *Flexible accelerated failure time model*
- *Difference (or ratio) of restricted mean survival time (RMST) or restricted mean time lost (RMTL)*
  - Restricted: measured until fixed time point τ

$$E\{\min(T,\tau) \,|\, \mathbf{Z}_i\} = \beta_0 + \beta^\top \mathbf{Z}_i$$
$$\log[E\{\min(T,\tau) \,|\, \mathbf{Z}_i\}] = \beta_0 + \beta^\top \mathbf{Z}_i$$

.99

Restricted mean survival time/time lost

- RMST: mean event-free time up to 12 weeks
- *RMTL: mean time lost within first 12 weeks.* RMTL=12-RMST



.100

In a causal competing risks analysis, selection of an appropriate causal estimand is difficult. The cause-specific hazard has often been suggested, but it suffers from the same selection mechanism as the marginal hazard. Contrary to the marginal analysis, a contrast on the scale of the cause-specific cumulative incidence neither reflects the size of the causal effect, unless the exposure has no causal effect on the competing risk. Otherwise, its value is influenced by the causal effect of the exposure on the competing risks. Stensrud *et al.* [48] recently suggested an estimand that can be used if the effect is separable into components that each only affect either the event of interest or the competing events.

We end by describing the quantities of interest for the examples that we covered in this course.

The examples revisited

- Example G: staphylococcus infection in hospital
  - Etiology (what if everyone would stay in hospital): marginal hazard, Kaplan-Meier. Assumption: competing risk discharge independent
  - Prediction (how many infections are observed in hospital): cause-specific cumulative incidence
- Example B: impact of cART on causes of death.
  - Competing risks
  - Etiology: cause-specific hazard(?)
    Prediction: cause-specific cumulative incidence; subdistribution hazard for regression model
- Example C: SI and AIDS as first event after HIV infection.
  - Competing risks. Marginal analysis possible if other event is ignored
  - Etiology: cause-specific hazard(?)
    Prediction: cause-specific cumulative incidence
- Example H-2: natural history in IDU en MSM.
  - Marginal distribution
  - Problem: competing risks are dependent

.101

19 Exercises part IV

1. **From cause-specific to subdistribution.** Suppose we have two competing event types, say 1 and 2, and two groups, say I and II. Suppose the cause-1-specific hazard is equal in both groups, but the cause-2-specific hazard is larger in group I. What can we say with respect the relative size of the subdistribution hazards for cause 1?
2. **Competing risks is Fine and Gray?** In a study on the effect of the use of β-blockers on prostate cancer (PCa) [22], the authors used a proportional subdistribution hazards model for PCa-specific death, with other causes of death as competing event type. They found that individuals that used β-blockers had a lower subdistribution hazard for PCa-specific death. On the other hand, these individuals had an increased subdistribution hazard for other causes of death, such as cardiac mortality. In the discussion they address the question of whether this increased hazard for other CODs can explain the lower hazard for PCa-specific death. Comment on the following statement:

   > To address this potential bias, we performed all analyses with the Fine and Gray competing risk regression model. In addition, we observed no increase in all-cause mortality among β-blocker users although other-cause mortality was higher, strengthening the interpretation of an association between the use of β-blockers and PCa-specific mortality.

3. **Treatment failure under interventions.** The introduction of combination antiviral therapy (cART) has greatly changed the disease course from HIV infection to AIDS and death. In the search for the best moment to start cART, we investigated whether individuals who start cART in the first year after HIV infection ("early starters") have a different risk of treatment failure than individuals who start later, i.e. during chronic infection [58].

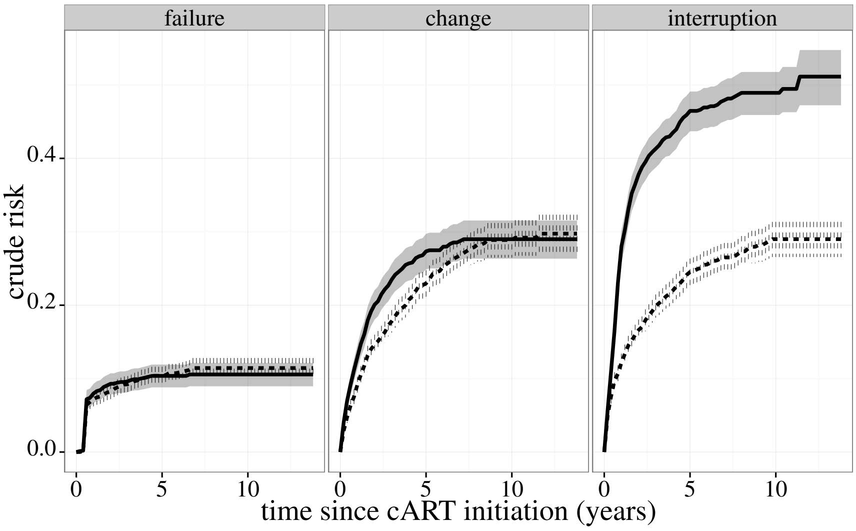


Figure 8: Cause-specific cumulative incidence (with 95% CI) by exposure group. Solid lines: early starters; dashed lines: late starters.

Treatment change and treatment interruption were treated as competing risks. In Figure 8, we plot the cause-specific cumulative incidence estimates for the three event types.
Contrary to expectation, we did not see a lower probability of treatment failure in the group of early starters. However, the early starters were much more likely to interrupt treatment. In order to investigate the influence of potential confounders, we fitted a proportional subdistribution hazards model that included risk group and age, amongst others. Results did not change much. The subdistribution hazard ratio for failure as end point for the early starters versus the late starters was 0.93 (95% CI 0.72-1.20). For treatment change and treatment interruption, the subdistribution hazard ratios were 1.06 (95% CI 0.91-1.24) and 1.54 (95% CI 1.33-1.79) respectively.
Comment on the following explanation for the lack of effect on treatment failure:

> It may be that individuals interrupt treatment because they are responding well to therapy, rather than simply because they are scheduled to do so. In that case, they would be less likely to fail treatment if they had not interrupted and the marginal hazard ratio of failing treatment in early versus chronic infection would be smaller than the one we report.

Part V
Software

20 Nonparametric estimation: Aalen-Johansen

In the text we denote R functions using the full name. In the code we make use of R's object orientation and only use the generic function.

20.1 Single event type

We first give a short introduction to functionality the `survival` package for the classical setting with a single event type. With right censored data, the outcome is represented by a column with the event/censoring times and a status column that tells whether the event was observed.

Right Censored Data

Representation Two columns: **time** since origin and **status** variable. `status=1` if event observed and `status=0` if event right censored

```
id time status
 1  7.7      1
 2  4.3      1
 3  5.6      0
 ...
```

In R via `Surv(time=time, event=status)`

Kaplan-Meier Main function: `survfit.formula`

```
survfit(Surv(time,status)~1, data=...)
```

.102

As example we use the data on time to AIDS/SI switch as first event. To remain in the classical survival setting, we combine both event types. The status column has three values; we use a logical expression to denote occurrence of an event. We show the information for the same four HIV infected individuals as in Section 9.1.

Four example individuals

`aidssi` data set (available in `mstate` package and in Stata)

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

Kaplan-Meier both event types combined:

```
KM.overall <- survfit(Surv(time,status!=0)~1,
                                    data=aidssi)
```

Note: `event` argument can be a logical expression (`status!=0`)
Kaplan-Meier per value of CCR5:

```
KM.ccr5 <- survfit(Surv(time,status!=0)~ccr5,
                                    data=aidssi)
```

.103

Numerical summary

Basic summary:

```
KM.overall

       n events median 0.95LCL 0.95UCL
[1,] 329    222    7.3    6.44    8.36
```

More detailed summary:

```
summary(KM.overall)

   time n.risk n.event survival std.err lower 95% CI upper 95% CI
  0.112    329       1    0.997 0.00303        0.991        1.000
  0.137    328       1    0.994 0.00429        0.986        1.000
  0.474    325       1    0.991 0.00525        0.981        1.000
      .      .       .        .       .            .            .
 12.936     41       1    0.217 0.02604        0.171        0.274
 13.361     22       1    0.207 0.02665        0.161        0.266
 13.936      1       1    0.000     NaN           NA           NA
```

Survival at 12 years obtained via

```
summary(KM.overall, time=12)
```

.104

In R

Uses the function `survfit.formula`, which calls `survfitKM`

```
survfitKM
 function (x, y, weights = rep(1, length(x)),
     stype = 1, ctype = 1,
     se.fit = TRUE, conf.int = 0.95,
     conf.type = c("log", "log-log", "plain", "none",
                                       "logit", "arcsin"),
     conf.lower = c("usual", "peto", "modified"),
     start.time, id, cluster, robust,
     influence = FALSE, type, entry = FALSE, time0 = FALSE)
```

.105

The scale at which the confidence intervals are computed is chosen via the argument `conf.type` in the `survfit.formula` function. The default is `conf.type="log"`, which chooses the cumulative hazard scale as in (7).

Plotting Kaplan-Meier survival curves

`plot.survfit` for first plot:

```
 function (x, conf.int,
   mark.time = FALSE, pch = 3, col = 1, lty = 1, lwd = 1, cex = 1,
   log = FALSE, xscale = 1, yscale = 1, xlim, ylim,
   xmax, fun, xlab = "", ylab = "", xaxs = "r",
   conf.times, conf.cap = 0.005, conf.offset = 0.012,
 conf.type = c("log", "log-log", "plain", "logit", "arcsin", "none"),
   mark.col, noplot = "(s0)", cumhaz = FALSE, firstx, ymin,
   cumprob = FALSE, ...)
```

`fun="event"` plots cumulative incidence (i.e. upwards from 0):

```
plot(KM.overall, fun="event")
```

`lines.survfit` adds curves .106

`ggsurvfit` package

- Extension of `ggplot2`.
  See `https://www.danieldsjoberg.com/ggsurvfit`
- Allows for number-at-risk table, automatic legend
- Basic survival function via

```
ggsurvfit(survfit2(Surv(time,status!=0)~ccr5,
                                   data=aidssi))
```

  Use `survfit2` for nicer labeling of groups
- `type="risk"` plots cumulative incidence
- CIs not plotted by default; use `+ add_confidence_interval()` .107

`survminer` package

- Extension of `ggplot2`.
  See `https://rpkgs.datanovia.com/survminer/index.html`
- Allows for number-at-risk table, automatic legend
- Basic survival function via

```
ggsurvplot(survfit(Surv(time,status!=0)~ccr5,
                                   data=aidssi))
```

- `fun="event"` plots cumulative incidence .108

Results Cox model for combined end point

```
coxph(formula = Surv(time, status != 0) ~ ccr5,
                                   data = aidssi)
```

Result:

```
         coef exp(coef) se(coef)     z       p
ccr5WM -0.701     0.496    0.186 -3.77 0.00016

Likelihood ratio test=16.5  on 1 df, p=4.95e-05  n= 324,
                        number of events= 220
   (5 observations deleted due to missingness)
```

.109

20.2 Aalen-Johansen estimator

Estimation of the cause-specific (cumulative) hazard is completely standard, because those with a competing event are treated as censored observations. Computation of the Aalen-Johansen estimator of the cause-specific cumulative incidence is a bit more involved. However, it is available in the major statistical programs.

Software to compute Aalen-Johansen estimator

- Stata: `stcompet` command[14]
- SAS: macros `%CUMINCID`[15]
  or `%CIF`[16]
- R, some options:
  - standard `survival` package
  - `cmprsk` or `prodlim` package
  - any package for multi-state models, e.g. `etm, mstate, msSurv` .110

[14] See `https://ideas.repec.org/c/boc/bocode/s431301.html`. See also [11]
[15] See `http://support.sas.com/kb/30/addl/fusion_30511_1_cumincid.sas.txt`
[16] See `http://support.sas.com/resources/papers/proceedings12/344-2012.pdf`

Aalen-Johansen estimator

Event column needs to be a *factor*:

```
aidssi$cause <- relevel(aidssi$cause, ref="event-free")
csiSurv <- survfit(Surv(time, event=cause)~1,
                                          data=aidssi)
```

Summary of estimate:

```
summ <- summary(csiSurv, times=seq(2,8,by=2))

summ
 time n.risk n.event Pr((s0)) Pr(AIDS) Pr(SI)
    2    300      15    0.953  0.00632 0.0404
    4    240      52    0.786  0.10322 0.1112
    6    170      54    0.603  0.17070 0.2259
    8    122      41    0.454  0.25430 0.2916
```

Confidence intervals: `summ$lower` and `summ$upper` .111

We use the same `survfit.formula` function as in the situation with a single event type. The difference is that in the specification within the `Surv` function, the `event` argument needs to be a factor variable. Note that the lowest level of the factor is assumed to represent censorings. Since `AIDS` is the lowest level, we need to reorder the levels. With this specification, the `survfitAJ` function is called from within `survfit.formula`. We obtain a detailed numeric description of the estimates using the `summary.survfitms` function[17].

When we apply the `summary.survfitms` function to the estimates, the columns `Pr(AIDS)` and `Pr(SI)` give the estimates $\widehat{F}_k$ for the first (AIDS) and the second cause (SI) respectively. The probability not to have experienced any of the events is in column `Pr((s0))`. No standard errors and confidence intervals are shown, but they have been computed. They can be obtained by selecting the respective components of the output of the `survfit.formula` or `summary.survfitms` function, e.g. `summ$lower` and `summ$upper`. By default, confidence intervals are computed on the log-scale as specified in (7). Estimates can be plotted using the `plot.survfit` and `lines.survfit` functions. The only difference from the classical survival setting is that `fun="event"` is the default, which makes the curves go up from 0 instead of go down from 1. For a curve that goes down from 1 one specifies[18] `fun=\(x) 1- x`.

Plotting

- `survival`: use same `plot` function
  - specific event types selected via `csiSurv["AIDS"]` etc.
  - default curve shown from 0 upwards `fun="event"`
- `ggsurvfit`: use `ggcuminc` function
  - specific event types selected via `outcome` argument
  - alternate and stacked display format not implemented

  ```
  ggcuminc(csiSurv, outcome=c("AIDS","SI"))
  ```
- `survminer`: use `ggcompetingrisks` function
  - less flexible than `ggcuminc` in `ggsurvfit` .112

[17] This function is explained in the same help file as `summary.survfit`.

[18] `fun="identity"` used to work as well. This may be a bug in the current version of `survival`.

21 Nonparametric Estimation of Subdistribution

Before we explain the creation of the time-varying weights that are used in the estimator of the subdistribution hazard, we first briefly describe the counting process formulation for time-to-event data.

21.1 Time-varying values; counting process format

In the counting process format, the follow-up time of an individual is split into intervals that are determined by characteristics that change during his follow-up. These can be final events or end of follow-up, but can also be covariables that change over time, states that he enters or leaves over time or weights that vary over time.

The simplest situation in which the counting process format is useful is when we need to correct for left truncation. Consider the follow-up data from three example individuals in slide 113. An individual's follow-up data is expressed by three variables instead of two: apart from the event or censoring time (`event.time`) and a variable denoting whether the event time is observed or right censored (`status`), a third variable is used that gives the time at which the individual came under observation (`entry.time`). The first individual experienced the event after 4.3 years, and had been in follow-up since her time origin. The second individual, who had also been in follow-up since his time origin, was censored after 5.6 years. The value 3.4 of the third individual describes that he entered the study and came under observation 3.4 years after the event that determined his time origin.

Left truncated data: analysis and data representation

Individuals only contribute while they are in follow-up

- *Data*: extra column, describing entry time in risk set

```
id     entry time    event time      status
 1            0.0           4.3           1
 2            0.0           5.6           0
 3            3.4           7.7           1
```

- In R: `Surv(entry.time, event.time, status)`
- Note: log-rank test function `survdiff` does not work with left-truncated data; use equivalence with score test from Cox model

```
> coxph(
  Surv(entry.time, event.time, status) ~ ccr5,
                                     ...)$score
```

.113

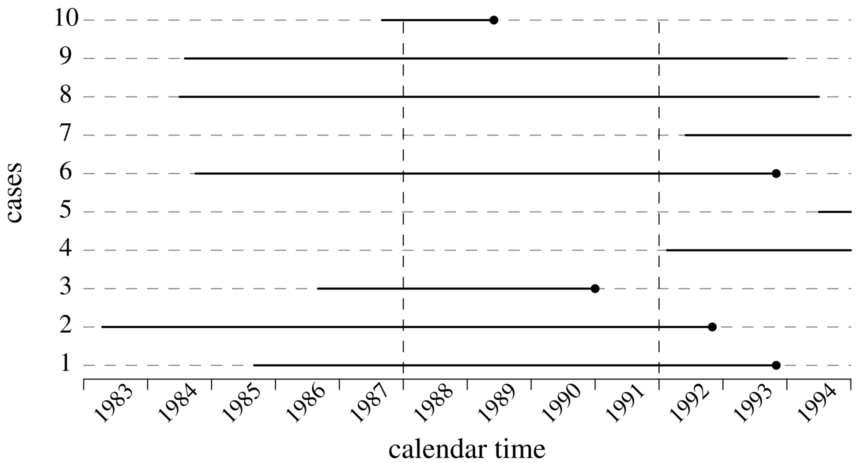
A similar way of presentation is used for time-varying variables. One of the simplest examples of a time-varying covariable is calendar time of follow-up. As an example, suppose we want to investigate whether the time from HIV infection to AIDS has changed over calendar time. Calendar time is categorized into three periods: before 1988, from 1988 to 1992 and after 1992[19]. One way of analysis is to use a fixed covariable "calendar period of HIV infection". However, individuals that became infected before 1988 and had a long follow-up also contribute to later calendar periods. An alternative approach is to use a time-varying covariable "calendar period of follow-up".

In slide 114 we describe the follow-up history over calendar time for ten individuals. Solid lines denote the time from HIV infection until AIDS or censoring per individual. For example, individual 1 became infected in 1985, at 1985.7 if the year is represented numerically, and developed AIDS 8.2 years later at 1993.9. In his personal time scale, his calendar period of follow-up changed at 2.3 years and 6.3 years, and he experienced the event after 8.2 years.

[19] This more or less represents changes in the availability of antiretroviral treatment.

In the data set, his history is split over three rows. At the end of the first period, he is still event-free and the value of his status variable is zero. One can also say that he is censored for follow-up in the first period at 2.3 years. This can be seen as a form of administrative censoring, as if we were performing the analysis at the end of 1988. He enters the second period at 2.3 years. This can be seen as a form of late entry that induces left truncation. He comes under observation for experiencing the event "AIDS between 1988 and 1992" at 2.3 years, whereas an individual that became infected at the same time 1985.7 but developed AIDS within 2.3 years would be missed for the second calendar time period.

Example time-varying covariable



.114

Data format

- start time, stop time and event status at stop time, plus covariable value in interval. Represented by three rows:

```
id      Tstart      Tstop  status    cal.period
 1           0        2.3       0        < 1988
 1         2.3        6.3       0     1988-1991
 1         6.3        8.2       1        > 1991
```

  Enters in risk set "1988-1991" ("> 1991") after 2.3 (6.3) years
- R: `Surv(Tstart, Tstop, status)` ~ `cal.period`
- Similar for discontinuous risk intervals and for time updated weights

.115

21.2 Representation of the weights

To incorporate information on time-varying weights, we use the counting process representation. As an example, consider three individuals with different end points from a larger data set in slide 116. Each has a different end point; the first individual was censored, the second experienced the event type of interest `1` and the third had the competing event `2`.

Example data

Three individuals from data set

id	entry.time	event.time	event.type
1	0.25460	0.63644	0
2	0.00000	0.64358	1
3	0.08005	0.25615	2

$\Longrightarrow$

id	Tstart	Tstop	status	weight.cens	weight.trunc	count	failcode
1	0.25460	0.63644	0	1.00000	1.00000	1	1
2	0.00000	0.64358	1	1.00000	1.00000	1	1
3	0.08005	0.25615	2	1.00000	1.00000	1	1
3	0.25615	0.31778	2	1.00000	1.02941	2	1
3	0.31778	0.37693	2	1.00000	1.02941	3	1
3	0.37693	0.38928	2	1.00000	1.02941	4	1
3	0.38928	0.46029	2	1.00000	1.02941	5	1
3	0.46029	0.50979	2	1.00000	1.02941	6	1
3	0.50979	0.64358	2	0.67849	1.07230	7	1
3	0.64358	0.64724	2	0.67849	1.07230	8	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

.116

We also show the data set in a representation that gives extra follow-up, with weights, to the individuals with a competing event. The follow-up information for the first two individuals does not change: they are at risk with a weight of one from their entry time until they are censored or experience the event of interest. The first row of the third individual refers to his actual follow-up period, again with a weight of one. After the event he remains in the risk set with a weight that changes over time. It needs to be evaluated at all later event times of type 1. Every new weight evaluation generates a new row. Suppose that the first seven observed event times of type 1 after 0.25615 were at 0.31778, 0.37693, 0.38928, 0.46029, 0.50979, 0.64358, and 0.64724. These times are in the **Tstop** column. Since calculations are only made at the event times, as value in the **Tstart** column we can choose any value between the previous event time and the value in **Tstop**. We choose the previous event time. The weights are split over two columns. Column **weight.cens** contains the censoring weights $\{\widehat{\overline{\Gamma}}(\texttt{Tstop}-)/\widehat{\overline{\Gamma}}(t_{(j)}-)\}$; $t_{(j)} = 0.25615$ is his event time, i.e. the value of `Tstop` in his first row. Column **weight.trunc** contains the weights $\{\widehat{\Phi}(\texttt{Tstop}-)/\widehat{\Phi}(t_{(j)}-)\}$ that correct for left truncation.

The censoring weights only change value if there is a censored observation between two subsequent event times. For example, individual 1 was censored at 0.63644 which changes the censoring weight between the event times 0.50979 and 0.64358. Similarly, there were late entries at 0.28737 and 0.56966 (from other individuals in the data). These cause **weight.trunc** to change value at the subsequent event times 0.31778 and 0.64358. Rows in which both columns do not change value can be collapsed without losing information. For example, the period from 0.25615 until 0.50979 can be represented by one row, with weights 1.0000 for **weight.cens** and 1.02941 for **weight.trunc**. The **status** column copies the event status from the original data set. The **count** column counts the row number within individuals. Hence the row with **count** equal to 1 reflects the period when the individual was in actual follow-up. The **failcode** column specifies the cause of interest.

21.3 Creation of the data set with weights

Software

Stata `stcrprep` command
SAS `%PSHREG` macro
R `finegray` function in `survival` package
`crprep` function in `mstate` package .117

In Stata, the data set with the time-varying weights can be computed via the `stcrprep` command[20]. In SAS the macro `%PSHREG` can be used [30]. In R, we can compute the weights using the `finegray` function in the `survival` package. An alternative is the `crprep` function, which is part of the `mstate` package. `stcrprep` and `crprep` can also compute truncation weights. The `finegray` function gives a slightly different data

[20]See `http://repec.org/usug2013/lambert.uk13.pdf`

representation: it creates a data set in which **Tstart** and **Tstop** are based on the censoring times, i.e. the jump times of $\widehat{\widehat{\Gamma}}$, instead of the event times. The relevant information is the same, though.

Assume that the data set for which we showed the first three individuals in slide 116 is called `dataset` in R.

The `crprep` function can be specified as follows:

```
crprep(Tstop="event.time", status="event.type", data=dataset,
   trans=1, cens=0, Tstart="entry.time", id="id", shorten=FALSE)
```

The required arguments `Tstop` and `status` have as value the column names that contain the event times and event types. The values 1 and 0 in `trans` (the event type of interest) and `cens` (the value to denote censored observations) are the defaults and can be left out. If there is left truncation, we specify the entry time via the `Tstart` argument. By default it is assumed to be zero for every individual, which is the situation without delayed entry. The `id` argument can be used if we want to keep the original individual identifiers in the weighted data set. If this argument is not specified, persons are given a number based on their row position in the original data set. By default rows in which the weights do not change are collapsed. This can be overridden via `shorten=FALSE`. Covariables can be copied to the weighted data set via the `keep` argument. The `crprep` function allows for the creation of the weighted data set for several event types at once by specifying them in the `trans` argument.

The `finegray` function uses a formula structure to specify the required weight calculations. It can only compute weights for one event type at a time. It does not yet allow for left truncation.

We apply the `finegray` and `crprep` functions to the `aidssi` data set. The columns **status** and **cause** give the same information on event type, but in a different format. We can use either of them. In `finegray`, all other variables are copied to the new data set. In `crprep`, we need to specify which variables we want to copy via the `keep` argument.

R: create data set with weights

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

```
finegray(Surv(time,cause)~., data=aidssi, etype="AIDS")
```

Note: `event-free` first level in `factor` specification

```
crprep(Tstop="time", status="status", data=aidssi,
                            trans=1, cens=0, keep="ccr5")
aidssi.w <- crprep(Tstop="time", status="cause",
            data=aidssi, trans="AIDS", cens="event-free",
                                            keep="ccr5")
```

Both event types at once via `trans=c("AIDS","SI")` .118

In slide 119 we show some rows from the output of the second specification of the `crprep` function, which was assigned to `aidssi.w`. The `failcode` column gives the event of interest. Individuals with `SI` as first event remain in the risk set with time-varying weights. An example is individual 8, who had an SI switch at 8.605. Note the **count** column, which gives the row number per individual.

Resulting data set (`crprep`)

	id	Tstart	Tstop	status	weight.cens	ccr5	count	failcode
3	3	0.000	2.234	AIDS	1.000	WW	1	AIDS
17	8	0.000	8.605	SI	1.000	WW	1	AIDS
18	8	8.605	8.638	SI	0.991	WW	2	AIDS
19	8	8.638	8.755	SI	0.982	WW	3	AIDS
.	.	.	.	.	.	.	.	.
78	14	0.000	5.054	event-free	1.000	WW	1	AIDS
79	15	0.000	10.196	AIDS	1.000	WM	1	AIDS

`finegray`:

- different column names **patnr, fgstart, fgstop, fgwt**
- **status** converted to numeric; extra **fgstatus** 0-1 column
- no **failcode** column, **count** column not default .119

21.4 The product-limit estimator

After we have added the weights to our data, we can use standard code that has been written for the Kaplan-Meier product-limit estimator. The only requirement is that the code allows for probability weights when computing the estimates.

PL-form: Kaplan-Meier with probability weights

Stata Specify `pweights` option in `stset` command; use standard `sts` command
SAS `PROC LIFEREG`
R `weights` argument in `survfit` function (`survival` package)

```
survfit(Surv(Tstart,Tstop,status=="AIDS")~1,
                  data=aidssi.w, weights=weight.cens)
```

- Additional left truncation: `weights=weight.cens*weight.trunc` .120

If we compute the weights for both event types in one go via the `trans` argument, the result combines the data sets needed for both analyses:

Both event types at once: resulting data set

	id	Tstart	Tstop	status	weight.cens	ccr5	count	failcode
3	3	0.000	2.234	AIDS	1.000	WW	1	AIDS
17	8	0.000	8.605	SI	1.000	WW	1	AIDS
18	8	8.605	8.638	SI	0.991	WW	2	AIDS
19	8	8.638	8.755	SI	0.982	WW	3	AIDS
.	.	.	.	.	.	.	.	.
78	14	0.000	5.054	event-free	1.000	WW	1	AIDS
79	15	0.000	10.196	AIDS	1.000	WM	1	AIDS
			⋮					
2768	3	0.000	2.234	AIDS	1.000	WW	1	SI
2769	3	2.234	2.322	AIDS	0.997	WW	2	SI
2770	3	2.322	2.982	AIDS	0.993	WW	3	SI
.	.	.	.	.	.	.		
2870	8	0.000	8.605	SI	1.000	WW	1	SI
2913	14	0.000	5.054	event-free	1.000	WW	1	SI
2914	15	0.000	10.196	AIDS	1.000	WM	1	SI
2915	15	10.196	10.448	AIDS	0.974	WM	2	SI
2916	15	10.448	10.467	AIDS	0.961	WM	3	SI
.	.	.	.	.	.	.	.	.

.121

The first part represents the data with weights for `AIDS` as event of interest. These rows have the value `AIDS` in the `failcode` column. The second part represents the data with weights in which `SI` is the event of interest. If we want the SI-specific cumulative incidence, we restrict to the rows that have `failcode` equal to `SI` and use

```
survfit(Surv(Tstart,Tstop,status=="SI")~1, data=aidssi.w,
                  weights=weight.cens, subset=failcode=="SI")
```

Both event types at once: estimation

```
csiPL <- survfit(
       Surv(Tstart,Tstop,status==failcode)~failcode,
              data=aidssi.w, weights=weight.cens)
csiPL$strata
## failcode=AIDS   failcode=SI
##           310           310
summary(csiPL["failcode=SI"],times=seq(2,10,by=2))
```

and obtain

```
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    2    302      13 0.959615  0.0110     0.938344     0.981369
    4    272      22 0.888783  0.0177     0.854687     0.924239
    6    218      34 0.774112  0.0240     0.728466     0.822619
    8    190      18 0.708440  0.0265     0.658364     0.762324
   10    161      11 0.665410  0.0279     0.612936     0.722377
```

or use `csiPL[2]` .122
The specification `~failcode` calculates estimates per value of `failcode`, i.e. for both end points separately. This specification is similar to the computation of the Kaplan-Meier per value of CCR5 in slide 103. Per value of `failcode`, the event of interest occurs when the `status` variable has the same value as the `failcode` variable, as specified in the logical statement `status==failcode`. Characteristics of each of the two estimates can be selected via square brackets, in which we define the value of `failcode` that we want. Instead of `csiPL["failcode=SI"]`, we can also use the ordering of event types as assumed by the `survfit` function. The ordering is the same as in the `strata` component of the output. We see that `SI` is the second event type and we can write `csiPL[2]`.

21.5 Log-rank tests

The log-rank test for equality of cause-specific hazards can be performed with code that performs the standard log-rank test. In R we use the `survdiff` function. For SI as end point we write

```
survdiff(Surv(time,status==2)~ccr5, data=aidssi)
```

and obtain

```
n=324, 5 observations deleted due to missingness.

           N Observed Expected (O-E)^2/E (O-E)^2/V
ccr5=WW 259       84     79.2     0.290      1.15
ccr5=WM  65       23     27.8     0.827      1.15

 Chisq= 1.1  on 1 degrees of freedom, p= 0.284
```

We use Gray's log-rank test to test whether the cause-specific cumulative incidences differ by CCR5 genotype. It has been implemented in the macro %CIF in SAS. In R, we can use the cmprsk package. We can test for a difference in progression to SI and AIDS by CCR5 genotype in the aidssi data via

```
with(aidssi, cuminc(time, status, group=ccr5)$Tests)
```

and obtain

```
           stat           pv df
1 13.149851204 0.0002875421  1
2  0.005817877 0.9392003078  1
```

The first rows is the test result for cause 1, i.e. AIDS, the second for SI. An alternative is to use code for the standard log-rank test with the extended data set that is used for estimation of the subdistribution hazard. Since the subdistribution hazards are estimated and compared per group level, the weights in the PL-form should be computed separately as well.

Gray's log-rank test

- Implemented directly, using original (unweighted) data set

  SAS: %CIF macro

  R: cuminc function in cmprsk package

- Use score test from proportional subdistribution hazards model
  - Compute censoring weights per subgroup (e.g. via strata argument in crprep function)

    ```
    aidssi.wCCR <- crprep(Tstop="time",
      status="cause", data=aidssi,
      trans=c("AIDS","SI"), cens="event-free",
      id="patnr", strata="ccr5")
    ```

  - Variance of the test statistic is different from the one suggested by Gray. Affects p-value

.123

In stcrprep (Stata) we use byg(ccr5). In %PSHREG (SAS) we use the by argument. In R we specify Surv(...)~ccr5 in the finegray function. Since the data is in counting process format, the survdiff function cannot be used. We use the score test from the proportional subdistribution hazards model. The coxph can incorporate time-varying weights in the same way as the survfit function does. We need to distill the value of the test statistic and calculate the p-value ourselves.

R example code

```
test.AIDS <- coxph(Surv(Tstart,Tstop,status=="AIDS")~ccr5,
                    data=aidssi.wCCR, weights=weight.cens,
                                 subset=failcode=="AIDS")
test.SI <- coxph(Surv(Tstart,Tstop,status=="SI")~ccr5,
                   data=aidssi.wCCR, weights=weight.cens,
                                   subset=failcode=="SI")
```

Score test statistic and p-value

```
## AIDS
c(test.AIDS$score, 1-pchisq(test.AIDS$score,1))
[1] 12.428139562  0.000422913
## SI
c(test.SI$score, 1-pchisq(test.SI$score,1))
[1] 0.007207246 0.932344471
```

.124

We see that the results are comparable to Gray's test as implemented in the cmprsk package.

22 Fine & Gray model

Proportional subdistribution hazards (Fine & Gray)

- Implemented directly (using robust estimator of standard error; no left truncation)

  Stata `stcrreg` command

  SAS `PROC PHREG` procedure (SAS/STAT $\geq$ 13.1)

  R `crr` function in `cmprsk` package;
  `FGR` function from `riskRegression` package provides a formula interface
- Compute weights and use weighted Cox model
  - `%PSHREG` in SAS can compute weights and fit model in one go
- None allows weight calculation to depend on covariables via regression model
- Translate to cumulative incidence: use standard code for prediction based on results of Cox model with weights

.125

There are a couple of programs that calculate the weights and fit the model in one go. In Stata from version 11 onwards, the `stcrreg` command can be used[21]. The `stcurve` command can be used to plot the cause-specific cumulative incidence for a combination of covariables [23]. In SAS from version 9.4 onwards, the `PROC PHREG` procedure can be used[22]. In R, we can use the `crr` function in the `cmprsk` package. The `FGR` function in the `riskRegression` package provides a formula interface to the `crr` function. Only censoring weights are computed; weights to correct for late entry are not incorporated. The weights are based on the product-limit statistic of the censoring times, possibly stratified by a categorical variable. All use the sandwich estimator of the standard error.

An alternative approach is to create the weights and use a procedure to fit a Cox model, in which we specify the weights. In Section 21.2 we mentioned how to create the weights in Stata (via `stcrprep`), in SAS (via `%PSHREG`) and in R (via `crprep` or `finegray`). None of the functions currently allow the weight function to depend on covariables via a regression model. The SAS macro by default fits the model in one go, but one can choose to only generate the data set. In Stata and R the Cox model is fitted as a second step. If we cluster by person identifier, we obtain the sandwich estimator for the standard error.

We show how the model is fitted in R, using the CCR5 example. We use the `crprep` function and have the `ccr5` variable included in the data set via the `keep` argument (if we use one overall weight function) or the `strata` argument (if the weight function depends on the CCR5 value). We called the resulting data sets `aidssi.w` and `aidssi.wCCR` respectively.

We use the standard `coxph` function to fit the model, but specify weights. If we do not want to use the sandwich estimator of the standard error, we need to specify `robust=FALSE`. After the model has been fitted, we can use the functions that have been written to digest and validate results from the `coxph` function.

23 Prediction

A regression model on the cause-specific hazard can be fitted using standard software for hazard based mdoels. If we want to fit a single model for multiple event types, we first

[21] See http://www.stata.com/stata11/stcrreg.html

[22] See https://support.sas.com/rnd/app/stat/papers/2014/competingrisk2014.pdf

need to create the stacked data set. One way to to this is by using the `crprep` function and specify all the event types in the `trans` argument. We obtain a data set with the weights that are needed to estimate the subdistribution hazard. However, by restricting to the first row (`count==1`), we can perform analyses on the cause-specific hazard. We may have to create type-specific covariables as well. Often, the most straightforward approach is to create them yourself. However, the `expand.covs` in the `mstate` package function may come in handy.

From cause-specific hazard regression to cumulative incidence

Stata `stpm2` for parametric hazard model; `stpm2cif` translates to cumulative incidence

SAS Macros `CumInc.sas` and `CumIncV.sas`

R recent version of `survival` package
`mstate` package

.126

To our knowledge, in Stata there is no command that translates results from a semi-parametric proportional cause-specific hazards model to the cumulative scale. However, the command `stpm2` fits proportional hazards models in which the baseline hazard is a smooth parametric function of time. If the baseline hazard is modeled flexibly enough, results are similar to those from the semiparametric model. The `stpm2cif` command translates estimates to the cause-specific cumulative incidence [27]. In SAS, estimates from a proportional cause-specific hazards model can be translated to the cause-specific cumulative incidence via the macros `CumInc.sas` and `CumIncV.sas`[23]. In R the `survival` package can be sued, but we prefer the `mstate` package. It allows to quantify the effect of covariables on the cumulative scale in more general Markov multi-state models (see also Chapter 3 of [15]).

To obtain predictions based on the proportional subdistributions hazards model, we create a data frame with all possible covariable combinations and use this in the `newdata` argument of the `survfit` function.

From Fine & Gray to cumulative incidence

```
fit.FG <- coxph(Surv(Tstart,Tstop,status==failcode) ~
              ccr.comb + strata(failcode), data=aidssi.w,
                 robust=FALSE, weights=weight.cens)
```

Predict for four possible combinations

```
indivs <- data.frame(
         ccr.comb=factor(c("WW","WW","WM.AIDS","WM.SI")),
                    failcode=c("AIDS","SI","AIDS","SI"))
pred.sdh <- survfit(fit.FG, newdata=indivs)
```

Results per combination, e.g. for SI and `WW`:

```
pred.sdh[2]
```

.127

We can use the selection function for the `survfit` output to summarize and plot the curves for a subset of the individuals. Since rows 2 and 4 in `newdata` contain the data for SI as end point, `pred.sdh[c(2,4)]` gives the estimates of the SI-specific cumulative incidences for both values of CCR5. In obtaining the figure from slide 90, we first computed and plotted the nonparametric estimates, after which the four curves based on the proportional subdistribution hazards model were added via

```
lines(pred.sdh[c(2,4)], fun="event", lty=1:2, lwd=2)
lines(pred.sdh[c(1,3)], lty=1:2, lwd=2)
```

[23]See `http://staff.pubhealth.ku.dk/~pka`

24 Computer Practicals

The data we use for the computer practicals on competing risks and multi-state models comes from the EBMT registry and contains information on 1977 patients who received an allogeneic bone marrow transplantation. Bone marrow transplantation is one of the ways to treat patients with leukemia. The process after allogeneic bone marrow transplantation, i.e. transplantation from a genetically non-identical donor, can be described via the states as in Figure 9. An adverse event is the acute reaction of the immune system in the donor graft against normal host tissues, which is called acute graft-versus-host disease (AGvHD). Another intermediate event is relapse, and death is the final event. A beneficial event is the recovery of platelet counts to normal levels.

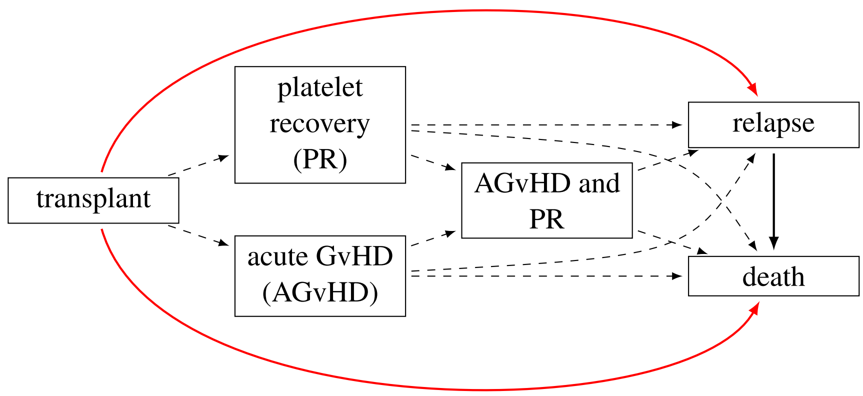


Figure 9: Example: events after bone marrow transplantation.

All these states have been used to model and predict the disease course after transplantation [56]. A simplified version was used in a tutorial on competing risks [41]: AGvHD was ignored and relapse and death were combined into one state.

In the computer practicals on competing risks we only consider relapse and death as competing risks from the disease-free state, while in the multi-state class we additionally consider relapse as a possible intermediate state on the pathway to death. Hence we leave out the states in the middle and can describe our model via the states that are connected via the solid arrows in Figure 9. We investigate the role of the EBMT risk score[24]. This score is a combination of donor-recipient match and age of the recipient, among other things. It divides patients into three risk groups. We quantify the effect of the EBMT score on all three transitions and we test whether the effects on the transitions are different. We also investigate whether progression from relapse to death depends on year of relapse. Treatment of relapse has improved over time, and some clinicians have gone so far as to suggest that having a relapse does not influence mortality anymore.

The data can be found in the `mstate` package. Make sure that you have installed `mstate`. We load the package and the data via

```
library(mstate)
data(ebmt1)
```

Time is measured in days from transplantation. The EBMT risk score is called **score**.

For visualisation, we use the standard plotting functionality in the `survival` and `mstate` packages. However, we will also show some functionality in the `ggsurvfit` package. This package builds upon `ggplot2`, which not only gives more beautiful visualisations, but it also often requires less coding. However, some type of visualisations are much harder to obtain.

[24]EBMT stands for European Group for Bone and Marrow Transplantation.

1. **A first look at the data** Have a look at the help file of the `ebmt1` data set to get information on the meaning of the variables and their possible values.
   Inspect the data by looking at the first couple of individuals or by viewing the complete data set. How many rows and columns does the data set have? Look at the type of the variables (count, continuous, binary, multinomial, ordinal).
   Summarize all the variables. Why are there 1521 `NA`'s in the variable **yrel**? How many relapses were observed and how many deaths?
2. **Estimation of overall survival** Compute the Kaplan-Meier estimator for relapse-free survival. Relapse-free survival means that the individual neither had a relapse nor died, i.e. both end points are combined as an event. Obtain the estimates of relapse-free survival at one and five years. Plot the estimate of the cumulative incidence (the complement of the Kaplan-Meier survival curve) using the `plot` function in the `survival` package, as well as using the corresponding functions in the `ggsurvfit` and `survminer` packages. Change the arguments until you get a plot that is ready to publish.
3. **A Cox model for overall survival** Fit a Cox model for the effect of the EBMT score on relapse-free survival. Locate the hazard ratios. What is your conclusion with respect to the effect of the EBMT score?
4. **Some data preparation** Before we can start with the analyses in which relapse and death are seen as competing risks, we need to do some data preparation. In the `ebmt1` data set, information with respect to relapse and death is given in separate columns. In a competing risks analysis, the usual data presentation is via single time and status columns.
   Add three more columns to the `ebmt1` data set. One column, to be called **time**, contains the time of the first of the two events, relapse and death, or the time of censoring if neither was observed. Use year instead of day as time unit. The second column, to be called **stat**, is a column that has the value 0 for right censored individuals, 1 if there was a relapse and 2 if the individual died without having had a relapse. Add another column that explains the meaning of 0, 1 and 2: `"Event-free"`,`"Relapse"` and `"Death"`.
   R Hint. The function `pmin` computes the minimum value of two of more columns of numeric variables. It is a vectorized function, meaning that the minimum per row is computed. It is a vectorized function, meaning that the minimum per row is computed. Since death never occurs before relapse, we can see which individuals are censored via the **srvstat** column. There are many ways to add the columns, but one efficient way is the following.

```
ebmt1 <- within(ebmt1,{
  time <- pmin(srv,rel)/365.25
  stat <-  ifelse(rel<srv,relstat,srvstat*2)
  type <- factor(stat,
                 labels=c("Event-free","Relapse","Death"))
  })
```

5. **Estimation of cause-specific cumulative incidence (Aalen-Johansen estimator)** Estimate the cause-specific cumulative incidence for relapse and relapse-free mortality. Save the result in an R object. What is the probability, with 95% confidence intervals (on the default log scale), to have a relapse within one year and within five years.
   Compare the estimates at 1 and 5 years of both the relapse-specific and the death-specific cumulative incidence with the relapse-free survival distribution, i.e. the distribution of time to remain free of both event types.
6. **Some plots** (a) Plot the estimated cause-specific cumulative incidence for each end point using the overlaid display format. Plot the 95% confidence intervals as well.

(b) Plot the estimated cause-specific cumulative incidence using the stacked format (without the confidence intervals). First plot the relapse-specific cumulative incidence, and plot the death-specific cumulative incidence on top of this curve.
(c) Plot the cause-specific cumulative incidence estimates for each end point using the alternate display format.

7. **Regression on the cause-specific hazard** Quantify the effect of the EBMT score on both events by fitting two separate cause-specific proportional hazards models. Store the computations in `fit.rel` and `fit.death` respectively. Does a higher value of EBMT risk score increase the cause-specific hazards? Are the effects of EBMT score comparable for both event types?

8. **Estimation of cause-specific cumulative incidence (PL-form)** In practical 4, we used the Aalen-Johansen estimator of the cause-specific cumulative incidence. Now we will use the product-limit form to compute the estimate.

   - Use either the `crprep` function or the `finegray` function to create the data set with weights for relapse as end point. Have a look at its help file. Store the data set as an object named `Webmt1`. Include the covariables **score** and **age** and also store the **type** column in the new data set.

     Inspect the first ten rows of the weighted data set. What do the newly created variables represent? How many rows does the newly created data set have?

   - Compute the PL-form of the estimator of relapse-specific cumulative incidence and plot the estimate of the relapse-specific cumulative incidence. Compare the estimates and confidence intervals.

9. **Impact of EBMT risk score** Compute the nonparametric estimator of the cause-specific cumulative incidence for each of the three EBMT risk scores and for both end points using the product-limit form.

   - If we want to calculate the product-limit form that is equivalent to the Aalen-Johansen form for each value of **score**, the weights based on the censoring distribution need to be calculated separately as well. Create a data set that has the score-specific weights and store it in `Webmt.score`.

   - Make plots to compare the results with the curves when we use the same weights per value of EBMT score, as was created in the data set `Webmt1`. Use one plot window for relapse and another one for death. What do you conclude with respect to the choice of the weight function?

   - The statistic $\widehat{\widehat{\Gamma}}$ can be interpreted as an estimate of the time to censoring distribution. Estimate and plot the time-to-censoring distribution for each value of **score**. Does the distribution depend on the value of the EBMT risk score? Does this observation correspond to the amount of dependence on the weight function in the estimate of the crude risk?

10. **Log-rank tests** Run and compare the three commands given below. Why is the output from the first and the second different, and what do these commands test? Why do the second and the third give the same value of the test statistic?

```
coxph(Surv(Tstart,Tstop,status==1)~score, data=
 subset(Webmt.score,failcode==1), weights=weight.cens)$score
coxph(Surv(Tstart,Tstop,status==1)~score,
        data=subset(Webmt.score,failcode==1&count==1))$score
survdiff(Surv(time,stat==1)~score,data=ebmt1)
```

11. **Impact of EBMT score on subdistribution hazards** We compute the cause-specific cumulative incidence based on a proportional subdistribution hazards model. We first look at the parameter estimates from this model.

- Quantify the impact of EBMT score on the subdistribution hazards for both event types. Compare the parameter estimates with the ones from the proportional cause-specific hazards model. Can you explain the difference?
- Use the results from the proportional subdistribution hazards model to estimate the cause-specific cumulative incidence for each value of risk score and both event types. Make a graph that compares these estimates with the nonparametric estimates. Use the overlaid format for the three values of risk score and the separate format for the two event types.

12. **Regression on the cause-specific hazard, combined analysis** Repeat the analyses, but now by fitting one proportional cause-specific hazards model for both competing risks at once. Fit the model in three ways:
    1. By including an interaction between the EBMT score and the type of end point
    2. By creating a single compound variable that combines the EBMT risk score and the type of end point. You can use the code
    3. By creating type-specific covariables.

    Do all approaches give the same estimates? If not, can you translate one output into the other? Compare the estimates with those from the separate analyses.
13. **Test for equality of effects** Test the null hypothesis that the hazard of medium risk, relative to low risk, is the same for both event types. And similarly for high risk, relative to low risk.
    Perform the likelihood ratio test of the null hypothesis that the effect of both medium and high risk is the same for both event types. The likelihood ratio test is performed via the `anova` function.

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Symbol Description

$F(t)$ cumulative incidence function/net risk, all event types combined
$F_k(t)$ cause-specific cumulative incidence function/crude risk
$\overline{F}$ $1-F$, "survival" function. Similarly: $\overline{F_k} = 1 - F_k$ etc.
$\mathrm{P}_{gh}(s,t)$ transition probability
T random variable that has F as distribution: $T \sim F$
T_k random variable that has F_k as distribution: $T_k \sim F_k$
C time of censoring
Γ distribution of censoring time, $C \sim \Gamma$
L entry time
Φ distribution of entry time, $L \sim \Phi$
$h(t)$ overall hazard, all event types combined
$h_k(t)$ subdistribution hazard
$\lambda_k(t)$ cause-specific hazard
$\lambda_{gh}(t)$ transition hazard
$r(t)$ observed number at risk
$r^*(t)$ number at risk in estimate of subdistribution hazard
$R(t)$ risk set
$\widehat{F}^{\mathrm{PL}}$ product-limit estimator of F (Kaplan-Meier)
$\widehat{F}^{\mathrm{EC}}$ ECDF estimator of F
$\widehat{F_k}^{\mathrm{AJ}}$ Aalen-Johansen estimator of F_k
$\widehat{F_k}^{\mathrm{PL}}$ product-limit estimator of F_k
$\widehat{F_k}^{\mathrm{EC}}$ ECDF estimator of F_k
$\widehat{\Lambda}$ Nelson-Aalen estimator of cumulative hazard
$t_{(i)}$ timing of i-th observed event (in numerical order)
$c_{(i)}$ timing of i-th observed censoring (in numerical order)
$v_{(i)}$ timing of i-th observed entry (in numerical order)
$d(t)$ number of events, of any type, at time t
$d_k(t)$ number of events of type k at time t
$d_{gh}(t)$ number of transitions from g to h at time t
m_i number of censorings at time $c_{(i)}$
w_i number of individuals entering the risk set at time $v_{(i)}$
$\widehat{N}$ virtual sample size, including missed individuals

Abbreviations

AIDS acquired immune deficiency syndrome
cART combination anti-retroviral therapy
CCR5 CC chemokine receptor 5
CI confidence interval
COD causes of death
EBMT European group for bone marrow transplantation
ECDF empirical cumulative distribution function
HCV hepatitis C virus
HIV human immunodeficiency virus
IDU injecting drug user
MSM men who have sex with men
NSI non syncytium inducing
PL product-limit
SI syncytium inducing